



# TIN CAN PORT AND APAPA PORT MODERNIZATION PROJECT

## CONSTRUCTION ENVIRONMENTAL AND SOCIAL MANAGEMENT PLAN

### PREPARED BY:



**ITB Nigeria LTD**

[www.itbng.com](http://www.itbng.com)



**Hitech Construction  
Company LTD**

[www.hitech-company.com](http://www.hitech-company.com)

Final Version

January 2026

**Document Details**

|                        |                                                                                                                                    |             |              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|
| <b>Document Title:</b> | Construction Environmental and Social Management Plan (CESMP) for the Tin Can Port and Apapa Port Modernization Project in Nigeria |             |              |
| <b>Status:</b>         | Revised Report submitted to Lenders                                                                                                |             |              |
| <b>Version:</b>        | 02                                                                                                                                 | <b>Date</b> | January 2026 |

**Document Revisions**

| Rev | Date     | Prepared by               | Revised by                  | Approved by                      |
|-----|----------|---------------------------|-----------------------------|----------------------------------|
| 00  | Nov 2024 | Carlos Silva, ESG Advisor | Phillip Geddes, HSE Manager | Ramzi Chidiac, Managing Director |
| 01  | Nov 2025 | Carlos Silva, ESG Advisor | Phillip Geddes, HSE Manager | Ramzi Chidiac, Managing Director |
| 02  | Jan 2026 | Carlos Silva, ESG Advisor | Phillip Geddes, HSE Manager | Ramzi Chidiac, Managing Director |



## Table of Contents

|                                                                                |           |
|--------------------------------------------------------------------------------|-----------|
| <b>List of Abbreviations.....</b>                                              | <b>7</b>  |
| <b>1. Introduction.....</b>                                                    | <b>8</b>  |
| 1.1 Project Background .....                                                   | 8         |
| 1.2 Background .....                                                           | 8         |
| 1.3 Project Overview .....                                                     | 9         |
| 1.3.1 Tin Can Port Modernization .....                                         | 10        |
| 1.3.2 Apapa Port Modernization.....                                            | 10        |
| 1.3.3 Ancillary Sites.....                                                     | 11        |
| 1.4 Presentation of the Project Entities .....                                 | 12        |
| 1.4.1 The Project Promoter, Buyer and Borrower .....                           | 13        |
| 1.4.2 The Contractor ITB – Hitech Joint Venture .....                          | 13        |
| 1.5 CESMP Objectives .....                                                     | 13        |
| <b>2. Organizational Structure and Policies.....</b>                           | <b>15</b> |
| 2.1 Organizational Structure and Responsibility .....                          | 15        |
| 2.2 All Employees .....                                                        | 20        |
| 2.3 Subcontractors .....                                                       | 21        |
| 2.4 Suppliers.....                                                             | 21        |
| 2.5 NPA Port Management / Operations .....                                     | 22        |
| 2.6 NESREA.....                                                                | 22        |
| 2.7 NPA Marine Department.....                                                 | 22        |
| 2.8 NPA Security Department .....                                              | 23        |
| 2.9 NPA Safety / EHS Unit.....                                                 | 23        |
| 2.10 Port Safety & Security Coordination Committee (PSSCC) .....               | 23        |
| 2.11 Emergency Services (NPA Fire Service + External Responders) .....         | 23        |
| 2.12 Terminal Operators.....                                                   | 23        |
| 2.13 National Environmental Consultant .....                                   | 24        |
| 2.14 Policies.....                                                             | 24        |
| 2.14.1 ITB Nigeria Limited .....                                               | 25        |
| 2.14.2 Hitech Construction Africa Limited.....                                 | 25        |
| <b>3. Construction Environmental and Social Management and Monitoring Plan</b> | <b>28</b> |
| 3.1 Monitoring Management.....                                                 | 28        |



|           |                                                                            |           |
|-----------|----------------------------------------------------------------------------|-----------|
| 3.1       | Construction Environmental and Social Management and Monitoring Plan ..... | 30        |
| <b>4.</b> | <b>Occupational Health and Safety Management .....</b>                     | <b>39</b> |
| 4.1       | OHS Overview .....                                                         | 39        |
| 4.1.1     | Personal Protective Equipment (PPE) .....                                  | 39        |
| 4.1.2     | Safety .....                                                               | 39        |
| 4.2       | Management of Work .....                                                   | 43        |
| 4.2.1     | Targets and Performance Monitoring of the project H&S .....                | 43        |
| 4.2.2     | Communication and Consultation with the Workforce .....                    | 45        |
| 4.2.3     | Financial Integrity and Ethical Community Engagement .....                 | 50        |
| <b>5.</b> | <b>Waste Management.....</b>                                               | <b>51</b> |
| 5.1       | Waste Management Procedures.....                                           | 51        |
| 5.1.1     | Methodological Approach .....                                              | 51        |
| 5.1.2     | Storage and Containment of Waste.....                                      | 52        |
| <b>6.</b> | <b>Emergency Responsiveness.....</b>                                       | <b>59</b> |
| 6.1       | Emergency Scenario Identification and Procedures .....                     | 59        |
| 6.1.1     | Falls from Heights: .....                                                  | 59        |
| 6.1.2     | Fire Outbreaks: .....                                                      | 59        |
| 6.1.3     | Hazardous Material Spill: .....                                            | 59        |
| 6.1.4     | Marine Incidents.....                                                      | 60        |
| 6.2       | Emergency Response Team (ERT) .....                                        | 60        |
| 6.3       | On-Site Medical Clinic .....                                               | 60        |
| 6.4       | Emergency Equipment .....                                                  | 60        |
| 6.5       | Incident Reporting and Investigation Procedure.....                        | 60        |
| 6.6       | Evacuation and Coordination with External Services.....                    | 62        |
| 6.7       | Inspections and Drills .....                                               | 62        |
| 6.8       | Emergency Contact Information .....                                        | 62        |
| <b>7.</b> | <b>Security management .....</b>                                           | <b>63</b> |
| <b>8.</b> | <b>Training.....</b>                                                       | <b>63</b> |
| 8.1       | Training Strategy.....                                                     | 63        |
| 8.1.1     | Needs Assessment: .....                                                    | 63        |
| 8.1.2     | Objectives and Learning Outcomes: .....                                    | 63        |
| 8.1.3     | Training Programs: .....                                                   | 63        |
| 8.1.4     | Training Delivery: .....                                                   | 64        |



|                        |                                                   |           |
|------------------------|---------------------------------------------------|-----------|
| 8.1.5                  | Training Records and Documentation:.....          | 64        |
| 8.1.6                  | Continuous Improvement:.....                      | 64        |
| 8.1.7                  | Types of Training Programs.....                   | 65        |
| 8.2                    | Training Programme .....                          | 66        |
| 8.2.1                  | Induction Training .....                          | 66        |
| 8.2.2                  | Regular Training .....                            | 67        |
| 8.2.3                  | Specialized Training.....                         | 68        |
| 8.2.4                  | Awareness Campaigns .....                         | 68        |
| <b>9.</b>              | <b>Supplier and Subcontractor Management.....</b> | <b>70</b> |
| 9.1                    | Suppliers delivery on site .....                  | 71        |
| 9.2                    | Non-Conformity System .....                       | 71        |
| 9.2.1                  | Identification:.....                              | 71        |
| 9.2.2                  | Categorization:.....                              | 71        |
| 9.2.3                  | Resolution: .....                                 | 71        |
| 9.2.4                  | Consequences:.....                                | 72        |
| 9.2.5                  | Continuous Improvement:.....                      | 72        |
| <b>10.</b>             | <b>Grievance Redress Mechanism .....</b>          | <b>73</b> |
| 10.1                   | Purpose and Scope.....                            | 73        |
| 10.2                   | GRM Process .....                                 | 74        |
| <b>11.</b>             | <b>Monitoring and Reporting.....</b>              | <b>77</b> |
| <b>12.</b>             | <b>Auditing .....</b>                             | <b>79</b> |
| 12.1                   | Audit Procedure .....                             | 79        |
| 12.1.1                 | Audit Planning .....                              | 79        |
| 12.1.2                 | Conducting the Audit.....                         | 79        |
| 12.1.3                 | Review of Audit Results .....                     | 80        |
| <b>13.</b>             | <b>Other Management Plans .....</b>               | <b>81</b> |
| <b>Appendixes.....</b> |                                                   | <b>83</b> |
| Appendix A:            | Safety Signs .....                                | 83        |
| Appendix B:            | Incident Report Form .....                        | 86        |
| Appendix C:            | COVID-19 Management .....                         | 87        |
| Appendix D:            | Waste Inventory Log .....                         | 90        |
| Appendix E:            | Waste Disposal Log.....                           | 91        |



|                                               |    |
|-----------------------------------------------|----|
| Appendix F: Waste Disposal Receipt Form ..... | 92 |
| Appendix G: Hazardous Waste Log.....          | 93 |
| Appendix H: Training Log.....                 | 94 |



## List of Abbreviations

|       |                                                        |
|-------|--------------------------------------------------------|
| CESMP | Construction Environmental and Social Management Plan  |
| CLO   | Community Liaison Officer                              |
| DSTA  | Daily Safe Task Analysis                               |
| E&S   | Environmental and Social                               |
| EPC   | Engineering, Procurement, and Construction             |
| EPFI  | Equator Principles Financial Institutions              |
| ERT   | Emergency Response Team                                |
|       |                                                        |
| GHG   | Greenhouse Gas                                         |
| GRM   | Grievance Redress Mechanism                            |
| IFC   | International Finance Corporation                      |
| LCO   | Labour Compliance Officer                              |
| MEP   | Mechanical, Electrical, and Plumbing                   |
| OECD  | Organization for Economic Co-operation and Development |
| RA    | Risk Assessment                                        |
| SME   | Small and Medium-sized Enterprise                      |
| VPSHR | Voluntary Principles on Security and Human Rights      |
| WHO   | World Health Organization                              |



## 1. Introduction

### 1.1 Project Background

The Nigerian maritime industry serves as a vital artery for the nation's economic growth and the Nigerian Port Authority (NPA) holds a pivotal role within this sector, managing key ports that facilitate the majority of Nigeria's international trade. To address the growing demand for efficient, safe, and globally competitive maritime infrastructure, the Federal Government of Nigeria, through the NPA, has initiated the modernization of the **Tin Can Island Port Complex (TCIPC)** and **Lagos Port Complex, Apapa** also referred as the "**Apapa Port**". These modernization efforts are being supported through financing from the **UK Export Finance (UKEF)** and aim to significantly improve port functionality, operational safety, and compliance with international standards.

Both ports are critical to Nigeria's economic development, yet their infrastructure has significantly deteriorated over the years, threatening not only the safety of port operations but also reducing operational efficiency. By modernizing these ports, Nigeria will secure its standing as a leading maritime hub in West Africa, ensuring that it can continue to support the nation's trade and economic activities for decades to come.

### 1.2 Background

The **Tin Can Island Port Complex (TCIPC)**, commissioned in 1977, and the **Lagos Port Complex, Apapa (LPC)**, also known as Apapa Quays, established in 1913, are two of the most important maritime facilities in Nigeria. Both ports play a central role in the country's international shipping activities, handling millions of tonnes of cargo annually. However, decades of operation without significant upgrades have left their infrastructure in dire need of rehabilitation. Cracking, structural deformation, erosion, and corrosion have compromised their operational efficiency and safety, requiring urgent modernization.

The **Tin Can Island Port Modernization Project** focuses on rehabilitating the Quay Wall and other associated infrastructure. The Project will span across 10 berths, covering approximately 2,560 meters along the southern shoreline of Tin Can Island. Following approximately 47 Years of Operation, the Quay Wall Structure has reached the expected material service life as it is showing signs of fatigue in the form of excessive settlement of slabs and deflection of the walls due to the corrosion of the steel sheet and erosion of the retained soil. These issues are currently impacting the terminal's operations across several berths within the complex, affecting port operability and limiting the capacity of cargo ships. Furthermore, the structural degradation poses safety risks for port activities, as the potential collapse of these elements could lead to accidents among terminal operator workers and damage to machinery.

Currently, the port handles approximately 22% of Nigeria's total cargo throughput, with an annual cargo volume of over 17.7 million of metric tonnes. However, the port has a total capacity of 30 million metric tonnes, which means it is currently operating at only 59% of its full capacity due to structural limitations.

As for Tin Can Port, the Federal Government of Nigeria, through the Nigerian Port Authority (NPA), is leveraging financing from the UK Export. The total value of the Apapa Port Modernization Project is estimated at \$328.08 million USD dollars. The Project is financed through a loan of 85% of the Project costs from the UK Export Finance (UKEF) and a contribution of the remaining cost by the Project borrower, the National Ports Authority (NPA).

The **Apapa Modernization Project** similarly aims to rehabilitate the Quay Wall and address structural issues that have developed over nearly a century of operation. Structural deformations and cracks, indicative of structural defects, appear on the Quay Apron Slab within a width range of 5m to 12m from the Quay face. They are also present along the reinforced concrete pavement deck and at specific locations on the Quay's capping



beams. This initiative is crucial to enhancing the functionality and efficiency of the port complex, as the existing infrastructure has deteriorated and in need of rehabilitation.

Currently, the port handles approximately 26.81% of Nigeria's total cargo throughput, with an annual cargo volume of over 21.8 million of metric tonnes. However, the port has a total capacity of 40.8 million metric tonnes, which means it is currently operating at only 53.43% of its full capacity due to structural limitations. Therefore, the modernization will focus on rehabilitating berths 4 to 13 along a Quay length of approximately 2 kilometers. This initiative is crucial to enhancing the functionality, efficiency, maritime safety, security and compliance to globally acceptable standards.

To undertake this Quay Wall rehabilitation, the Federal Government of Nigeria, through the Nigerian Port Authority (NPA), is leveraging financing from the UK Export. The total value of the reconstruction as part of the Apapa Port Modernization Project is estimated at \$414.78 million USD dollars. The Project is financed through a loan of 85% of the Project costs from the UK Export Finance (UKEF) and a contribution of the remaining cost by the Project borrower, the National Ports Authority (NPA). The total cost of the Project includes rehabilitation, basic and detailed engineering, civil and mechanical works, and operational costs.

Both modernization Projects will introduce upgraded infrastructure that is designed to last for at least 50 years, using reinforced concrete structures with high durability standards. Alongside the Quay Wall rehabilitation, the Projects will include the upgrade of essential port facilities such as bollards, fenders, safety ladders, and apron slabs. The modernization efforts will ensure that the ports can safely accommodate larger vessels, improve cargo-handling efficiency, and enhance the ports' role in Nigeria's maritime trade.

As part of the environmental and social assessment process, both Projects have been classified as **Category B** under the **2012 (Revised 2016) OECD Common Approaches**, meaning they are expected to have limited adverse impacts, most of which can be mitigated.

The proposed Construction Environmental and Social Management Plans (CESMP) for both Projects will ensure that these impacts are properly managed, with a strong emphasis on engaging local stakeholders, communities, and affected parties throughout the Project life cycle.

### 1.3 Project Overview

The Tin Can Island Port and Apapa Port, located approximately 3 kilometres apart in Lagos State, Nigeria, both sit along the Badagry Creek and are undergoing modernization to enhance their operational efficiency and maritime safety. Given their proximity, the modernization of these two major ports is being conducted in parallel to optimize resources and time. Both Projects will share the same ancillary sites, thereby reducing logistical challenges, minimizing traffic congestion, and enhancing the coordination of construction activities. This collaborative approach provides significant cost and efficiency advantages, as the Projects benefit from shared resources, such as equipment, personnel, and material transport through the nearby barge jetty.

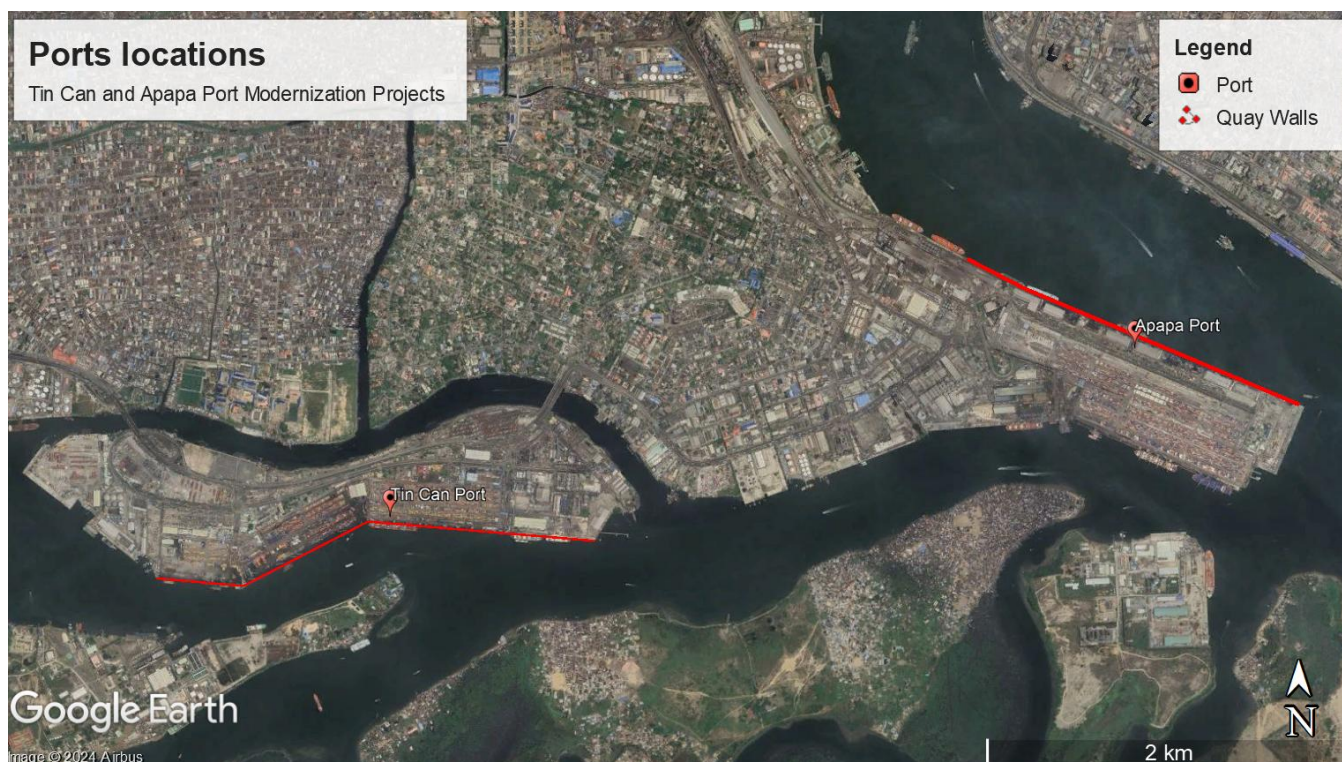


Figure 1-1: Ports Locations

### 1.3.1 Tin Can Port Modernization

The Tin Can Port Quay Wall rehabilitation will consist of 10 berths of approximately 2,560 meters total located in the southern shoreline of Tin Can Island in Lagos, Nigeria. Access to the island is facilitated by the Creek Road to the east and the Apapa Oworoshoki road to the west, both of which lead to the port complex's entrance, where a security control station managed by the NPA is located. The island is surrounded by navigable lagoons of the Badagry Creek. The Quay Wall's footprint that will be rehabilitated starts in Berth 1 at Josepdam Port Service Ltd (6.431597° N; 3.360528° E) and ends at berth 10 in Five Star Logistics Limited (6.429761° N; 3.338772° E). The shoreline comprises a front combi-wall consisting of circular steel piles filled with concrete and connected to each other with AZ-sections steel sheet piles.

The new Quay wall which has a design life of 50 years, includes installing a Combi-Wall System of 1,600 mm diameter steel-cased piles, rear barrettes, and intermediate bored piles to stabilize the structure and reclaim eroded soil. The construction process will be conducted berth-by-berth, ensuring terminal operations continue during the four-year construction phase of the Project. At the Quay Wall, the construction site will cover an area of 40 to 50 meters wide from the shoreline in sections of 200 meters across the Quay Wall's footprint.

### 1.3.2 Apapa Port Modernization

For the Apapa Port, the repair works will begin from Berth 4 (6.445492°N, 3.378982°E) and stop at the last Berth 14 (6.438310°N, 3.395447°). The total Quay Wall length is approximately 2.0km. Additionally, the Berth Facilities (i.e. –Bollards, Fenders, Safety Landers and Apron Slab) will also be upgraded along with a Rail Foundation for STS Gantry Cranes, where possible. The proposed Quay Wall is also designed for a 50-year service life, with reinforced concrete structures meeting Eurocode durability standards. Key features include a 16-meter draft depth to accommodate larger vessels, reinforced concrete members with a minimum compressive strength of C40/50 (40 MPa), and a crack width limit of 0.1mm for deck beams and slabs. Structural steel members will have a corrosion allowance based on water exposure, with S355-grade steel used



for piles. The Quay Wall will withstand operational loads such as crane operations, berthing, and mooring forces, designed according to BS 6349 standards. The rehabilitation plan includes reinforced concrete, water stops, and expansion joints to ensure durability and minimize maintenance needs.

### 1.3.3 Ancillary Sites

For the construction phase of both port modernization projects, the contractor will establish 2 ancillary sites, which are locations or facilities that are secondary or supplementary to the construction site or main activities, providing support or complements. These sites are located in already-existing industrial zones that do not require earthworks for establishment.



Figure 1-2: Location of Ancillary Sites

The General Support Yard will be providing essential infrastructure and facilities to support the rehabilitation efforts of the Projects. With an area of 4.8 hectares and located approximately 500 meters from the northern extent of Tin Can Island (6.435177°N, 3.342207°E) this yard will provide various activities essential to the Project's success. The yard will feature:

- Mixing batching plant for concrete
- Warehouse and storage spaces for construction materials, equipment, tools, and other supplies
- Heavy Equipment Yard, facilitating the storage construction machinery and equipment.
- Workshop to enable the repair and maintenance of these assets
- Wash bay for vehicles and machinery connected to a decantation dam
- Canteen area for workers
- Drainage system for both storm water and wastewater
- Septic tank to collect domestic sewage water and store it for primary treatment.
- Two (2) underground boreholes for water supply
- Welfare facilities for the workers which include:

- PPE Equipment and Overall Jump Suits
- Lockers and Changing Room Facilities
- Washing/Shower Facilities
- Toilet Facilities
- First Aid Appliances and Clinic
- Drinking Water
- Transportation between Job Site and Facility Location
- Canteen Area with Various Food Vendors
- Designated Praying Area
- Standby generators Power units (800 KVa) to be used for situations such as power outages. The site will be connected to the national grid.
- Administrative buildings with offices for the staff with a kitchen
- Security Station with personnel to provide security to all Project sites and facilities.
- A clinic with a nurse available. The nurse will be also responsible for first aid cases and general check-ups for all staff and workers
- All waste will be stored on site in allocated areas and will be collected and disposed by a subcontractor.
- Toilets for all staff and visitors

Additionally, a Laydown Area, named the Savaric Yard located in Ibafor (6.435835°N, 3.323500°E), will be used for storage and laydown of the Project construction activities. It is an industrial site previously used by another construction company, Julius Berger, and is located next to a private residential complex on the east, that belongs to the same owner. On the west, there are two (2) oil tank farms owned by Obat Oil and Rahamaniyya.

The Storage Area will be used to store heavy equipment and as a laydown area where construction materials and equipment can be stored, organized, and easily accessed. It is equipped with a barge jetty, storm water and sewage wastewater drainage systems, electricity supply and a potable water plant. The entrance is secured by hired personal of the Savaric Company.

To avoid delays caused by traffic as the access roads to Tin Can and Apapa Ports are normally congested, a barge will be used to transport materials and steel to the construction sites. This jetty will be only used for the Projects.

Construction activities are proposed to take place from Monday to Friday (7:30 am to 5.:30) and Saturday, (from 7.30 am to 2:30pm). No night-time work is planned under normal circumstances. Any exceptional activities requiring extended hours (e.g., continuous concrete pours or critical marine operations) will be subject to prior approval by the National Port Authority and communicated to affected stakeholders in advance. Recruitment will take place during mobilization and site establishment phase post Financial Close.

***Detailed Project descriptions of each port are included in the ESIA studies of each port.***

#### 1.4 Presentation of the Project Entities

The project is under an Engineering, Procurement, Construction, and Financing (EPC+F) with the Federal Republic of Nigeria, structured as follows.

#### 1.4.1 The Project Promoter, Buyer and Borrower



The Nigerian Ports Authority (NPA), the project promoter, buyer and borrower, is the government agency responsible for managing, operating, and developing ports in Nigeria, which includes the Lagos Port Complex and Tin Can Island Port in Lagos; Calabar Port, Delta Port, Rivers Port at Port Harcourt, and Onne Port. The agency, established in 1954, works under the supervision of the Federal Ministry of Transportation. The NPA plays a crucial role in the modernization and development of ports across the country, including the Tin Can Island Port in Lagos.

#### 1.4.2 The Contractor ITB – Hitech Joint Venture



**ITB Nigeria Limited (The JV Managing Partner)** was established in 1995 to address the growing demand for innovation and improved engineering expertise in Nigeria. With over two decades of experience in the construction industry, the teams of builders and civil engineering contractors have established ITB as the leading provider of top-quality construction services across the country. The company's successful track record of delivered projects and strong relationships with clients have reinforced its position as a leader in the market. Pursuing international best practices, **ITB is ISO certified for Quality, Occupational Health and Safety and Environmental Management Systems**. It has completed projects which have been **EDGE and LEED certified**, with commitment and dedication to utmost attention for quality, safety and a greener environment.



Founded in 1988, **Hitech Construction Africa Limited (JV partner)** has become a prominent actor in Building and civil Engineering in West Africa. The company has broadened its footprint across the region, notably in Nigeria, Togo, and Benin, with a strategic foothold in the UK through their London office. Harnessing advanced engineering technologies, Hitech is committed to driving innovation and implementing international standards in its construction projects, including the **ESF of the International Finance Corporation and the World Bank EHS Guidelines for development projects**. The diversified project portfolio encompasses infrastructure, roadworks to marine engineering & construction. With a workforce of over 2,500 professionals, the company ensures first-rate quality and safety, as well as rigorous environmental standards.



**Dar Al-Handasah Consultants** (Shair and Partners) is a privately-owned international multidisciplinary consulting organization in engineering, architecture, planning, environment, project and construction management, facilities management, and economics, founded in 1956. Its principal design head offices and technical support facilities are located in Beirut, Lebanon; Cairo, Egypt; London, UK; Pune, India; and Amman, Jordan; with a network of 47 permanent local branch offices throughout the Middle East, Africa, Asia and Europe of which 23 are on the African Continent. Dar has been appointed by ITB - Hitech to develop the design of the quay wall and technical engineering studies of the project.

### 1.5 CESMP Objectives

The Construction Environmental and Social Management Plan (CESMP) for the Tin Can Port Modernization Project in Nigeria aims to achieve the following objectives:



- **Ensure Compliance:** Adhere the project construction phase to all relevant environmental, social, and legal obligations, including IFC Performance Standards, Equator Principles IV, and Nigerian regulations.
- **Impact Mitigation:** Implement effective measures to minimize or eliminate negative environmental and social risks and impacts identified in the Supplementary ESIA and the E&S Site Evaluation Report of the project during construction activities.
- **Community and Worker Protection:** Safeguard the health, safety, and welfare of both local communities and project workers, ensuring that construction activities do not adversely affect their well-being.
- **Monitoring and Reporting:** Establish robust monitoring and reporting mechanisms to track the environmental and social performance of the project, allowing for timely corrective actions as needed.
- **Stakeholder Engagement:** Facilitate ongoing and transparent communication with all relevant stakeholders, including local communities, to address concerns and ensure their involvement throughout the construction process.

These key objectives are designed to guide the contractor in executing construction activities responsibly and sustainably, aligning with international standards and contributing positively to the local environment and society. The main aspects to be covered in the CESMP include the following:

- **Biodiversity Preservation:** Ensure minimal disruption to local ecosystems during construction.
- **Waste Management:** Implement efficient waste disposal practices and recycling systems.
- **Energy Efficiency:** Promote sustainable energy sources and reduce energy consumption.
- **Water Conservation:** Minimize water usage and manage stormwater runoff effectively.
- **Community Engagement:** Involve local communities in project planning and decision-making.
- **Labor Rights and Safety:** Prioritize worker safety, fair wages, and adherence to labour laws.
- **Government and Authorities:** Coordinate with relevant authorities for permits and compliance.



## 2. Organizational Structure and Policies

The Contractor recognizes that an environment, health and safety friendly workplace along with environment, safety and health-conscious personnel are core of driving the business sustainability. The contractor is committed to ensuring environmentally sustainable and responsible operations to achieve the highest standards of performance and compliance. This includes implementing best practices in environmental stewardship, health and safety management, and social responsibility. The Contractor's approach integrates these principles into all aspects of the project, ensuring that operations not only meet but exceed regulatory requirements and contribute positively to the surrounding communities and environment. Hitech has an ESMS and Policy framework aligned with IFC in place. ITB is ISO 45001, 9001 and 14001 certified as well EDGE, LEED and COREN (Nigerian Engineering Excellence).

### 2.1 Organizational Structure and Responsibility

The Contractor will have ultimate responsibility for implementing the provisions of the ESMP. This role will include the management of environmental impacts, monitoring of subcontractors' performance as well as development of mechanisms for dealing with environmental and social problems.

The contractors will also ensure that the activities of its subcontractors during establishment, construction and decommissioning phases are conducted in accordance with 'good practice' measures and the project requirements, implementation of which will be required through contractual documentation. In order to facilitate this, and to demonstrate commitment to the ESMP. The contractor will conduct regular internal site inspections and audits, the results of which will be documented.

The organization structure of the project EHS Team is presented in the figure below. This structure might be subject to changes based on project needs during implementation.

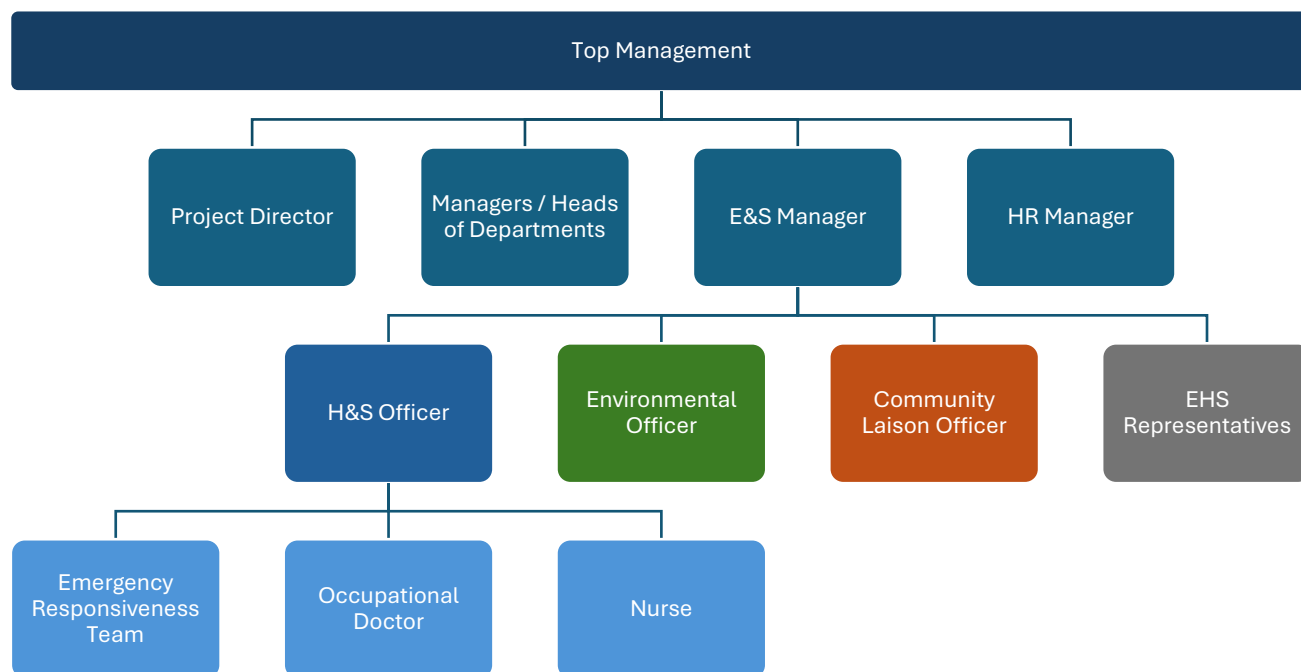


Figure 2-1 EPC Contractor's Organizational Structure

Table 2-1: Roles and Responsibilities

| Role             | Responsibility                                                                                                |
|------------------|---------------------------------------------------------------------------------------------------------------|
| Project Director | <ul style="list-style-type: none"> <li>Oversee the entire project and ensure alignment with CESMP.</li> </ul> |



| Role        | Responsibility                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <ul style="list-style-type: none"> <li>• Approve agreements with subcontractors and partners</li> <li>• Provide strategic leadership and ensure resources are allocated efficiently.</li> <li>• Report on project progress and compliance to stakeholders.</li> <li>• Ensure the support of all Managers and top management to the E&amp;S Department</li> <li>• Authorize the internal acceptance of handover packages after confirming that all safety and operational preconditions required by NPA, the PSSCC, and Terminal Operators have been met.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| E&S Manager | <ul style="list-style-type: none"> <li>• Responsible for developing, implementing, and overseeing the ESMP project.</li> <li>• Ensure project activities comply with legal and regulatory environmental and social requirements.</li> <li>• Facilitate the participation and consultation of stakeholders, including local communities, in the planning and implementation of environmental and social measures.</li> <li>• Establish EHS objectives and collect KPIs on ongoing basis for every department and construction site.</li> <li>• Review and approve all checklists, monitoring logs, inspection logs, analysis results and registries related to environmental and social activities of the project.</li> <li>• Plan and coordinate the QHSE Audits to be done on quarterly basis. This includes audits to the subcontractors.</li> <li>• Undertake air quality, noise, vibration measurements as per the stipulated frequency at the identified measurements points.</li> <li>• Oversee the EHS Training Program implantation</li> <li>• Coordinate, supervise the subcontractors HSE Managers, its KPIs, EHS objectives and activities as per the agreement to ensure their compliance with the project CESMP and contractor policies.</li> <li>• Develop and validate partnerships and agreements with subcontractors, NGOs, and third parties.</li> <li>• NPA coordination for handoff, construction/operational phase coordination and eventual operational phase</li> <li>• Ensure the GRM is properly implemented.</li> <li>• Coordinate EHS surveys to gather information from workers.</li> <li>• Identify external trainers with credentials for specific trainings and coordinate these trainings.</li> <li>• Oversee CSR initiatives of the project.</li> <li>• Prepare EHS Bulletins to be shared on monthly basis to external stakeholders.</li> <li>• Coordinate EHS readiness checks prior to each formal handover of a berth or construction zone, including inspection of barriers, signage, access routes, emergency arrangements, spill controls, and marine-side risk controls.</li> </ul> |



| Role                      | Responsibility                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <ul style="list-style-type: none"> <li>• Evaluate and report on the E&amp;S performance of the project on ongoing basis to the Lenders and the NESREA</li> <li>• Review all internal and external non conformities as well as recommendations to propose corrective actions with designated dates and responsible parties.</li> <li>• Prepare the E&amp;S Self-Monitoring Report for Lenders as per the facility agreement.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Health and Safety Officer | <ul style="list-style-type: none"> <li>• Develop, implement, and oversee health and safety policies and procedures for the project.</li> <li>• Ensure the implementation of mitigation measures to minimize health and safety risks.</li> <li>• Monitor health and safety performance and compliance with regulations.</li> <li>• Oversee the health of workers in coordination with an occupational doctor and the nurse, ensure all workers receive a medical examination.</li> <li>• Ensure PPE is available and provided for all workers and visitors.</li> <li>• Conduct regular inspections and audits to assess the effectiveness of health and safety measures.</li> <li>• Prepare weekly observation logs</li> <li>• Coordinate the development of risk assessments at every construction area/department in coordination with Top Management.</li> <li>• Develop Safe Work Procedures for all identified activities of high risks.</li> <li>• Oversee the programme of daily toolbox talks, awareness campaigns and trainings.</li> <li>• Coordinate activities undertaken by the Emergency Responsiveness Team</li> <li>• Responsible for incident and emergency management</li> <li>• Prepare emergency scenarios at all project sites and undertake emergency drills.</li> <li>• Evaluate the health and safety performance of the project</li> <li>• Analyse trends and near misses to improve the health and safety performance.</li> <li>• Conduct pre-handover HSE inspections and prepare Handover Readiness Reports documenting PPE requirements, workfront demarcation, safe access routes, exclusion zones, and emergency response locations.</li> </ul> |
| Environmental Officer     | <ul style="list-style-type: none"> <li>• Implement mitigation measures outlined in the ESMP to minimize environmental impacts.</li> <li>• Monitor and assess environmental performance and compliance with environmental regulations.</li> <li>• Conduct environmental inspections and audits to evaluate the effectiveness of mitigation measures. Internal inspections will be carried out regularly by the Contractor's E&amp;S team and reported to the Employer. In addition, independent environmental audits will be undertaken in line with Nigerian regulatory requirements and lender standards. These independent audits will provide third-party verification of compliance and effectiveness of mitigation measures, with findings reported to both the Employer and the Lenders. Environmental inspections and audits will be undertaken internally by the Contractor and complemented by quarterly Environmental Compliance Reports prepared by a NESREA-accredited independent firm, along with periodic NESREA inspections. In addition, the project will undergo ISO audit cycles, labour inspections by the relevant national regulators, and a biannual external audit of the ESMS.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



| Role                          | Responsibility                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <ul style="list-style-type: none"> <li>• Liaise with regulatory bodies and stakeholders to address environmental concerns.</li> <li>• Prepare environmental reports and documentation required for permits and approvals.</li> <li>• Coordinate water analysis measurements of underground water and stormwater runoff.</li> <li>• Supervise soil and water in the project area and ensure implementation of measures for stormwater, spill management, erosion and sedimentation.</li> <li>• Prepare weekly observations logs and environmental near misses</li> <li>• Evaluate construction activities are not affecting ecological receptors in the project area of influence.</li> <li>• Keep registries of wildlife interactions and update the biodiversity registry with identified species in coordination with a biodiversity expert.</li> <li>• Supervise the waste management and traceability with the subcontractor as per the cradle-to-grave waste management system.</li> <li>• Undertake audits to the waste collection and disposal partner.</li> <li>• Develop and implement awareness campaigns with the workers and stakeholders related to waste management, pollution prevention, biodiversity conservation and ecosystem services.</li> </ul> |
| Community Liaison Officer     | <ul style="list-style-type: none"> <li>• Implement community engagement activities outlined in the ESMP to promote dialogue and consultation with local communities.</li> <li>• Coordinate project E&amp;S disclosure with stakeholders</li> <li>• Monitor social impacts of the project and implement measures to address community concerns.</li> <li>• Coordinate implementation and monitoring of the livelihood mitigation activities</li> <li>• Coordinate the Grievance Redress Mechanism to address community complaints and feedback.</li> <li>• Collect and analyse social and livelihood KPIs on monthly basis</li> <li>• Coordinate CSR and community development programs and initiatives to enhance local well-being.</li> <li>• Arrange traditional ceremonies needed for the project activities</li> <li>• Prepare regular reports on community engagement activities and outcomes.</li> </ul>                                                                                                                                                                                                                                                                                                                                                        |
| Emergency Responsiveness Team | <ul style="list-style-type: none"> <li>• Develop and implement emergency response plans.</li> <li>• Conduct regular training and emergency drills.</li> <li>• Coordinate and manage emergency responses.</li> <li>• Provide first aid and coordinate with external emergency services.</li> <li>• Maintain clear communication during emergencies.</li> <li>• Investigate incidents and implement corrective actions.</li> <li>• Document and report all emergency incidents and responses.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| EHS Representatives           | <ul style="list-style-type: none"> <li>• Assist in implementing mitigation measures related to environmental, health, and safety aspects of the project.</li> <li>• Monitor environmental, health, and safety performance within their respective work area</li> <li>• Participate in inspections and audits to identify areas for improvement and ensure compliance.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Role                          | Responsibility                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <ul style="list-style-type: none"> <li>Promote awareness of environmental, health, and safety issues and encourage adherence to standards.</li> <li>Report any concerns or non-compliance to the appropriate authorities.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Occupational Doctor           | <ul style="list-style-type: none"> <li>Provide medical expertise and advice on occupational health issues affecting project personnel.</li> <li>Coordinate partnership with local hospitals for emergencies and medical examinations.</li> <li>Conduct health assessments and screenings to monitor the health status of workers exposed to occupational hazards.</li> <li>Develop health promotion programs and initiatives to improve worker well-being and prevent work-related illnesses or injuries.</li> <li>Collaborate with project management and health and safety personnel to ensure effective health management practices.</li> <li>Coordinate activities undertaken by the Nurse on site.</li> </ul>              |
| Nurse                         | <ul style="list-style-type: none"> <li>Provide medical care and first aid services to project personnel in case of injuries or emergencies.</li> <li>Maintain health records and administer medications as prescribed.</li> <li>Assist in health education and promotion activities, such as workshops on hygiene and disease prevention.</li> <li>Collaborate with the Health and Safety Manager and Occupational Doctor to ensure the health and well-being of project personnel.</li> </ul>                                                                                                                                                                                                                                  |
| Human Resources Manager       | <ul style="list-style-type: none"> <li>Manage the hiring and recruitment of the project workforce as per the Human Resources Management Plan</li> <li>Ensure staff orientation includes section on guiding principles for effective worker – management relationship in accordance with provision of PS2</li> <li>Develop and update the contractor's human resource policy and procedure to capture provisions of labour and working conditions per PS2</li> <li>Coordinates with E&amp;S Manager in planning for training on the project's ES requirements for staff and partner</li> <li>Ensure all employees have a contract, social security as per the national regulations and receive a medical examination.</li> </ul> |
| Head of Departments/ Managers | <ul style="list-style-type: none"> <li>Oversee department-specific operations.</li> <li>Ensure implementation of CESMP within departmental activities.</li> <li>Monitor and report on departmental compliance with environmental, health, and safety standards.</li> <li>Coordinate with other departments to align with overall project objectives.</li> <li>Supervise all compliance of Site Supervisors with the CESMP</li> <li>Report near misses and observations</li> <li>Develop Risk Assessments and Safe Work Procedures in coordination with the E&amp;S Manager</li> <li>Participate in weekly HSE Meetings to discuss environmental and social matters.</li> <li>Cooperate during QHSE audits.</li> </ul>           |



| Role             | Responsibility                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Site Supervisors | <ul style="list-style-type: none"> <li>• Supervise and coordinate all phases of activity in the area;</li> <li>• Promote E&amp;S awareness through a proactive attitude on the part of all employees;</li> <li>• Support the E&amp;S Department in their activities and procedures.</li> <li>• Put an immediate and definitive end to any risky situation or behaviour.</li> <li>• Ensuring the implementation and full application of the project's CESMP</li> <li>• Correcting or ensuring the correction of any HSE breaches noted or reported.</li> <li>• Undertake daily risk analysis with workers on site as per the planned activities</li> <li>• Ensure that all staff and visitors have HSE induction training before entering the construction site area.</li> <li>• Contribute to or participate in the investigation of all reportable accidents and incidents.</li> <li>• Take responsibility for the HSE performance of the project.</li> <li>• Ensuring that weekly HSE Toolbox Talks are carried out and documented</li> <li>• Ensuring that a weekly HSE inspection is carried out and documented.</li> <li>• Taking and documenting the necessary disciplinary measures for breaches of HSE rules, regulations, procedures and programmes</li> </ul> |

Implementation of the ESMP will be required through contractual documentation. All subcontractor agreements will include explicit clauses requiring compliance with the ESMP, CESMP, and applicable international standards (including IFC Performance Standards, World Bank Group EHS Guidelines, and Nigerian regulatory requirements).

To facilitate this and demonstrate commitment:

- **Contractual flow-down:** Subcontractor contracts will contain binding E&S clauses covering labour, health and safety, waste management, pollution prevention, and community engagement.
- **Oversight and accountability:** The main contractor will conduct regular internal site inspections and audits of subcontractor performance, with results documented and submitted to the Employer/Engineer.
- **Corrective actions:** Non-compliance will trigger corrective action requests, with repeated breaches subject to penalties or contract termination.
- **Capacity building:** Subcontractors will receive induction and training on ESMP requirements before mobilization.

All employees are responsible for working in accordance with the documented environmental and social procedures and instructions, specific responsibilities defined in individual job descriptions and instructions. The EPC Contractor is responsible for making sure that all subcontractors are following the CESMP guidelines and procedures.

## 2.2 All Employees

All employees are responsible for working in accordance with the documented environmental procedures and instructions, specific responsibilities defined in individual job descriptions and instructions.

- Comply with the contractor E&S Policies and Procedures



- Are individually and collectively responsible for working safely and in respect of the environment, use the PPE at all times and following the health and safety, biodiversity, waste management and emergency response procedures.
- Supporting the environmental and social policy goals set for the Project in the daily performance of their work.
- Participate in the HSE Induction, Toolbox talks and trainings provided by the E&S Team
- Report near misses and non-compliances to their supervisor or directly to the E&S Team
- Complying with the specific requirements of the Project Environmental and Social Management Plan and its supporting documents.
- Notifying their immediate manager, supervisor, or Environmental and Social Manager regarding any spills, equipment malfunctions, or unsafe situations that could have a negative environmental or social impact.

### 2.3 Subcontractors

The following listing provides the overview of responsibilities from the subcontractors associated with the project:

- Allocate HSE Manager to be in ongoing coordination and communications with the contractor E&S Manager.
- Inclusion of environmental and social mitigation measures in construction activities and budgets as specified in the project CESMP.
- Responsible for ensuring the project adheres to national law and the policy of the lender, including social and environmental safeguards.
- Comply with the project CESMP and Contractor E&S policies and procedures.
- Day-to-day project management and coordination: project administration; technical oversight (engineering and environmental); review, supervise and approval of the work of the employees.
- Environmental and social supervision to ensure that all mitigation is provided as specified in the project ESMP by their own staff.
- Monitor the environmental and social compliance of the project according to the CESMP.

At this stage, it is expected that subcontractors will be engaged for wastewater and waste collection and disposal. Any subcontractor shall go through the requirements and engagement process explained in the **Contractor and Supplier Management Plan (CSMP)**.

### 2.4 Suppliers

The following list provides an overview of the key requirements needed by the project from the supplier:

- Comply with the supplier agreement and policies
- Comply with the delivery protocols on site as per the supplier agreement
- Comply with labour laws, ensuring no forced or child labour, and safe working conditions.
- Handle and dispose of packaging and waste responsibly, aligning with the project's waste management plan.
- Promote local sourcing where possible to support local economic development.
- Health and Safety Compliance: Ensure safe production and handling of materials in line with safety standards.
- Maintain clear communication with the project team on supply status and issues.
- Cooperate with environmental and social audits and provide required documentation.

They must follow the contractor policies and code of conduct as stipulated in the CSMP.



## 2.5 NPA Port Management / Operations

- Provide overall oversight of port operations and construction interfaces.
- Approve work sequencing, construction-zone demarcation, and berth-by-berth handovers.
- Chair the Port Safety & Security Coordination Committee (PSSCC) and review monthly EHS and security performance.
- Ensure construction activities align with port-wide schedules, navigational safety, and emergency readiness.
- Validate readiness for new berth handovers or high-risk works.
- Issue operational notices when construction impacts port traffic, marine operations, or safety protocols
- Lead and approve the formal handover of each berth from operational status to construction readiness, and later from construction completion back to operational control, ensuring compliance with port-wide safety, navigational, and operational requirements.

## 2.6 NESREA

The National Environmental Standards and Regulations Enforcement Agency (NESREA) is a governmental parastatal under the Federal Ministry of Environment (FMEnv) in Nigeria, responsible for enforcing environmental regulations and promoting compliance with national standards. It plays a pivotal role in monitoring, auditing, and certifying environmental performance across various facilities and projects.

- Ensure quarterly environmental monitoring is conducted by NESREA-approved consultants and review submitted reports.
- Oversee the annual renewal of licenses or permits for air quality, waste management, and toxic substances.
- Require and review three-yearly facility audits to determine the environmental status of facilities, ensuring reports are uploaded to the NESREA Portal.
- Collaborate with the Federal Ministry of Environment to conduct bi-annual impact mitigation monitoring and assess the effectiveness of mitigation measures.
- Oversee third-party audits of ISO-certified management systems, ensuring compliance with ISO 14001, 9001, and 45001 standards.

## 2.7 NPA Marine Department

- Coordinate vessel movements adjacent to active construction berths.
- Issue clearance for barges, floating plant, and marine equipment.
- Update VTS/Harbour Master on temporary restrictions, reduced channel width, or marine safety zones.
- Participate in joint marine risk assessments and daily coordination with the contractor.
- Review barge and tugboat movements to avoid conflicts with terminal operations.
- Coordinate with **Lagos Channel Management (LCM)** when construction affects navigational depths or channel alignment.
- Provide navigational and marine safety clearance for each handover, validating temporary marine restrictions, channel width, buoy placements, and barge operating conditions.



## 2.8 NPA Security Department

- Enforce access control into construction zones in line with ISPS requirements.
- Validate daily access lists for workers, subcontractors, and vehicles.
- Conduct joint security patrols with the Contractor Security Team.
- Ensure segregation of operational areas from construction zones.
- Lead incident-response coordination for any security event involving construction activities.

## 2.9 NPA Safety / EHS Unit

- Conduct joint inspections, toolbox participation, and interface walkthroughs at boundary zones.
- Verify CESMP, ICMP, TMP and Emergency Response Plan alignment with port procedures.
- Review high-risk work near terminals or quay edges.
- Validate emergency access routes during shifting workfronts.
- Participate in incident investigations and review contractor reporting.

## 2.10 Port Safety & Security Coordination Committee (PSSCC)

- Provide high-level oversight of cross-entity EHS, security and interface performance.
- Validate readiness for new berth activation or high-risk work fronts.
- Ensure consistency between construction scheduling and port safety obligations.
- Review monthly reports, critical incidents, and systemic risks.
- Approve major temporary adjustments affecting operational continuity.
- Validate readiness for handovers between operational and construction phases, ensuring that all safety, environmental, and security requirements have been met before activation or return of a berth/workfront.

## 2.11 Emergency Services (NPA Fire Service + External Responders)

- Provide coordinated emergency response to fire, spill, medical or marine incidents.
- Participate in joint emergency drills aligned with the Emergency Response Plan.
- Review and validate emergency access routes as construction zones shift.
- Support incident stabilization at interface areas between construction and operations.
- Review and approve emergency access routes and readiness measures (hydrants, muster points, fire paths) before any berth handover or change of construction zone.

## 2.12 Terminal Operators

- Coordinate terminal operations (vessel calls, truck flows, cargo handling) with berth-by-berth construction sequencing.
- Approve access to concession areas and validate temporary route changes.
- Maintain safe separation between terminal activities and construction zones.
- Share daily and weekly operational schedules relevant to the contractor's workfront.
- Participate in joint inspections, incident reviews, and interface coordination (SICG).



- Participate in berth-by-berth handover processes, confirming that operational impacts, cargo flows, and terminal safety requirements are addressed before granting construction access or resuming operations.

### 2.13 National Environmental Consultant

The Environmental Consultant plays a central role in ensuring that the project remains compliant with FMEnv and NESREA requirements by conducting quarterly environmental monitoring, verifying ESMP implementation, and preparing evidence-based Environmental Compliance Reports. As an independent, NESREA-certified specialist, the consultant designs the monitoring plan, selects sampling locations based on risk and receptor sensitivity, oversees accredited laboratory analysis, documents non-compliances, proposes corrective actions, and engages with regulators and affected stakeholders to ensure transparency and early identification of issues.

- Prepare the **Quarterly Environmental Compliance Monitoring Plan** including sampling locations, parameters, logistics, and monitoring methodology.
- Conduct field monitoring of air, water, soil (if applicable), noise, vibration, waste, erosion, biodiversity, and emissions using calibrated instruments and approved methods.
- Ensure sampling quality assurance, including GPS-tagging, photographic evidence, chain-of-custody procedures, and use of FMEnv/NESREA-accredited laboratories.
- Verify ESMP implementation, assessing the effectiveness of mitigation measures by contractors and subcontractors.
- Identify non-compliances or environmental risks and propose corrective and preventive actions aligned with Nigerian regulations and IFC/EP good practice.
- Engage with stakeholders, including regulators, community focal points, HSE units, and on-site personnel, to capture environmental concerns and grievances.
- Prepare and submit the Quarterly Environmental Compliance Report (Q-ECR) to FMEnv and NESREA, including laboratory results, monitoring data, geo-referenced evidence, and a corrective action plan.
- Participate in regulatory inspections or IMM missions and support the project in responding to FMEnv/NESREA observations.
- Maintain comprehensive records and an audit-ready archive of monitoring data, certificates, photos, and logs.

### 2.14 Policies

As this project is being delivered through the **ITB–Hitech Joint Venture**, both companies' Policy Frameworks collectively govern how the project is managed. The Joint Venture applies the **stricter or more comprehensive requirement** across ITB's and Hitech's systems to ensure full alignment with the IFC Performance Standards, the World Bank Group EHS Guidelines, Nigerian regulatory obligations, and lender expectations. This integrated approach ensures that all environmental, social, governance, labour, safety, ethical conduct, and community-related commitments are implemented consistently across project activities, with both partners jointly responsible for monitoring, reporting, and maintaining compliance throughout the duration of the works.



### 2.14.1 ITB Nigeria Limited

ITB Nigeria Limited has established an ESMS aligned with the IFC PS, the World Bank Group EHS Guidelines, the UN Guiding Principles on Business and Human Rights, and all applicable Nigerian laws and regulations. The ESMS outlines the responsibilities of ITB's HSE and ESG functions, including monitoring, reporting, incident management, and supervision of subcontractors and third-party activities to ensure compliance with regulatory and lender expectations. ITB develops project-specific management plans wherever required, proportionate to project risks and impacts.

In line with international good practice and recent lender due diligence requirements, ITB has adopted a Policy Framework to guide ethical, safe, and responsible operations across all activities and sites. This Framework sets out the principles, behavioural expectations, and environmental and social requirements applicable to all employees, subcontractors, suppliers, labour providers, service partners, and any party acting on behalf of ITB.

It reflects global standards and supports a culture of transparency, compliance, and sustainable development. The core components of the **ITB Policy Framework (2025)** include:

- **Governance, Ethics & Compliance Policies:** ITB maintains governance systems through its Anti-Bribery & Anti-Corruption (ABC) Policy, Anti-Money Laundering & Counter-Terrorist Financing (AML/CTF) Policy, Equality, Diversity & Inclusion (EDI) Policy, Child Labour & Young Workers Policy, Slavery & Human Trafficking Policy, Subcontractor & Third-Party Policy. These policies reflect ITB's commitment to ethical business practices, non-discrimination, human rights, and global financial integrity standards.
- **Environmental & Social Management Policies:** Under its ESMS, ITB implements policies covering environmental protection, pollution prevention, resource efficiency, biodiversity conservation, community health and safety, cultural heritage, and emergency preparedness and response. These policies align with IFC Performance Standards (PS1, PS3, PS4, PS6, PS8) and the WBG EHS Guidelines.
- **Labour & Workforce Policies:** ITB's labour policies include Occupational Health, Training & Competency, Road Transport Safety, Rigging and Lifting, Safe Work at Height, Smoke-Free, and Alcohol and Drugs policies. These instruments ensure a safe working environment in line with IFC PS2, Nigerian labour regulations, and ILO conventions.
- **Supply Chain & Procurement Policies:** The Supply Chain & Procurement Policy and the Subcontractor and Third-Party Policy require all suppliers and subcontractors to adhere to ITB's standards on ethics, labour rights, environmental protection, safety, and community relations.
- **Security & Social Risk Policies:** ITB's Security Policy is aligned with the Voluntary Principles on Security and Human Rights (VPSHR) and promotes responsible interactions with workers and communities.

Through this comprehensive Policy Framework, ITB ensures consistent, ethical, and compliant operations across its construction activities. It establishes clear expectations for all workforce members and partners, reinforcing ITB's commitment to safety, accountability, and responsible project delivery in Nigeria.

### 2.14.2 Hitech Construction Africa Limited

Hitech has an established an Environmental and Social Management System (ESMS) in line with the IFC Performance Standards, the World Bank HSE guidelines and regulations, the Equator Principles IV as well as national regulations where the project takes place. This system details all the responsibilities of the HSE Unit, monitoring, reporting, and auditing procedures to ensure compliance with the regulatory framework. As part of this ESMS, Hitech develops project specific management plans tailored to the impacts and needs of the project.



HITECH is committed to the highest standards by implementing a Policy Framework that ensures consistent and ethical operations across all its endeavors. This framework serves as a guiding document, outlining principles, procedures, and responsibilities that all stakeholders adhere to. By establishing clear guidelines, HITECH ensures a culture of accountability, transparency, and excellence, contributing to the quality and reliability of its services and interactions. The following points summarize the **Policy Framework** components:

- **HSE Policy:** HITECH Construction's HSE Policy is dedicated to upholding the safety of workers and the protection of the environment while adhering to ISO 45001 and ISO 14001:2015 standards, and the UN's sustainable development goals. The policy outlines commitments to providing safe working conditions, preventing pollution, protecting biodiversity, and involving workers in HSE decision-making processes. It focuses on continuous improvement and the sustainable use of resources, emphasizing the integration of HSE into all activities.
- **Quality Policy:** HITECH Construction Company Limited aims to be an internationally recognized provider of quality engineering solutions, emphasizing innovation and the use of international designs and materials. The company is committed to complying with ISO 9001:2015 standards and other international criteria, ensuring quality in **construction** and engineering projects such as roads, bridges, and airports. The policy focuses on meeting stakeholder expectations, risk-based thinking, continual improvement, customer focus, and compliance with statutory and industry requirements.
- **Alcohol and Drugs Policy:** HITECH Construction enforces a strict no tolerance policy regarding the use and abuse of alcohol and drugs on its sites to maintain safety and health. The policy includes random alcohol and drug **screenings**, a baseline blood alcohol concentration (BAC) requirement of 0.00 for on-site personnel, and stringent disciplinary actions for violations. This policy is designed to prevent workplace hazards associated with impairment and ensures all operators are free from the influence of substances.
- **Gender-Based Violence (GBV) and Sexual Harassment Policy:** This policy outlines the organization's commitment to preventing and addressing gender-based violence and sexual harassment in the workplace, including reporting mechanisms and support services for victims.
- **Smoke-Free Policy:** This policy prohibits smoking in project areas to protect workers and the environment from the harmful effects of tobacco smoke.
- **HIV and AIDS Workplace Policy:** HITECH Construction Company Limited's HIV and AIDS Workplace Policy ensures support for workers living with HIV and AIDS, protecting their rights and providing access to healthcare without discrimination. Key principles include non-discrimination, confidentiality of medical information, provision of necessary support services, and regular awareness and prevention programs to educate and reduce stigma related to HIV/AIDS. The management is responsible for upholding these principles and ensuring all workers are treated fairly and with respect.
- **Human Resources Policy:** The Human Resources Policy at HITECH Construction Company Limited outlines guidelines for the recruitment, training, management, and development of the workforce. It emphasizes equal employment opportunities, skill development through ongoing training, fair wages compliant with local regulations, and the protection of worker rights including unionization and collective bargaining. Management and workers are jointly responsible for implementing this policy to maintain a fair, respectful, and professional work environment.
- **Community Investment Guideline:** The *Community Investment Guideline* by HITECH Construction Company Limited aims to foster socio-economic development in local communities through engagement in community programs and initiatives. This includes prioritizing local employment, building capacities within the community, contributing to infrastructure development, and maintaining



open communication with local leaders to ensure the projects serve mutual interests. The contractor is tasked with actively engaging and benefiting the community, ensuring sustainable local development.

- **Site Clearance Policy:** HITECH's Site Clearance Policy ensures safe and efficient site clearance activities with a strong emphasis on environmental protection and biodiversity conservation. The policy mandates proper clearance of hazards and debris, supervised vegetation clearance following biodiversity surveys, training for safe handling of materials, proper waste disposal, and rigorous documentation and monitoring of all activities by the Environmental Officer. Site supervisors and the Environmental Officer ensure that clearance procedures are adhered to, maintaining safety and environmental standards.
- **Non-Conformance Policy:** The *Non-Conformance Policy* at HITECH Construction Company Limited addresses deviations from project standards, policies, or regulations. It stipulates the identification and documentation of non-conformances, immediate corrective actions, root cause analysis to prevent recurrence, and mandatory reporting to management and regulatory authorities. All workers are responsible for reporting non-conformances, and management is tasked with ensuring that appropriate corrective actions are implemented effectively.
- **Anti-Bribery and Anti-Corruption Policy:** Hitech is committed to conducting all of its business activities ethically, transparently, and in full compliance with Nigerian and international anti-corruption laws. This policy sets out the principles and expectations that guide the company's efforts to prevent, detect, and respond to bribery and corruption in all forms. This policy is applicable to all individuals and entities associated with Hitech, including permanent and temporary employees, part-time staff, interns, management and board members, contractors, subcontractors, suppliers, vendors, agents, consultants, joint venture partners, and any other parties acting on behalf of or in collaboration with the company.
- **Anti-Money Laundering (AML) Policy:** This Anti-Money Laundering (AML) Policy sets out the principles and procedures to prevent Hitech from being used to facilitate money laundering, terrorism financing, or transactions involving the proceeds of crime. Hitech strictly prohibits accepting or processing funds that are suspected to originate from criminal activities, concealing the origins of illegally obtained funds, facilitating transactions or services that assist money laundering or terrorism financing, and doing business with individuals or entities listed on national or international sanctions lists.

**HITECH Code Of Conduct:** This Code of Conduct outlines the ethical standards and behavioral expectations that apply to all HITECH employees, subcontractors, suppliers, consultants, and third-party partners. It forms part of the company's Environmental and Social Management System (ESMS) and reflects our commitment to legal compliance, responsible business, and sustainable development. Every person operating under or on behalf of HITECH is expected to adhere to these standards across all project sites and corporate activities.



### 3. Construction Environmental and Social Management and Monitoring Plan

The following section outlines the key aspects of the Construction Environmental and Social Management Plan (CESMP) for the Tin Can Port Modernization project in Nigeria. Each aspect is evaluated based on its potential impacts, proposed mitigation measures, and detailed monitoring parameters to ensure compliance with international standards and Nigerian regulations. The CESMP is designed to mitigate environmental and social impacts during the construction phase, addressing concerns related to air quality, noise, vibration, water, soil, waste management, traffic, and community health. This comprehensive approach ensures that the contractor conducts all activities responsibly, minimizing risks to the environment and surrounding communities

*Table 3-1: Summary of the Criteria Adopted to determine the absolute significance of the impacts assessed*

| Impact Intensity |          |        | Extent of the Impact |       |          | Duration of the Impact |        |      |
|------------------|----------|--------|----------------------|-------|----------|------------------------|--------|------|
| Low              | Moderate | Strong | Punctual             | Local | Regional | Short                  | Medium | Long |

For a detailed overview of the potential impacts associated with the project, please refer to the ESIA which includes an analysis and description of the impacts

#### 3.1 Monitoring Management

The plan presented also includes the monitoring component; which has been developed with the following objectives:

- Evaluating the effectiveness of the proposed mitigation measures and the protection of the ambient environment, workers and the community as per prescribed/ applicable standards for the Project
- Identify the need for improvements in the management plan
- Verify Compliance with regulations and community obligations
- Allow Comparison against baseline conditions and assess the changes in environmental quality in the Project area of influence

Please note the following with regards to the Monitoring Plan:

- The Monitoring Plan is established to include Performance Indicators.
- The primary monitoring parameters for each receptor include:
  - Frequency of measurements
  - Sampling Locations and Type
  - Methods to be adopted for monitoring
  - Parameters to be assessed
  - Respective responsibilities
- Reporting
  - The monitoring officer is required to provide their reports and feedback with ample time for response and change mechanism to ensure that actions are adequately integrated across the project cycle based on reporting



- The reporting mechanism for the monitoring plan will provide a clear indication whether the measures articulated in the ESMP are being implemented or if there is clear lack of compliance with the ESMP mitigation measures.

The ESMMP is high level matrix that summarizes for the environmental and social management team for the project the primary monitoring KPIs that can assess the environmental and social performance of the project. It also provides clear guidance on the monitoring activities that need to be regularly conducted. Knowledge of the ESMMP, just like the ESMP, must also be with other project development personnel such as the construction site manager, the QA team associated with the engineering of the project, and the project manager/director.

The ESMMP is separated for the construction and operation of the project, since in the majority of operations, monitoring of specific environmental receptors will be a consistent and regular activity that will require a standalone matrix for monitoring and follow up. However, the indicators for decommissioning are joint with the construction phase as they include similar impacts that require monitoring.

## 3.1 Construction Environmental and Social Management and Monitoring Plan

Table 3-2: Construction Environmental and Social Management and Monitoring Plan

| N | Aspect                          | Potential Impact                                                                                                                                                                                                                                        | Impact Magnitude | Mitigation Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Monitoring Parameters and Frequency                                                                                                                                                                                                                                                                                                                            | Location, Thresholds, Responsibility                                                                                                                                                                              | KPIs                                                                                                                     | Target                                                                                 | Residual Impact |
|---|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------|
| 1 | Soil, Surface and Water Quality | Potential to impact soil and sediment quality through soil compaction, erosion, fuel and lubricant spills, improper waste disposal, and contamination from hazardous materials, which can harm nearby ecosystems and human health                       | Moderate         | All the construction material will be stored in dedicated storage areas;<br>A designated machinery and equipment storage area will be developed for the Project;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Parameters:<br>• Project areas are used and organized as per the layout<br>Frequency:<br>• Daily Inspections                                                                                                                                                                                                                                                   | Locations:<br>• Storage areas for machinery and construction material<br>Thresholds:<br>• Storage shall not exceed the allocated areas unless approved by Project Manager<br>Responsibility:<br>• Site Supervisor | • Number of non-conformities                                                                                             | • Zero non-conformities                                                                | Not-Significant |
| 2 |                                 |                                                                                                                                                                                                                                                         |                  | Storage of construction materials, fuels, lubricants, and hazardous materials in dedicated areas with secondary containment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Parameters:<br>• Inspections of storage areas<br>• Ensure proper material storage, containment functionality<br>• no spills<br>Frequency:<br>• Weekly                                                                                                                                                                                                          | Locations:<br>• Construction sites<br>• machinery storage<br>• fuel storage<br>• hazardous material storage<br>Responsibility:<br>• Site supervisor<br>• Environmental officer                                    | • Number of incidents of improper material storage or spills<br>• Number of non-conformities related to waste management | • Zero unresolved incidents.                                                           |                 |
| 3 |                                 |                                                                                                                                                                                                                                                         |                  | Removal of temporary structures, surplus materials, and waste upon completion of work<br>Use of construction and demolition waste for site filling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Parameters:<br>• Waste quantities and traceability through the waste logs<br>Frequency:<br>• Weekly inspections of waste storage and disposal practices<br>• Quarter Audits                                                                                                                                                                                    | Locations:<br>• Entire project site<br>• Waste storage areas<br>Thresholds:<br>• Waste cannot exceed the allocated area limits<br>Responsibility:<br>• Site supervisor<br>• Environmental officer                 | • Volume of waste removed and disposed of properly<br>• Related Grievances                                               | • 100% compliance with waste removal protocols<br>• Zero unresolved related grievances |                 |
| 4 |                                 | Potential Impact on coastal morphology, affecting shoreline shape, sediment dynamics, erosion patterns, beach profiles, intertidal zones, and marine habitats, with potential impacts on sediment distribution, seabed stability, and soil salinization | Moderate         | Establish drainage structures and stormwater system to minimize erosion and maintain natural features.<br>Stormwater shall be separated from sanitary (domestic) and industrial wastewater streams in accordance with the WWMP. Stormwater from potentially contaminated areas (e.g., workshops, fuel storage, wash bays, batching/washout areas) shall be captured and routed through silt traps and/or pre-treatment units (grease traps / oil-water separators) prior to discharge to the stormwater network.<br>Surface runoff from process areas or potential sources of contamination should be prevented, or where not practical, runoff from process and storage areas should be segregated from potentially less contaminated runoff. | Parameters:<br>• Inspections and erosion observations<br>• Drainage functionality; evidence of cross-connection between stormwater and wastewater<br>Frequency:<br>• Conduct weekly erosion inspections of erosion-prone areas.<br>• Evaluate reassessed site-specific measures continuously whenever needed based on weekly observation logs and near misses. | Location:<br>• Shoreline next to project sites.<br>Responsibility:<br>• Environmental Officer                                                                                                                     | • Number of erosion incidents.                                                                                           | • Zero unresolved incidents                                                            | Minor           |

| N | Aspect | Potential Impact                                                                                                                                                                                                                                                                                | Impact Magnitude | Mitigation Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Monitoring Parameters and Frequency                                                                                                                                                                                                                                                                           | Location, Thresholds, Responsibility                                                                                                                                                                                                                                                               | KPIs                                                                                                                                                                                                               | Target                                                                                             | Residual Impact |
|---|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------|
|   |        |                                                                                                                                                                                                                                                                                                 |                  | Continuously monitor shoreline changes and adapt management strategies accordingly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                    |                                                                                                    |                 |
| 5 |        |                                                                                                                                                                                                                                                                                                 |                  | Establishment of a Stormwater system. Stormwater runoff from process/maintenance areas shall be routed through silt traps and, where relevant, grease traps/oil-water separators before entering the stormwater network.<br>No untreated wastewater (domestic or industrial) shall be discharged to Badagry Creek / Lagos Lagoon<br>Leverage rainwater and flood resources to establish a long-term leaching mechanism. This helps reduce soil salinity over time                                                                                                                                                                                                                        | Parameters:<br>• Water Quality Analysis<br>• Suspended Solids (SS), Oil & Grease (and visual checks for sheen/debris) at stormwater discharge points.<br>Frequency:<br>• Biannually Water Quality Analysis of stormwater after grease traps<br>• Daily Inspections of the grease traps and stormwater system. | Location:<br>• General Support Yard, Laydown Yard and Construction Site.<br>Thresholds:<br>• Total Dissolved Solids (ppm) should not exceed 1000 mg/l<br>Responsibility:<br>• Environmental Officer                                                                                                | • Near misses<br>• Non-Conformities<br>• Environmental Pollution Incidents                                                                                                                                         | • Zero unresolved incidents of water pollution                                                     |                 |
| 6 |        |                                                                                                                                                                                                                                                                                                 |                  | Analyse soil properties before disposal to ensure compatibility with the seabed.<br>Establish and Supplier Management Plan to ensure sustainable sources of material and undertake monitoring and audits to suppliers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Parameters:<br>• Conduct tests prior to soil disposal to control grain size distribution, chemical composition, salinity levels, organic matter content.<br>• Registries of sand disposal<br>Frequency:<br>• Every disposal action or event.                                                                  | Location:<br>• Designated soil disposal sites.<br>Thresholds:<br>• Compliance with environmental soil compatibility standards.<br>• Maintain a 5% variation from baseline results.<br>Responsibility:<br>• Environmental Officer<br>• Supplier                                                     | • Percentage of soil tests that meet compatibility requirements.                                                                                                                                                   | • 100% of soil tests show compatibility with disposal requirements.                                |                 |
| 7 |        |                                                                                                                                                                                                                                                                                                 |                  | Maintenance or repair of mechanized fleet to primarily take place in dedicated workshop facilities located at the Savaric Yard (Ibafon) and the General Support Yard within Tin Can Port, each with bunded, impervious floors, oil traps and a wash bay connected to a closed septic tank. Septic tanks will be managed in line with the Project Wastewater Management Plan and emptied periodically by a licensed, LAWMA-approved contractor; no direct discharge to soil or surface water will be allowed. Oil-water separators and grease traps should be installed and maintained as appropriate at refuelling facilities, workshops, wash bays, fuel storage and containment areas. | Parameters:<br>• Frequency of inspections.<br>• Condition and functionality of oil traps and separators.<br>• Presence of any spills or leaks.<br>Frequency:<br>• Weekly inspections, with monthly comprehensive reviews.                                                                                     | Locations:<br>• Off-site workshop facilities.<br>• Refuelling facilities.<br>• Wash bays and storage areas.<br>Thresholds:<br>• No visible oil spills or leaks.<br>• Proper functionality of oil traps and separators at 100%.<br>Responsibility:<br>• Site supervisor<br>• Environmental officer. | • Number of inspections conducted.<br>• Number of incidents of oil spills or leaks.<br>• Percentage of oil traps and separators functioning properly.<br>• Number of maintenance activities completed on schedule. | • 100% completion of scheduled inspections.<br>• Zero unresolved incidents of oil spills or leaks. |                 |
| 8 |        | Flooding, Erosion and Sedimentation Impact. Land reclamation may increase the risk of flooding, erosion, and sedimentation by altering the natural coastline, disrupting sediment transport patterns, reducing coastal habitats, and disturbing the balance of sediment deposition and erosion. | Minor            | Erosion Control: Sediment control measures are instead focused on localized, controlled in-water activities<br>Sediment Management: Seabed disturbance will be minimal, targeted, and performed using a barge-mounted long-reach excavator under controlled conditions. Diver supervision and real-time positioning (RTK/DGPS) will ensure precise placement and minimize turbidity. If small                                                                                                                                                                                                                                                                                            | Parameters:<br>• Erosion rates<br>• sediment levels<br>• coastal stability<br>• flood incidents.<br>Frequency:<br>• Monthly inspections<br>• quarterly reviews.                                                                                                                                               | Locations:<br>• Waterfront construction site<br>Thresholds:<br>• no significant flooding.<br>Responsibility:<br>• Environmental Officer<br>• Coastal Engineer.                                                                                                                                     | • sediment levels<br>• effectiveness of coastal structures.                                                                                                                                                        | • Zero unresolved erosion and flood incidents                                                      | Non-Significant |

| N  | Aspect | Potential Impact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Impact Magnitude | Mitigation Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Monitoring Parameters and Frequency                                                                                                                                                                                                                                                                                                                                                 | Location, Thresholds, Responsibility                                                                                                                                                         | KPIs                                                                                                                                                        | Target                                                                                                                                                 | Residual Impact |
|----|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
|    |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  | quantities of sediment are displaced during mattress installation, they will be contained within the immediate work area. Should any pockets of contaminated sediment be encountered, these will be segregated, placed in lined skips or barges, and transported by licensed contractors to LASEPA-approved facilities. Continuous monitoring will ensure no release of contaminated material into the lagoon.<br>Flood Management: Develop flood management plans and install drainage systems.                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                              |                                                                                                                                                             |                                                                                                                                                        |                 |
| 9  |        | impact the Marine Environment, Hydrology, and Ecosystems through habitat destruction, introduction of pollutants, alteration of water flow patterns, sediment accumulation, and potential loss of protective coastal features, affecting marine life and ecosystem services.                                                                                                                                                                                                                                                                                                                                                                                                    | Moderate         | Habitat Protection: Identify and protect critical marine habitats.<br>Pollution Prevention: Implement spill prevention plans and use eco-friendly materials, controlled placement techniques, spill-prevention measures, and water-quality monitoring<br>Flow Management: Maintain natural water flow patterns during construction.<br>Pollution prevention and sediment controls are governed by the CESMP and SWMP; no separate BMP applies during construction.<br>Wastewater controls (domestic septic tanks; industrial pre-treatment via grease traps/oil-water separators/sedimentation pits; stormwater segregation) shall be implemented in accordance with the WWMP to prevent contaminated discharges to the lagoon | Parameters:<br>• Water quality, sediment levels, marine habitat conditions<br>• water flow patterns.<br>Frequency:<br>• Surface water (lagoon/creek): Conductivity, temperature, pH, turbidity, dissolved oxygen, bacteriological quality – biannually<br>• Effluent/yard discharge checks: SS, COD, BOD – biannually<br>• Routine: weekly visual checks for sheen/turbidity/debris | Locations:<br>• Construction sites<br>• adjacent marine areas.<br>Thresholds:<br>• Water quality and sediment levels within safe limits.<br>Responsibility:<br>Environmental Officer         | <ul style="list-style-type: none"> <li>Number of pollution incidents</li> <li>sediment and water quality levels</li> <li>habitat health reports.</li> </ul> | <ul style="list-style-type: none"> <li>Zero pollution incidents</li> <li>no significant habitat destruction.</li> </ul>                                | Minor           |
| 10 |        | Mobilisation of pollutants from contaminated land or sediments can occur if earthworks disturb legacy contamination. Baseline studies of the Lagos Lagoon and Tin Can Island area indicate the presence of heavy metals (e.g., Cu, Pb, Zn, Cd) and hydrocarbons in sediments due to long-term industrial activity.<br>Only very limited excavation will take place in this Project, restricted to the controlled removal of damaged quay wall elements, localized seabed preparation for scour protection, and minor excavation inside the existing construction yards. No bulk earthworks or land reshaping will occur. All works take place on existing paved port platforms. | Minor            | Soil Testing: Conduct pre-construction soil tests to identify contamination.<br>Containment Measures: Use liners and barriers to prevent pollutant spread.<br>Safe Handling Procedures: Implement procedures for safe handling and disposal of contaminated soil.<br>Protective Gear: Ensure workers use appropriate protective gear.                                                                                                                                                                                                                                                                                                                                                                                          | Parameters:<br>• Levels of heavy metals and hydrocarbons in soil and water.<br>Frequency:<br>• Weekly soil and water quality tests during earthworks.                                                                                                                                                                                                                               | Locations:<br>• Earthwork sites<br>• contaminated land areas.<br>Thresholds:<br>• Pollutant levels within safety limits.<br>Responsibility:<br>• Environmental Officer<br>• Site Supervisor. | <ul style="list-style-type: none"> <li>Number of pollutant level exceedances</li> <li>compliance with handling procedures.</li> </ul>                       | <ul style="list-style-type: none"> <li>Zero exceedances of pollutant levels</li> <li>100% compliance with handling and disposal procedures.</li> </ul> | Minor           |

| N  | Aspect               | Potential Impact                                                                                                                                                                                            | Impact Magnitude | Mitigation Measures                                                                                                                                                                                                                                                                                                                                                       | Monitoring Parameters and Frequency                                                                                                                                                                   | Location, Thresholds, Responsibility                                                                                                                                                                                         | KPIs                                                                                     | Target                                                                                | Residual Impact |
|----|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------|
| 11 | Air Quality          | Potential Impact ambient air quality through construction activities like demolition, concrete pouring, and heavy equipment operation, as well as transportation of materials by trucks.                    | Minor            | Dust Control: Implement water spraying and other dust suppression techniques.<br>Covering Materials: Cover materials during transport to minimize dust release.<br>Regular Maintenance: Maintain equipment to ensure optimal performance and reduced emissions.<br>Emissions Control: Use low-emission vehicles and machinery.                                            | Parameters:<br>• Dust levels (PM10, PM2.5), emission levels (NOx, CO2).<br>Frequency:<br>• Continuous air quality monitoring<br>• weekly inspections.                                                 | Locations:<br>• Construction sites<br>• material transport routes.<br>Thresholds:<br>• Dust and emission levels within acceptable air quality standards.<br>Responsibility:<br>• Environmental Officer<br>• Site Supervisor. | • Number of dust and emission level exceedances<br>• frequency of equipment maintenance. | • Zero exceedances of dust and emission levels                                        | Minor           |
| 12 |                      | The project may lead to increased dust emissions from various construction activities and ancillary sites, potentially affecting air quality                                                                | Minor            | Dust Suppression: Implement water spraying and other dust control measures.<br>Covering Materials: Cover stockpiles and trucks to minimize dust.<br>Paved Roads: Use paved or treated roads to reduce dust from vehicle movement.                                                                                                                                         | Parameters:<br>• Dust particle levels (PM10, PM2.5).<br>Frequency:<br>• Continuous dust monitoring<br>• weekly reviews.                                                                               | Locations:<br>• Construction sites<br>• ancillary sites.<br>Thresholds:<br>• Dust levels within acceptable air quality standards.<br>Responsibility:<br>• Environmental Officer<br>• Site Supervisor.                        | • Number of dust exceedances<br>• effectiveness of dust suppression measures.            | • Zero dust level exceedances                                                         | Minor           |
| 13 |                      | Increase in air pollution through elevated vehicle emissions, including nitrogen oxides, volatile organic compounds, and particulate matter, which can adversely affect human health                        | Minor            | Emission Standards: Enforce strict emission standards for vehicles.<br>Alternative Fuels: Promote the use of alternative fuel vehicles.<br>Vehicle Maintenance: Regularly maintain vehicles to ensure optimal performance and lower emissions.                                                                                                                            | Parameters:<br>• Levels of nitrogen oxides (NOx), volatile organic compounds (VOCs), and particulate matter (PM10, PM2.5).<br>Frequency:<br>• Continuous air quality monitoring<br>• monthly reviews. | Locations:<br>• Main transport routes<br>• project sites.<br>Thresholds:<br>• Pollutant levels within environmental safety standards.<br>Responsibility:<br>• Environmental Officer<br>• Fleet Manager.                      | • Number of pollutant level exceedances<br>• vehicle maintenance records.                | • Zero exceedances of pollutant levels<br>• 100% completion of scheduled maintenance. | Minor           |
| 14 |                      | combustion activities, including operation of diesel power generators and equipment, are key sources of emissions that can adversely affect human health through pollutants like NO2, SO2, PM10, and PM2.5. | Minor            | Emission Control: Use emission control devices like diesel particulate filters and catalytic converters.<br>Low-Emission Equipment: Utilize low-emission or alternative fuel equipment, if possible.<br>Regular Maintenance: Conduct regular maintenance of generators and equipment to ensure optimal performance.<br>Dust Control: Implement dust suppression measures. | Parameters:<br>• Levels of NO2, SO2, PM10, and PM2.5.<br>Frequency:<br>• Continuous air quality monitoring, quarterly reviews.                                                                        | Locations:<br>• Generator sites<br>• equipment operation areas.<br>Thresholds:<br>• Pollutant levels within environmental safety standards.<br>Responsibility:<br>• Environmental Officer<br>• Maintenance Supervisor.       | • Number of pollutant level exceedances<br>• maintenance records.                        | • Zero exceedances of pollutant levels<br>• 100% completion of scheduled maintenance. | Not significant |
| 15 | Noise and Vibrations | Impact on ambient noise quality from port construction activities and ancillary sites.                                                                                                                      | Minor            | Noise Barriers: Install temporary noise barriers around construction zones.<br>Low-Noise Machinery: Utilize equipment with noise reduction features.<br>Work Scheduling: Schedule noisy activities during off-peak hours.<br>Monitoring and Communication: Regularly monitor noise levels and inform the community about construction schedules.                          | Parameters:<br>• Ambient noise levels (decibels).<br>Frequency:<br>• Continuous noise monitoring and monthly reviews.                                                                                 | Locations:<br>• Construction sites<br>• ancillary sites<br>• surrounding areas.<br>Thresholds:<br>• Noise levels within environmental safety standards.<br>Responsibility:<br>• Environmental Officer<br>• Site Supervisor.  | • Number of noise level exceedances<br>• community noise complaints.                     | • Zero noise level exceedances                                                        | Minor           |



| N  | Aspect | Potential Impact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Impact Magnitude | Mitigation Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Monitoring Parameters and Frequency                                                                                                                                | Location, Thresholds, Responsibility                                                                                                                                                                  | KPIs                                                                                | Target                                                                           | Residual Impact |
|----|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------|
|    |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  | Work Scheduling: Noisy activities (e.g., pile driving, demolition, heavy equipment use) will be scheduled during off-peak hours to minimise disruption to both workers and nearby communities. For communities, this reduces exposure to construction noise during sensitive times such as evenings, nights, and weekends. Advance notice will be provided to affected residents and businesses, and noise monitoring will be undertaken at the site boundary to ensure compliance with permissible limits.                                                                                                                                                                             |                                                                                                                                                                    |                                                                                                                                                                                                       |                                                                                     |                                                                                  |                 |
| 16 |        | Potential Impact on Human Health Associated with Increased Noise Levels from Construction Activities to other terminal operators and contractors in the area                                                                                                                                                                                                                                                                                                                                                                                      | Minor            | Noise Barriers: Install temporary noise barriers around construction sites.<br>Quiet Equipment: Use machinery with noise reduction features.<br>Work Scheduling: Schedule noisy activities during times that least impact other operators and contractors.<br>Communication: Inform nearby operators and contractors about the construction schedule.                                                                                                                                                                                                                                                                                                                                   | Parameters:<br>• Noise levels (decibels).<br>Frequency:<br>• Continuous noise monitoring.                                                                          | Locations:<br>• Construction sites<br>• adjacent terminals.<br>Thresholds:<br>• Noise levels within occupational safety limits.<br>Responsibility:<br>• Site Supervisor<br>• Health & Safety Officer. | • Number of noise level exceedances<br>• complaints from operators and contractors. | • Zero noise exceedances<br>• Zero unresolved complaints related to noise levels | Minor           |
| 17 |        | Generate underwater noise and vibrations from activities such as pile driving, and vessel transportation, potentially impacting marine life and ecosystems.                                                                                                                                                                                                                                                                                                                                                                                       | Moderate         | Noise Abatement Technologies: Use noise dampening techniques and equipment like bubble curtains around pile driving sites.<br>Scheduling: Conduct noisy activities during times least sensitive to marine life.<br>Speed Restrictions: Implement speed limits for vessels to minimize underwater noise.                                                                                                                                                                                                                                                                                                                                                                                 | Parameters:<br>• Underwater noise levels<br>• marine life behaviour.<br>Frequency:<br>• Continuous noise monitoring<br>• bi-weekly marine life surveys.            | Locations:<br>• Pile driving sites<br>• vessel routes.<br>Thresholds:<br>• Underwater noise levels within safe limits for marine life.<br>Responsibility:<br>• Environmental Officer                  | • Number of noise exceedances<br>• observed disturbances in marine life behaviour.  | • Zero noise exceedances                                                         | Minor           |
| 18 |        | Impact nearby community structures due to ground vibrations from construction activities, potentially leading to discomfort, stress, and safety risks, particularly for sensitive structures. Construction activities such as pile driving, compaction, and demolition may generate ground vibrations that could affect nearby community structures and cause discomfort or stress to residents. Sensitive structures (e.g., informal housing, clinics, schools, or older buildings with compromised foundations) may be particularly vulnerable. | Major            | Scientific Analysis: Prior to commencement, a vibration risk assessment will be conducted, including:<br>Pre-construction structural surveys of buildings within the zone of influence;<br>Establishment of vibration thresholds based on international standards (e.g., DIN 4150, BS 7385) and IFC EHS Guidelines;<br>Modelling of expected vibration levels from planned equipment and methods.<br>Vibration Control: Implement vibration monitoring and control techniques.<br>Sensitive Structure Surveys: Conduct pre-construction surveys of nearby sensitive structures.<br>Construction Scheduling: Schedule construction activities to minimize impact during sensitive times. | Parameters:<br>• Vibration levels<br>• structural integrity of nearby buildings.<br>Frequency:<br>• Continuous vibration monitoring<br>• weekly structural surveys | Locations:<br>• Construction sites<br>• nearby sensitive structures.<br>Thresholds:<br>• Vibration levels within acceptable limits.<br>Responsibility:<br>• Site Supervisor<br>• Structural Engineer. | • Number of vibration exceedances<br>• number of structural damage incidents.       | • zero unresolved structural damage incidents.                                   | Moderate        |

| N  | Aspect    | Potential Impact                                                                                                                                                                                                                                                                                                          | Impact Magnitude | Mitigation Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Monitoring Parameters and Frequency                                                                                                                                                                                                                                                      | Location, Thresholds, Responsibility                                                                                                                                                               | KPIs                                                                                                      | Target                                                                          | Residual Impact |
|----|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------|
|    |           |                                                                                                                                                                                                                                                                                                                           |                  | Community Communication: Inform the community about construction activities and schedules.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                    |                                                                                                           |                                                                                 |                 |
| 19 | Waste     | Construction activities will generate various types of waste, including construction debris, excavated soil, packaging waste, chemical waste, metal scraps, wood waste, plastic waste, electrical and electronic waste, and hazardous waste, which could potentially impact the local environment and community receptors | Moderate         | Waste Management Plan: Develop and implement a comprehensive waste management plan.<br>Segregation and Recycling: Segregate waste streams and maximize recycling.<br>Proper Disposal: Ensure proper disposal of hazardous and non-hazardous waste.<br>Training Programs: Train workers on waste management practices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Parameters:<br>• Waste types and volumes<br>• recycling rates<br>• disposal records.<br>Frequency:<br>• Weekly waste audits<br>• monthly reviews.                                                                                                                                        | Locations:<br>• Construction sites<br>• waste storage areas.<br>Thresholds:<br>• Compliance with waste management regulations.<br>Responsibility:<br>• Environmental Officer<br>• Site Supervisor. | • Waste segregation rates<br>• recycling rates<br>• number of non-compliance incidents.                   | • 100% waste segregation<br>• zero unresolved non-compliance incidents.         | Minor           |
| 20 | Hydrology | Spills or Leaks Resulting in Contamination of Groundwater and Surface Water. The fuels or hydraulic fluids used in the equipment have the potential to spill, which can lead to the release of hydrocarbons. Contamination Risk: The soil, groundwater, and surface water in the area can become contaminated by these    | Moderate         | Spill Prevention Plans: Develop and implement spill prevention and response plans.<br>Secondary Containment: Use secondary containment systems for fuel and hydraulic fluid storage.<br>Regular Maintenance: Regularly inspect and maintain equipment to prevent leaks.<br>Emergency Response Training: Train staff in spill response and containment procedures.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Parameters:<br>• Soil and water quality (hydrocarbon levels)<br>• equipment condition.<br>Frequency:<br>• After spill: sampling for hydrocarbons/chemicals as needed following any spill event<br>• weekly equipment inspections.<br>• Maintain spill logs and corrective action records | Locations:<br>• Storage areas<br>• construction sites.<br>Thresholds:<br>• Hydrocarbon levels within safe limits.<br>Responsibility:<br>• Environmental Officer<br>• Site Supervisor.              | • Number of spills or leaks<br>• hydrocarbon levels in soil and water<br>• equipment maintenance records. | • Zero unhandled spills or leaks<br>• 100% completion of scheduled maintenance. | Minor           |
| 21 |           | Activities such as quay demolition, drilling and piling, grouting, backfilling, and raw material storage and handling can lead to increased turbidity in nearby water bodies due to the resuspension of bottom sediment, production of fine particulate matter, and inadequate sediment management methods.               | Moderate         | Silt curtains will be used only where safe and operationally feasible, considering port currents, vessel movements, navigation lanes, and berthing operations. If conditions do not allow deployment, equivalent sediment control measures will be applied — including precise mechanical excavation, timing during low current periods, and turbidity monitoring.<br>Dust Suppression: Use water spraying and dust control methods.<br>Sediment Management: Seabed disturbance will be minimal, targeted, and performed using a barge-mounted long-reach excavator under controlled conditions. Diver supervision and real-time positioning (RTK/DGPS) will ensure precise placement and minimize turbidity. If small quantities of sediment are displaced during mattress installation, they will be contained within the immediate work area. Should any pockets of contaminated sediment be encountered, these will be segregated, placed in lined skips or barges, and transported by licensed contractors to LASEPA-approved facilities. Continuous monitoring will ensure no release of contaminated material into the lagoon.<br>Material Handling Protocols: Ensure proper storage and handling of raw materials. | Parameters:<br>• Water turbidity<br>Dust levels<br>Frequency:<br>• Weekly water turbidity tests<br>Daily dust monitoring.<br>Effluent discharges (where applicable): SS, COD, BOD – biannually                                                                                           | Locations:<br>• Construction sites<br>• adjacent water bodies.<br>Thresholds:<br>• Turbidity and dust levels within safe limits.<br>Responsibility:<br>• Environmental Officer<br>Site Supervisor. | • Number of turbidity exceedances<br>• dust level readings.                                               | • Zero turbidity exceedances<br>• dust levels within acceptable ranges.         | Minor           |

| N  | Aspect       | Potential Impact                                                                                                                                                                                                                              | Impact Magnitude | Mitigation Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Monitoring Parameters and Frequency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Location, Thresholds, Responsibility                                                                                                                                                                                                                                                                                                                                                                                                                                     | KPIs                                                                                                                                                                                        | Target                                                                                                                                                         | Residual Impact |
|----|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
|    |              |                                                                                                                                                                                                                                               |                  | All concrete washout water, drilling/casing slurry water (if any), and wash water shall be managed via WWMP controls (sedimentation/decantation pits + controlled disposal) to prevent turbidity and contaminated runoff                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                             |                                                                                                                                                                |                 |
| 22 | Biodiversity | construction activities may disrupt ecosystem services such as soil stability, nutrient cycling, and habitat provision, while sediment runoff and chemical pollutants can degrade water quality and dust and emissions can impact air quality | Minor            | <p>Nutrient Management: Preserve and enhance nutrient cycling through soil conservation practices.</p> <p>Habitat Protection: Identify and protect key habitats.</p> <p>Pollution Prevention: Proper handling and disposal of chemical pollutants.</p> <p>Dust Suppression: Water spraying and dust control methods.</p> <p>Emission Control: Use low-emission equipment and vehicles.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Parameters:</p> <ul style="list-style-type: none"> <li>• Soil stability</li> <li>• nutrient levels</li> <li>• habitat conditions</li> <li>• sediment levels</li> <li>• water quality</li> <li>• air quality (dust and emissions).</li> </ul> <p>Frequency:</p> <ul style="list-style-type: none"> <li>• monthly water and air quality tests.</li> </ul>                                                                                                                                                                                                                            | <p>Locations:</p> <ul style="list-style-type: none"> <li>• Construction sites</li> <li>• adjacent water bodies</li> <li>• air monitoring stations.</li> </ul> <p>Thresholds:</p> <ul style="list-style-type: none"> <li>• Sediment levels</li> <li>• pollutant concentrations</li> <li>• air quality indices within safe limits.</li> </ul> <p>Responsibility:</p> <ul style="list-style-type: none"> <li>• Environmental Officer</li> <li>• Site Supervisor.</li> </ul> | <ul style="list-style-type: none"> <li>• nutrient level</li> <li>• habitat health, sediment and pollutant levels</li> <li>• air quality readings.</li> </ul>                                | <ul style="list-style-type: none"> <li>• no net loss</li> </ul>                                                                                                | Minor           |
| 23 |              | Disturbance of Marine Ecology due to land reclamation, infrastructure development, release of sediment and construction noise and vibrations.                                                                                                 | Minor            | <ul style="list-style-type: none"> <li>• Implement quiet construction methods and ensure equipment is well maintained to minimize underwater and ambient noise and vibration.</li> <li>• Schedule works to avoid unnecessary disturbance during peak tidal or high-current conditions where practicable.</li> <li>• Prevent uncontrolled release of sediments, debris, cement residues, or pollutants from quay-side works through good housekeeping, spill prevention, and containment measures.</li> <li>• Implement sediment and runoff controls in accordance with the Soil and Water Management Plan (SWMP), including drainage management, interception of runoff, and prevention of direct discharge to marine waters.</li> <li>• Wastewater and wash water from quay-side activities shall be managed in accordance with the WWMP (segregation of streams; pre-treatment where relevant; no untreated discharge).</li> </ul> <p>No in-water construction, dredging, or land reclamation activities are planned; therefore, no standalone Biodiversity Management Plan applies during the construction phase.</p> | <p>Parameters:</p> <ul style="list-style-type: none"> <li>• Visual inspection of quay-side works and adjacent water surface for turbidity, debris, or sheen</li> <li>• Construction noise and vibration observations</li> <li>• Review of environmental observation logs and near-miss records</li> <li>• Evidence of any wastewater discharge pathways to marine waters (none permitted)</li> </ul> <p>Frequency:</p> <ul style="list-style-type: none"> <li>• Weekly inspections during active works</li> <li>• Ad-hoc inspections following incidents or heavy rainfall</li> </ul> | <p>Locations:</p> <ul style="list-style-type: none"> <li>• Active quay rehabilitation areas</li> <li>• Adjacent marine waters</li> </ul> <p>Thresholds:</p> <ul style="list-style-type: none"> <li>• No visible excessive turbidity, debris, or pollution</li> <li>• Noise and vibration within applicable construction limits</li> </ul> <p>Responsibility:</p> <p>Environmental Officer<br/>Site Supervisor</p>                                                        | <ul style="list-style-type: none"> <li>• Number of sediment or pollution incidents</li> <li>• Number of noise or vibration complaints</li> <li>• Environmental near-miss records</li> </ul> | <ul style="list-style-type: none"> <li>• Zero significant marine pollution incidents</li> <li>• No sustained exceedance of sediment or noise limits</li> </ul> | Minor           |
| 24 | Social       | Road Traffic and Transport Risk of Project Vehicle Traffic Accidents due to increased traffic volume during construction on                                                                                                                   | Moderate         | Traffic Management Plan: Implement traffic control measures, designated routes, and schedules.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Parameters:</p> <ul style="list-style-type: none"> <li>• Traffic incidents</li> <li>• compliance with traffic plans</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Locations:</p> <ul style="list-style-type: none"> <li>• Construction sites</li> <li>• main transport routes</li> </ul>                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>• Number of traffic incidents</li> </ul>                                                                                                             | <ul style="list-style-type: none"> <li>• Number of unresolved traffic incidents is zero</li> </ul>                                                             | Moderate        |

| N  | Aspect | Potential Impact                                                                                                                                                                                                                                                               | Impact Magnitude | Mitigation Measures                                                                                                                                                                                                                                                                                                                                       | Monitoring Parameters and Frequency                                                                                                                                                                                                                                                                                                                                                                                                                   | Location, Thresholds, Responsibility                                                                                                                                                                                                                                                                                                                                                            | KPIs                                                                                                                                                                                                      | Target                                                                                                                            | Residual Impact |
|----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------|
|    |        | employees, nearby residents, and businesses                                                                                                                                                                                                                                    |                  | Driver Training: Ensure all drivers are trained in safe driving practices.<br>Vehicle Maintenance: Regular maintenance checks for all project vehicles.<br>Community Awareness: Inform nearby residents and businesses about traffic changes.                                                                                                             | <ul style="list-style-type: none"> <li>vehicle condition reports.</li> </ul> Frequency: <ul style="list-style-type: none"> <li>Daily checks and weekly reviews.</li> </ul>                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>community areas.</li> </ul> Responsibility: <ul style="list-style-type: none"> <li>Project Manager</li> <li>Traffic Safety Officer.</li> </ul>                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>compliance rate with traffic plans</li> <li>vehicle maintenance records.</li> </ul>                                                                                |                                                                                                                                   |                 |
| 25 |        | Occupational health and safety issues arose construction activities including physical hazards, chemical hazards, confined spaces, exposure to dust and noise.                                                                                                                 | Moderate         | Implement Safety Protocols: Regular training, PPE, and safety gear.<br>Chemical Handling Procedures: Proper storage and use of chemicals.<br>Confined Spaces Protocol: Ensure safe entry and exit procedures.<br>Dust and Noise Control: Dust suppression and noise barriers.                                                                             | Parameters: <ul style="list-style-type: none"> <li>Safety incidents</li> <li>chemical exposure</li> <li>dust levels</li> <li>noise levels.</li> </ul> Frequency: <ul style="list-style-type: none"> <li>Weekly inspections and monthly reviews.</li> </ul>                                                                                                                                                                                            | Locations: <ul style="list-style-type: none"> <li>Construction sites</li> <li>chemical storage areas</li> <li>confined spaces.</li> </ul> Thresholds: <ul style="list-style-type: none"> <li>Incident rates and exposure levels within safety limits.</li> </ul> Responsibility: <ul style="list-style-type: none"> <li>Health &amp; Safety Officer</li> <li>Site Supervisor.</li> </ul>        | <ul style="list-style-type: none"> <li>Number of incidents</li> <li>compliance with safety protocols</li> <li>dust and noise measurements.</li> </ul>                                                     | <ul style="list-style-type: none"> <li>100% compliance with safety protocols</li> </ul>                                           | Minor           |
| 26 |        | Community health and safety may be impacted by physical injury risks, exposure to hazardous materials, respiratory effects from air emissions, spread of communicable diseases, changes to diet and subsistence foods, and strain on local health services and infrastructure. | Minor            | Safety Protocols: Strict adherence to prevent injuries.<br>Hazardous Material Management: Proper handling and disposal.<br>Air Quality Control: Dust suppression and emission reduction.<br>Health Programs: Screenings and vaccinations.<br>Diet Support: Ensure nutritious food availability.<br>Infrastructure Upgrade: Improve local health services. | Parameters: <ul style="list-style-type: none"> <li>Injury incidents</li> <li>hazardous exposure</li> <li>air quality</li> <li>disease rates</li> <li>food availability</li> <li>health service usage.</li> </ul> Frequency: <ul style="list-style-type: none"> <li>Monthly assessments, quarterly reviews.</li> </ul>                                                                                                                                 | Locations: <ul style="list-style-type: none"> <li>Construction sites</li> <li>communities</li> <li>health facilities.</li> </ul> Responsibility: <ul style="list-style-type: none"> <li>Health &amp; Safety Officer</li> <li>Environmental Officer</li> <li>Community Relations Officer.</li> </ul>                                                                                             | <ul style="list-style-type: none"> <li>Injury incidents</li> <li>exposure levels</li> <li>air quality indices</li> <li>disease rates</li> <li>food availability</li> <li>health service usage.</li> </ul> | <ul style="list-style-type: none"> <li>Zero unhandled injuries</li> </ul>                                                         | Minor           |
| 27 |        | Disruption on fishing activities by damaging fishing grounds, polluting waters, and altering fish habitats, leading to potential declines in fish populations and reduced income for local fishermen.                                                                          | Moderate         | Environmental Monitoring: Regular water quality checks and assessments of fish populations, controlled placement techniques, spill-prevention measures, and water-quality monitoring<br>Engage with Local Fishermen: Develop compensation plans and alternative livelihood programs.                                                                      | Parameters: <ul style="list-style-type: none"> <li>Water quality (pH, turbidity, contaminants)</li> <li>fish population health.</li> </ul> Frequency: <ul style="list-style-type: none"> <li>Surface water: pH, turbidity, dissolved oxygen, bacteriological quality – biannually</li> <li>If a spill occurs: hydrocarbons/chemicals – as needed.</li> <li>Routine: weekly visual inspections near active works for sheen/turbidity/debris</li> </ul> | Locations: <ul style="list-style-type: none"> <li>Fishing grounds near the construction site</li> <li>nearby marine habitats.</li> </ul> Thresholds: <ul style="list-style-type: none"> <li>Water quality parameters within safe limits</li> <li>no significant decline in fish populations.</li> </ul> Responsibility: <ul style="list-style-type: none"> <li>Environmental Officer</li> </ul> | <ul style="list-style-type: none"> <li>Number of water quality tests conducted</li> <li>incidents of water pollution</li> <li>fish population health reports.</li> </ul>                                  | <ul style="list-style-type: none"> <li>Zero unhandled incidents of water pollution</li> </ul>                                     | Minor           |
| 28 |        | Potential Impacts on Socioeconomic opportunity, job generation, stimulation of economic growth and infrastructure improvement.                                                                                                                                                 | Positive         | -                                                                                                                                                                                                                                                                                                                                                         | -                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -                                                                                                                                                                                                                                                                                                                                                                                               | -                                                                                                                                                                                                         | -                                                                                                                                 | Positive        |
| 29 |        | Potential Infringement on Workers' Human Rights if not working under safe and fair working conditions                                                                                                                                                                          | Moderate         | Implement a Robust Health and Safety Program: Regular training, provision of safety gear, and strict adherence to safety protocols.                                                                                                                                                                                                                       | Parameters: <ul style="list-style-type: none"> <li>Number of safety incidents.</li> <li>Number of grievances filed.</li> <li>Compliance rates with labour laws.</li> </ul> Frequency:                                                                                                                                                                                                                                                                 | Locations: <ul style="list-style-type: none"> <li>Construction sites.</li> <li>Worker housing and facilities.</li> </ul> Thresholds:                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Incident rates</li> <li>grievance numbers</li> <li>compliance rates</li> </ul>                                                                                     | <ul style="list-style-type: none"> <li>Zero unhandled safety incidents</li> <li>all grievances resolved within a month</li> </ul> | Moderate        |

| N  | Aspect              | Potential Impact                                                                                                                                                                                                                                                                                                                 | Impact Magnitude | Mitigation Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Monitoring Parameters and Frequency                                                                                                                                                                                                                                                                                                                                                                                                | Location, Thresholds, Responsibility                                                                                                                                                                                                                                                                                                                                    | KPIs                                                                                                                                                                                                                                                                                                                               | Target                                                                                                   | Residual Impact |
|----|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------|
|    |                     |                                                                                                                                                                                                                                                                                                                                  |                  | <p>Fair Working Conditions: Ensure compliance with labour laws, provide fair wages, and establish clear grievance mechanisms.</p> <p>Regular Audits and Inspections: Conduct frequent audits and inspections to ensure compliance with safety and labour standards.</p>                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Monthly reviews and quarterly comprehensive audits.</li> </ul>                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Zero tolerance for safety violations.</li> <li>100% compliance with labour laws and lender standards.</li> </ul> <p>Responsibility:</p> <ul style="list-style-type: none"> <li>Project Manager, Health &amp; Safety Officer, HR Manager.</li> </ul>                                                                              |                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>100% compliance with labour laws and lender standards.</li> </ul> |                 |
| 30 |                     | Local procurement serves as a direct investment in the immediate economy, benefiting local trading and service providers and contributing to profitability and income growth                                                                                                                                                     | Positive         | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -                                                                                                                                                                                                                                                                                                                                                                                                                                  | -                                                                                                                                                                                                                                                                                                                                                                       | -                                                                                                                                                                                                                                                                                                                                  | -                                                                                                        | Positive        |
| 31 |                     | The influx of workers can result in competition for resources, pressure on existing infrastructure (such as healthcare, sanitation, and housing), and potential social conflicts between locals and non-locals. It will also generate job creation for local communities and an increase in local businesses' economic activity. | Moderate         | <p>Develop a Local Hiring Policy: Prioritize hiring from local communities to ensure job creation benefits them directly.</p> <p>Improve Infrastructure: Invest in upgrading local healthcare, sanitation, and housing to accommodate increased demand.</p> <p>Community Engagement Programs: Organize regular meetings with local communities to address concerns and foster good relationships.</p> <p>Resource Management Plan: Implement measures to manage the consumption of local resources effectively.</p> <p>Conflict Resolution Mechanism: Establish a system to address and resolve conflicts between workers and locals.</p> | <p>Parameters:</p> <ul style="list-style-type: none"> <li>Resource consumption (water, electricity).</li> <li>Health and sanitation facility usage.</li> <li>Housing occupancy rates.</li> <li>Number of social conflict incidents.</li> <li>Employment rates and local business activity.</li> </ul> <p>Frequency:</p> <ul style="list-style-type: none"> <li>Monthly monitoring with quarterly comprehensive reviews.</li> </ul> | <p>Locations:</p> <ul style="list-style-type: none"> <li>Local communities</li> <li>Project housing</li> <li>Healthcare centres</li> <li>Sanitation facilities</li> <li>Local businesses</li> </ul> <p>Responsibility:</p> <ul style="list-style-type: none"> <li>Community Relations Officer</li> <li>Project Manager</li> <li>Local Government Authorities</li> </ul> | <ul style="list-style-type: none"> <li>Resource usage rates (percentage of capacity used).</li> <li>Healthcare and sanitation facility occupancy rates.</li> <li>Number of reported social conflicts.</li> <li>Local employment rates.</li> <li>Economic activity metrics for local businesses (e.g., revenue changes).</li> </ul> |                                                                                                          | Minor           |
| 32 | Resource Efficiency | Excessive water consumption leading to strain on local supply; inefficient energy use resulting in higher greenhouse gas emissions and costs.                                                                                                                                                                                    | Moderate         | <p>Install water-saving fixtures.</p> <p>Reuse non-potable water for dust suppression and equipment washing, if needed.</p> <p>Ensure metered supply.</p> <p>Enforce idling controls for vehicles and machinery.</p>                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Parameters:</p> <p>Record monthly water and diesel consumption.</p>                                                                                                                                                                                                                                                                                                                                                             | <p>Locations: All construction sites, worker camps, equipment yards, and offices. Responsibility: Terminal ES Manager and Coordinator</p>                                                                                                                                                                                                                               | Water Consumption and Diesel consumption                                                                                                                                                                                                                                                                                           | Minimize water and diesel consumption                                                                    | Not significant |



## 4. Occupational Health and Safety Management<sup>1</sup>

### 4.1 OHS Overview

During the construction activities, the contractor will ensure the following:

#### 4.1.1 Personal Protective Equipment (PPE)

- Conduct regular hazard analysis to determine the correct PPE needed and require PPE use.
- Earmuffs to be provided to workers where exposure to high noise is expected.
- There must be adequate first aid kit distribution across all sub-project sites and mapping of all available trained personally on personal aid kits.
- There must be a differentiation established during the training for the workers on the PPE that is expected to be compulsory during all times during the construction of the project (safety boots, high visibility vest, hard hat) and PPE that is activity specific based on the Risk Assessment (RA) conducted for the activity (gloves, safety goggles, safety ventilation masks, safety jackets and gears etc.)

#### 4.1.2 Safety

- Creating a culture of positive reinforcement is. The occupational safety measures cannot simply be pre-planned and reactive.
- Share safety information across organizational boundaries through activities such as occasional interdivisional tailgate sessions
- Help workers know their physical limitations and emphasize the role fitness plays in safety.
- Encourage managers to support safe work practices. Doing resource management work in remote areas is dangerous. Workers need adequate safety equipment, support from nearby reserve rangers, and managers who seriously consider suggestions for improving safety.
- **Daily** toolbox talk should be given to workers in order to share any information about OHS.
- Require two-person lifts for heavy and cumbersome items.
- Ensure adequate measures in place to limit spread of infections and viruses.
- All workers will be provided health insurance.
- Sensitize workforce to HIV/AIDS prevention and treatment.
- Ensure hygienic and safe environments for workers, and adequate sanitation facilities.
- Ensure food suppliers and processing facilities meet minimum requirements prescribed by the Nigerian Authorities.

##### 4.1.2.1 Risk Assessment

Risk assessment is required for all construction and commissioning tasks undertaken and are required to be signed off. Accordingly, all high-risk activities to be carried out on site will require the Safe Work Procedures and a copy provided to the Health and Safety Officer for review and acceptance prior to any work commencing. This risk assessment will include the identification of possible hazards, possible risks and control measures.

- The contractor must prepare Risk Assessments (RA) for all activities on site. In the risk assessments done for every project site, whenever activities of high risk are identified, safe work procedures are followed.

---

<sup>1</sup> Refer to the **Occupational Health and Safety Management Plan** standalone document for more details.





- The RA is an activity that is expected to be conducted in collaboration between the department manager, site supervisors and foreman with the health and safety staff of the project, the primary review is by the occupational health and safety manager and final approval from the Project Manager.
- There needs to be sufficient pre-planning of activities so that there are no delays for approvals.
- No activity should be conducted, routine or non-routine that does not have a RA approval and is not appropriately reported on.
- The RA is not a replacement for the respective monitoring required for the health and safety of workers engaged on the project.
- The Daily Safe Task Analysis (DSTA) will involve carrying out a briefing each morning of activities, where tasks are reviewed and specific roles are assigned. Potential hazards are discussed, and strategies to mitigate these risks are developed, including the use of protective gear and regular equipment checks. Safety measures and emergency procedures are emphasized, and all details are documented with participants signing to acknowledge their understanding. Regular inspections are conducted throughout the day to ensure compliance and address any new risks, fostering a proactive and safe working environment.

A Risk Matrix is a tool used in risk management to define the level of risk by considering the probability or likelihood of harm occurring and the severity of the consequence. This matrix helps in prioritizing risks and determining the appropriate mitigation strategies for a project or operation. The matrix is constructed with types of losses on the vertical axis and levels of probability on the horizontal axis. Each cell in the matrix is a combination of these factors, which provides a risk score that falls into different risk levels: Low (F), Medium (M), High (E), and Very High (TE).



Table 4-1: Standardized Risk Matrix Table

| RISK MATRIX                                                                   |                                                                                                                                  |                                                                               |                                                                                                                  |                                                                            |                                                                                    |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Standardised risk matrix                                                      | Effect of danger/consequence - (When an event has more than one "type of loss", choose the "consequence" with the highest score) |                                                                               |                                                                                                                  |                                                                            |                                                                                    |
| Type of loss                                                                  | 1 - Insignificant                                                                                                                | 2 - Minor                                                                     | 3 - Moderate                                                                                                     | 4 - Major                                                                  | 5 - Catastrophic                                                                   |
| (S/S)<br>Personal injury<br>(health/safety)                                   | First aid kit / Exposure to a minor health risk                                                                                  | Medical treatment / Exposure to a major health risk                           | Accident with stoppage / Reversible impact on health                                                             | Single death or loss of quality of life / Irreversible impact on health    | Multiple deaths / Ultimately fatal impact on health                                |
| "(EI)<br>Environmental impact                                                 | Minimal environmental damage - incident L1                                                                                       | Material damage to the environment - L2 incident repairable in the short term | Serious environmental damage - L2 incident repairable within the duration of the project                         | Major environmental damage - L2 incident repairable after project duration | Extreme environmental damage - irreversible L3 incident                            |
| "(BI/MD) Business Interruption/Property Damage and Other Consequential Losses | No interruption to operation                                                                                                     | Brief interruption of operations                                              | Partial shutdown                                                                                                 | Partial loss of business                                                   | Significant or total loss of business                                              |
| "(I/ R)<br>Legal and regulatory                                               | Low-level legal problem                                                                                                          | Minor legal problem; non-compliance and breaches of the law                   | Serious breach of the law; investigation/reporting to authorities, prosecution and/or moderate sanction possible | Major infringement of the law; prosecution and considerable penalties      | Considerable penalties and prosecutions. Several prosecutions and prison sentences |
| "(R/S/C)<br>Impact on reputation / Social / Community                         | Slight impact: the public may be aware, but they're not worried.                                                                 | Limited impact - local public concern                                         | Considerable impact - regional public concern                                                                    | National impact - national public concern                                  | International impact - international public attention                              |
| PROBABILITY                                                                   | RISK ASSESSMENT                                                                                                                  |                                                                               |                                                                                                                  |                                                                            |                                                                                    |
| 5 - Almost certain                                                            | 11 (M)                                                                                                                           | 16 (E)                                                                        | 20 (E)                                                                                                           | 23 (TE)                                                                    | 25 (TE)                                                                            |
| 4 - Probable                                                                  | 7 (M)                                                                                                                            | 12 (M)                                                                        | 17 (E)                                                                                                           | 21 (TE)                                                                    | 24 (TE)                                                                            |
| 3 - Possible                                                                  | 4 (F)                                                                                                                            | 8 (M)                                                                         | 13 (E)                                                                                                           | 18 (E)                                                                     | 22 (TE)                                                                            |
| 2 - Unlikely                                                                  | 2 (F)                                                                                                                            | 5 (F)                                                                         | 9 (M)                                                                                                            | 14 (E)                                                                     | 19 (E)                                                                             |
| 1 - Rare                                                                      | 1 (F)                                                                                                                            | 3 (F)                                                                         | 6 (M)                                                                                                            | 10 (M)                                                                     | 15 (E)                                                                             |
| INTERPRETATION OF RISK LEVEL                                                  |                                                                                                                                  |                                                                               |                                                                                                                  |                                                                            |                                                                                    |
| RISK ASSESSMENT                                                               | RISK LEVEL                                                                                                                       |                                                                               | RISK MATRIX GUIDELINES                                                                                           |                                                                            |                                                                                    |
| 21 to 25                                                                      | (TE) - Very High                                                                                                                 |                                                                               | Eliminate, avoid, implement specific action plans/procedures to manage and monitor                               |                                                                            |                                                                                    |
| 13 to 20                                                                      | (E) - High                                                                                                                       |                                                                               | Proactive management                                                                                             |                                                                            |                                                                                    |
| 6 to 12                                                                       | (M) - Medium                                                                                                                     |                                                                               | Active management                                                                                                |                                                                            |                                                                                    |
| 1 to 5                                                                        | (F) - Low                                                                                                                        |                                                                               | Monitor and manage where necessary                                                                               |                                                                            |                                                                                    |



### How to Use the Risk Matrix

1. **Identify Potential Risks:** Evaluate each activity, process, or situation and determine the potential types of losses it could cause, such as personal injury, environmental impact, or property damage.
2. **Evaluate the Consequence:** For each identified risk, assess the consequence if the risk were to occur. Use the consequence scale ranging from 1 (Insignificant) to 5 (Catastrophic).
3. **Determine the Probability:** Estimate how likely it is that the identified risk will occur, ranging from 1 (Rare) to 5 (Almost Certain).
4. **Calculate the Risk Score:** Find the intersection of the consequence and probability on the risk matrix to determine the risk score.
5. **Assign a Risk Level:** Use the risk score to categorize the risk as Low (F), Medium (M), High (E), or Very High (TE).
6. **Implement Risk Management Actions:**
  - **Very High (TE):** Immediate actions are required to eliminate or control the risk. This could involve redesigning the process or implementing strict safety protocols.
  - **High (E):** Proactive management is needed. This includes detailed planning, regular monitoring, and mitigation strategies.
  - **Medium (M):** Active management is needed. Conduct regular monitoring and ensure risk controls are in place.
  - **Low (F):** Monitor and manage the risk as necessary. No immediate action is required, but it should be kept under observation.
7. **Monitor and Review:** Continually monitor the risks and the effectiveness of the implemented controls, and adjust as necessary to ensure safety and compliance.

#### 4.1.2.2 Training for OHS

- Safe lifting training
- Educate workers about tick-borne diseases and methods to prevent exposure to ticks
- Train workers on the benefits, limitations, use, and care of PPE
- Train all workers on first aid kit usage (with clear designation of specific wards who will be the first aid focal points for the project during construction)
- Train all workers on virus infection management (COVID 19)
- Training on risk avoidance, and managing safety during times of emergency
- Training on the Emergency Response plan and the Emergency response protocol for different scenarios

#### 4.1.2.3 Daily Safe Inspection Analysis or Daily Safe Task Analysis (DSTA)

- The Daily Safe Inspection Analysis (DSIA) is a procedure where a form is prepared by the site supervisor or manager daily before the commencement of activities with the participation of all workers on site.
- The primary goal is to identify potential risks associated with each task planned for the day and outline control measures for each identified risk. H&S Officer is responsible for collecting these forms and considering their findings in their weekly observation logs, which are then submitted to the E&S Manager.



- The DSIA also serves as a tool to identify health and safety hazards before starting work on a site or prior to any new activity that presents H&S risks, ensuring that exposure to hazards is mitigated.

Table 4-2: Daily Safe Inspection Analysis Template

| Task/ Activity | Potential Hazard | Control Measures/ Precautions |
|----------------|------------------|-------------------------------|
|                |                  |                               |
|                |                  |                               |
|                |                  |                               |
|                |                  |                               |

## 4.2 Management of Work

### 4.2.1 Targets and Performance Monitoring of the project H&S

The project will monitor and measure the H&S performance of its construction activities on a regular basis to assess its compliance with legal requirements, its policy commitments and conformance to the requirements of the CESMP. This shall be conducted through the establishment of routine monitoring activities, site inspections, internal and external audits and a process to implement corrective action to address non-conformances. The records of the results of periodic evaluations shall be retained.

The requirements for measuring and monitoring key performance indicators (KPIs) associated with specific H&S topics are documented in each management plan.

Monitoring requirements will take into account monitoring required by law, including the limits and thresholds prescribed by these requirements. In the case where limits required by law differ from the limits defined in acceptable or best practice standards (e.g., IFC and World Bank Standards), the stricter of the two will be applied.

Monitoring may either be conducted using internal resources or, if specialist expertise is required, by appointed external qualified service providers. Monitoring results shall be reviewed, analysed, and interpreted by the project management in order to determine the appropriate course of action.

Performance-based monitoring results will be communicated during regular project meetings, and decisions made accordingly. Non-conformances noted in monitoring activities will be resolved through the corrective and preventive action process discussed in the non-conformity and corrective actions section below.

#### 4.2.1.1 Targets and Metrics

The OHS key performance indicators are listed in the table below.

Table 4-3: OHS KPIs

| KPI         | Description                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Near Misses | Instances where an incident could have resulted in an accident or injury but was avoided, providing an opportunity for proactive safety measures. |

| KPI                                                   | Description                                                                                                                                                    |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Accidents with Lost Work Time</b>                  | Accidents that result in an employee being unable to perform their regular duties for a period of time, impacting productivity and operations.                 |
| <b>Accidents Without Lost Work Time</b>               | Accidents that occur but do not result in any lost work time, often involving minor injuries or incidents that are quickly resolved.                           |
| <b>Occupational Illness Cases with Lost Work Time</b> | Cases where employees suffer from work-related illnesses that lead to absence from work, affecting both health and productivity.                               |
| <b>Days Lost</b>                                      | The total number of days employees are unable to work due to accidents or occupational illnesses, reflecting the severity of incidents.                        |
| <b>Days Lost Due to Work Stoppages</b>                | Days lost because of work stoppages caused by accidents, safety concerns, or other emergency situations, impacting project timelines.                          |
| <b>First Aid Cases</b>                                | Instances where first aid is administered for minor injuries or illnesses on-site, preventing further complications and ensuring quick recovery.               |
| <b>Medical Consultations</b>                          | The number of consultations provided to workers by medical professionals for any health-related issues, whether work-related or not.                           |
| <b>Covid 19 Cases</b>                                 | Reported cases of Covid-19 among employees, impacting both health and operations due to quarantines, medical care, and preventive measures.                    |
| <b>Fatality</b>                                       | An incident that results in the death of an employee or visitor, representing the most severe type of workplace accident and requiring thorough investigation. |
| <b>Property Damage Cases</b>                          | Incidents where company property or equipment is damaged due to accidents or unsafe practices, potentially affecting operations and costs.                     |
| <b>Vehicle Incidents</b>                              | Accidents involving company vehicles or equipment, including collisions or breakdowns, potentially impacting project efficiency and safety.                    |
| <b>Accident Frequency Rate</b>                        | A metric that calculates the frequency of accidents per a standard number of working hours, helping to assess the effectiveness of safety measures.            |
| <b>Emergency Drills</b>                               | Scheduled drills conducted to prepare workers for potential emergencies, ensuring readiness and improving response times during real incidents.                |

This KPIs shall be collected by the H&S Officer **every month**. Each project area must appoint a EHS representative to provide this data to the project H&S Officer.

#### 4.2.1.2 Routine Monitoring and OHS Inspection

Every project area in coordination of the site supervisors is responsible for the daily monitoring of H&S performance at sites during project activities and will plan and perform regular site Safety Walks & Talks and works inspections.



As part of H&S monitoring, the H&S Officer or officer will conduct weekly inspections of all areas of Project sites and processes to identify H&S issues (e.g., leaks, spills, waste storage, hazardous substance storage, community issues, fire equipment, lifting equipment, PPE use, and other general housekeeping).

The intended result of the H&S inspection process is to ensure compliance and awareness of legal requirements as well as internal the project requirements on health, safety and environmental issues.

These inspections are not a substitute for safety or environmental inspections of facilities, installations and/or equipment established by competent authorities, regulators and/or manufacturers.

For non-conformities with high and imminent potential risk, the Health and Safety Officer will request to immediately bring the activity to a standstill until the necessary measures to guarantee proper safety conditions have been taken and shall inform the E&S Team

#### *4.2.1.3 Non-conformities and Corrective Actions*

All Project personnel are responsible for reporting H&S Incidents / Non-Conformities (NCs) immediately upon discovery to the appropriate emergency contact according to the applicable procedure detailed in the Emergency Preparedness and Response Plan and Occupational Health and Safety (OHS) Plan.

If the Project Manager or Health and Safety Manager is not available, H&S Incidents/NCs should be directed to the project Health and Safety Officer. The project Health and Safety Officer will maintain a log of all reported H&S Incidents/NCs and track the investigation and correction (as needed) for all reported Incidents/NCs within the project. The Health and Safety Manager or the project Manager must sign off on the resolution of the Incident/NC. Incidents/NCs that have not been signed off on must be discussed in site meetings and reported in **monthly** reports.

Communication of the results of the investigation/ corrections for reported hazards to Project personnel is the responsibility of the appropriate the project Area Manager.

General steps to manage non-conformities and determine corrective actions are:

- 1) Define the problem and its scope;
- 2) Contain the problem;
- 3) Identify the root cause;
- 4) Plan the corrective action;
- 5) Implement the plan;
- 6) Ensure the corrective actions work (i.e., re-inspect).

#### *4.2.1.4 H&S Performance Meetings*

The project's O&M and H&S representatives perform periodical (**weekly** recommended) performance review meetings.

#### *4.2.2 Communication and Consultation with the Workforce*

The project believes that involving the workforce in identifying and controlling risks is crucial to reducing accidents and improving H&S performance. The workforce has first-hand experience of site conditions and is often the first to identify potential problems that must communicate.



To ensure involvement of the entire workforce, and although the international language for the project is English, where needed the project may need to make special arrangements for workers who have little or no understanding beyond the local language. It is the responsibility of the H&S Manager to ensure that work documentation, training materials, and safety signage is available in languages generally understood by the workforce.

The following sub-sections set out the principal means of communicating, to and acquiring worker engagement, in order that risks on site can be managed in the most effective way.

- 1) Site safety signage
- 2) Bulletin boards
- 3) Daily pre-task briefings
- 4) Safety talks (toolbox talks / tailboard meetings)
- 5) Safety campaigns
- 6) Handling design changes during the project

#### *4.2.2.1 Site safety signage*

The project shall evaluate the site and assess the requirements for safety signage to meet the minimum standards set out below:

- Signage to meet local legislation
- Signs to identify the type of PPE required
- Signs to identify any site-specific hazards or requirements
- Location of first aid provision and first aiders
- Emergency master points
- Site office

The requirements for site signage shall be identified as the result of the site evaluation and mobilization process. All site signage must be in English and meet local standards.

#### *4.2.2.2 Bulletin boards*

Bulletin boards shall be strategically located, generally in the welfare facilities and/or main offices. They shall be monitored and updated by the H&S Manager to ensure the information is current and correct. As a minimum, they shall contain:

- All project parties' H&S Policies
- Site Rules
- Copy of the any formal notification required by local legislation
- Copies of any form posters mandated by local legislation
- Emergency Response Plan / Procedures: Evacuation arrangements and key contact information
- First Aid Arrangements
- Performance Statistics and incident information
- Safety Committee minutes of meeting
- Updated information relating to reported incidents
- Applicable Safety Alerts and Information
- Insurance Certificate

#### *4.2.2.3 Daily pre-task briefings/ DSIA/ DSTA*

Pre-task briefings or DSTA or DSIA shall be undertaken daily before the start of every shift rotation, by the relevant line management representative supported, when necessary, by a representative from the H&S team.

The pre-task briefing should inform technicians of their planned daily tasks, ensuring correct documentation is provided, manuals are discussed and understood, shift handover information is provided, and ensuring checks of equipment and tools, including PPE and radio's. It should also allow the opportunity to ask questions.

#### *4.2.2.4 Safety talks (toolbox talks / tailboard meetings)*

All the project employees and subcontractors shall receive Safety Talks appropriate to the work activity be performed. Such talks are undertaken on a proactive basis to highlight hazards related to the work activity along with the general precautions and requirements to prevent incident, injury or ill health. These will include details of any relevant Safety Alert(s) or Safety Awareness(s) in place and reference operational restrictions relevant to site. The site will maintain a record of safety talks performed at site.

#### *4.2.2.5 Safety campaigns*

The project will initiate periodic proactive 'H&S Awareness' program(s) and initiative(s), e.g., safety campaigns. Subcontractors are expected to participate fully in these activities, as well as develop and implement their own as necessary, as per the guidelines included in their own health and safety management systems or H&S Plans.

#### *4.2.2.6 Handling design changes during the project*

The project recognizes that unforeseeable changes can occur during the lifecycle of the project, and that hazards can arise as a result. If changes occur, it is important that emerging risks are assessed, and any control measures are effective.

#### *4.2.2.7 Contractor Management*

Contractors will be required to have arrangements for assessing the competence and resources of individuals or companies under their governance, including assessing and assuring their H&S performance. They must require anyone employed or contracted by them, to be aware of, agree to and comply with this Plan.

Contractors are responsible for the provision and implementation of appropriate arrangements for ensuring the health and safety of all those affected by their work. Competent personnel shall undertake works and deliver supervision and management necessary for the execution of work within their scope of responsibility. Contractors are responsible for ensuring that individuals or contractors under their governance are provided with adequate training in the use of any equipment related to their work. Contractors must maintain competence and training records and must ensure that only competent personnel are authorized to use such equipment.

Contractors must familiarize themselves with any site-specific hazards and any precautions required in respect thereof. A contractor must ensure that such risks are made known to all individuals under their governance.

The contractor is responsible for reporting accidents, dangerous situations and near miss incidents to the H&S Manager.

#### 4.2.2.8 Pre-Site activity

Prior to the commencement of any contracted work on site the following arrangements must as a minimum be addressed to ensure the Contractor is fully prepared in complying with the H&S requirements:

- During the tender or procurement phase, the subcontractor will be issued with H&S requirements for subcontractors, copy of the site H&S Plan (or a plan template if the site plan is not available yet) and any other H&S documentation deemed relevant to the subcontractor's work package.
- For a site's regular maintenance contractor, a pre-start meeting will be held with the contractor. This meeting shall explain in detail the H&S requirements for the site.
- Subcontractors must ensure that all required training is identified and that qualifications are in place and documented prior to mobilization.
- Applicable subcontractor safety documentation related to the scope will be obtained by the project and reviewed to ensure compliance to the project H&S requirements prior to acceptance on site.
- This Plan must serve as the basis for the subcontractor safety planning.

Before moving construction activities to a new berth, ITB-Hitech will obtain PSSCC validation confirming that emergency access, navigation safety, and port-wide traffic arrangements remain adequate. This verification will be documented and attached to the monthly CESMP updates.

#### 4.2.2.9 During site operation

It will be ensured that adequate measures are implemented to monitor the H&S performance of each subcontractor during the site activity.

As a minimum, this will require the following to be undertaken by H&S Manager:

- Ensure the provision of induction training for all subcontractors' personnel on site.
- Ensure that operational and H&S meetings are set up.
- Provide feedback to subcontractor or individuals where appropriate.

#### 4.2.2.10 De-mobilization

It will be ensured that adequate measures are implemented to monitor the H&S performance of each contractor and their subcontractors during the de-mobilization phase. As a minimum, this will require the following issues to be undertaken by the subcontractor:

- Remove properly all their waste materials, chemicals, and equipment from site.
- Ensure the lay down area provided for the subcontractor use is restored to its previous condition.
- Provide to the project any required documentation (i.e., records, certification, manuals etc.).
- Participate, under request, in a closeout meeting with the H&S Manager.

#### 4.2.2.11 Exchange of H&S information between the project and contractors

The project will use the same communication and consultation mechanisms used for its own workforce for handling information between the project and contractors.

For a site's regular maintenance contractor, the H&S Manager should have regular meetings with contractor representatives to discuss contractor H&S performance and exchange ideas in terms of improvement

opportunities. Specific details of the processes in place for exchange of information between the project and contractors relating to the projects are set out below:

- **Monthly** meetings to be held with the contractor to discuss overall performance of the H&S aspects.
- **Weekly** meeting to be carried out in site for monitoring purposes.

#### 4.2.2.12 *Liaisons with Government Authorities and Emergency Services*

All enquiries from government authorities shall be referred to the Project Manager, unless another person has been named as the point of contact.

All correspondence received from the enforcement authorities must be copied to the Project Manager for escalation accordingly to any local the project legal entity protocol. In certain circumstances, it may be relevant to provide a copy of the correspondence to all involved parties.

- Any proposed reply and communication with the enforcement authorities must be approved by the Project Manager and by the relevant person in the project legal entity.
- Any correspondence, including e-mails, received from the inspector must be forwarded to the Project Manager.
- Any letters received from the regulator should be clearly marked with the date of receipt and forwarded to the relevant Project Manager immediately.
- Any response to the Regulator should be discussed and agreed with the relevant Operation Management or project manager i.e., the project region and the project legal country entity.

#### 4.2.2.13 *Planned regulatory visits*

Government authorities must always receive the full cooperation from site personnel to allow the authorities to do their job. If government authorities hand down a restriction or require improvement, they shall be reported up the chain of command for its immediate implementation. Interference or obstruction of a government official could lead to personal legal liability in many countries.

Government authorities must never be permitted to inspect the site alone and must always be accompanied by a designated person who knows the site and meets the necessary training requirements.

On arrival at the site, authorities must always sign in at the site office if this practice is established and they will receive a safety induction. They will be provided with appropriate PPE, if they have not brought their own equipment.

During any such visits the Site Supervisor and/or H&S Officer must take notes of issues seen and statements made by the enforcement authority inspector, including details of related personnel if needed to allow subsequent further investigation of the circumstances leading to any observed event.

Prior to the visit itself, try to establish the following information:

- The purpose of the visit.
- Which Inspector(s) will be undertaking the visit
- Anything or anyone that the Inspector(s) specifically wish to see or meet.

Unplanned regulatory visits

- At the earliest opportunity, the most senior member of the site must inform the Project Manager.

- If any unsafe behaviours are identified during the visit, they must be dealt with there and then relative to the risk (i.e., if there is immediate danger apply an immediate stop work).
- At the end of the visit, hold a de-brief with the Inspector and discuss
  - any concerns
  - any actions to be followed up
  - anything positive they observed
  - any next steps proposed by the Inspector.
- If any significant issues are raised as part of a regulator visit, whoever escorted the Inspector must immediately inform by phone the relevant Project Manager – followed by an e-mail.

#### 4.2.3 *Financial Integrity and Ethical Community Engagement*

ITB Nigeria Limited maintains strict adherence to its Anti-Money Laundering and Counter-Terrorist Financing (AML/CTF) Policy (ITB-GOV-POL-002) and Anti-Bribery and Anti-Corruption (ABC) Policy (ITB-GOV-POL-001) in all community engagement activities. All community development projects, social investments, and stakeholder engagement initiatives will undergo thorough due diligence to ensure funds are used legitimately and transparently. Project managers must verify the identity and legitimacy of community representatives receiving funds or resources, with particular attention to beneficial ownership structures of community organizations. Any suspicious patterns such as unusual payment requests, redirection of funds to third parties, or complex financial arrangements inconsistent with project objectives must be reported immediately through the whistleblower channel ([whistleblower@itb.com](mailto:whistleblower@itb.com)). The ESMP implementation team will conduct quarterly reviews of all community-facing financial transactions to ensure compliance with these policies while maintaining detailed records for audit purposes.



## 5. Waste Management<sup>2</sup>

### 5.1 Waste Management Procedures

#### 5.1.1 Methodological Approach

The fundamental methodology by which the WMP is based on the standard traditional waste hierarchy which follows a logical flow to prevent the generation of solid waste and if unavoidable what are the most preferred methods for handling the solid waste generated. There is no scenario where the construction of this kind of infrastructure can follow more zero-waste styled hierarchy, and thus the adherence to the basis thought steps of the hierarchy can reduce the volume of waste that ends up in the landfill or dumpsites.

### The Traditional Waste Hierarchy Diagram



Figure 5-1: The traditional waste hierarchy diagram

To simplify the content of the figure above, consider the following:

- **Prevention**
  - Minimize waste generation at the source.
- **Reduce**
  - Decrease the amount of waste produced.
- **Reuse**
  - Use items more than once before discarding.
- **Recycle**
  - Process materials to make new products.
- **Recovery**
  - Recover a segment of the material through different means
- **Disposal**
  - Safely dispose of waste that cannot be managed otherwise

<sup>2</sup> Refer to the **Waste Management Plan** and **Wastewater Management Plan** standalone documents for more details.



### 5.1.2 Storage and Containment of Waste

After the generation of waste, the following step is the storage of waste until collection. Movement of waste onsite is to be kept to a minimum. All the following instructions must be followed when storing waste prior to collection from disposal trucks:

- Warning signals must be installed near waste storage and containment areas (where collection will also happen). This includes but is not limited to:
  - “No unauthorized access” – near waste storage and containment areas
  - “Sharp Objects” – sign near waste storage bins and containers
  - “Hand Gloves must be worn” – basic OHS sign across the project sites
  - “Hazardous Substances and/or waste” – near all hazardous material storage areas
- Waste must be segregated into the appropriate containers and skips highlighted below and stored appropriately.
- Transportation and movement of waste onsite needs to be kept to an absolute minimum.
- Transportation of waste on-site from larger generation/containment areas to waste storage areas is conducted by the contractor and not the waste subcontractor.
- Workers responsible for general site housekeeping and cleanliness must be appropriately trained on handling all types of waste, especially hazardous and medical waste as they will need to know how to interact with this material.
- Housekeeping workers for construction site, are not to be responsible or involved in the movement and transport of waste from one location to another.
- Any spills that occur during the collection or storage process must adhere to the spill management procedures highlighted in the emergency responsiveness section and utilized the spill kit adequately.
- All waste that is stored onsite must be logged through Appendix D. Maintaining the waste inventory up to date is a key performance indicator for waste management.
- The inspection of the storage areas and containment areas is to be conducted by the Environmental Officer.

#### 5.1.2.1 Storage Containers

The following provides a breakdown of the proposed storage containers for the waste generated from the Project:

- **Color-coded Bins**
  - These are utilized for smaller storage of material and not large volumes or quantities.
  - Green bins would be provided for recyclable material.
    - Size = **240 Litres**
  - Blue Bins would be provided for waste that is non-recyclable and mixed by nature, that could include construction waste but that would not include steel or cement.
    - Size = **240 Litres**
  - Yellow bins, accompanied by clear warning signs, would be provided for hazardous waste including all types and items mentioned previously such as paints, solvents, chemicals etc.
    - Size = **120 Litres**
  - Yellow Drums, would be utilized for the storage of liquid hazardous waste until disposal and should have clear colour coding.



- Red bins would be provided for all medical waste onsite and would be near the onsite clinic unit.
  - Size = **60 Litres**
- **Skips and Containers**
  - These are utilized for adequate storage of larger bulkier items and construction waste material and items. There are different types that come in different shapes and dimensions depending on the need and space availability, in addition to the frequency of collection and disposal of waste.
  - Open-top skips will be used for the majority of the construction and demolition waste, with segregation for the typology.
    - **10 cubic meters** would be the primary size for skips across the construction sites.
  - Covered Skips would be provided in smaller sizes and focused on more valuable waste material generated from the construction activity that require segregation from the open-top skip or require protection from water/weather to not be affected or further contaminated.
    - **6 cubic meters** would be provided for the protected skips

The storage area of waste should be in designated areas that are isolated from surface drainage systems. For hazardous waste containers and storage, they must have impermeable base, banded and secured containers until removal by the contractor. These containers must have a bund of a capacity of at least 110% the volume of the container.

The storage and containment of hazardous and chemical waste must take into consideration the following:

- Containers for accumulating hazardous waste must be labelled with the words “Hazardous Waste” and there must be available a list of the contents of the container (and the approximate concentration or amount of the hazardous substance), either on the label or on a sign adjacent to the container.
- Complete names must be used on this list, not chemical formulae or initials.
- This list must accompany the waste when it is taken to the hazardous waste storage area. All other labels on the containers must be removed or obliterated. If the identity of a waste is not known, the team producing that waste will bear the cost, if any, of having the waste analysed.
- Empty containers (including centrifuge tubes) that previously held hazardous substances are legally not hazardous waste once the contents have been removed by ordinary means. Emptying and bulking the contents of the centrifuge tubes is the responsibility of the lab that generated them. They should not be included with the actual hazardous waste. Although empty containers are not hazardous waste, they should be cleaned as thoroughly as possible before depositing in the “Co-mingle” recycling bins. Containers may be reused for waste as long as there are no incompatibility issues.
- Unwanted paint and paint related materials can be (if available) be sent to a recycling centre where possible or otherwise it is considered hazardous waste and disposed as such.
- Material safety data sheets (MSDS), material inventory, and emergency contact numbers will be maintained and displayed in storage areas and safety board.
- While not strictly hazardous waste, broken glass and sharp implements must be discarded in appropriate containers that have warning labels and not placed in general trash where they would pose a danger to housekeeping personnel.

#### 5.1.2.2 Segregation of Waste

##### Construction and Demolition Waste:



- **Concrete, Bricks, and Inert Materials:** Segregate these materials in designated open-top skips. They can be crushed and reused as fill or road base material.
- **Metals:** Separate ferrous and non-ferrous metals and store them in dedicated green bins or skips for recycling. This will be coordinated by the subcontractor and can be collected by scrap merchants only if they are trained.
- **Wood and Timber:** Collect wood waste in a specific skip for recycling or repurposing.
- **Plastics:** Segregate clean and uncontaminated plastic materials in green bins.

#### Recyclable Materials:

- **Paper and Cardboard:** Place in green bins to ensure they remain dry and clean for recycling.
- **Glass:** Use separate bins within the green category, ensuring they are not mixed with other recyclables.

#### Hazardous Waste:

- **Chemical Waste (Paints, Solvents, Adhesives):** Store in labelled, sealed yellow containers or drums. Ensure that the storage area is ventilated and protected from sunlight.
- **Used Oil and Lubricants:** Store in dedicated, labelled yellow drums with secondary containment to prevent leaks.
- **Asbestos and Lead-Based Materials:** Use double-bagging with heavy-duty, labelled yellow bags, and store in covered skips. Follow strict safety protocols for handling and disposal.
- **Batteries:** Place in specific yellow bins designed for hazardous electronic waste, ensuring terminals are covered to prevent short circuits.

#### Municipal Solid Waste:

- **Organic Waste:** If a composting facility is available, separate organic waste such as food scraps and place them in designated compost bins.
- **General Waste:** Non-recyclable and non-hazardous materials should be placed in blue bins.

#### Medical Waste:

- **Sharps (Needles, Syringes):** Use puncture-resistant red sharps containers.
- **General Medical Waste:** Store in designated red bins or bags that are leak-proof and clearly labelled.

#### Additional Considerations

1. **Labelling and Signage:**
  - All bins, skips, and containers should be clearly labelled with the type of waste they are intended for. Signs should include relevant safety warnings, especially for hazardous materials.
2. **Accessibility and Location:**
  - Waste collection points should be easily accessible to workers but located away from sensitive areas, such as water sources, to prevent contamination. Hazardous waste storage should be in a secure, restricted area with appropriate safety measures.
3. **Containment and Protection:**
  - Use covered skips or containers for waste that may generate dust, odours, or leachate. Implement secondary containment for hazardous liquid waste to prevent spills and leaks.



#### 4. Compliance and Safety Protocols:

- Ensure that all waste handling complies with local regulations and international best practices. Provide personal protective equipment (PPE) and training for workers handling hazardous materials.

##### 5.1.2.3 Preventing leakage

#### Use of Tarpaulins:

- Cover open-top skips and bins with heavy-duty tarpaulins to prevent waste materials from being blown away by the wind.



Figure 5-2: Skip covered with tarpaulin

- This should be based on determine the frequent wind direction.
- Secure the tarpaulins tightly with ropes or bungee cords to prevent them from becoming loose or torn.

#### Windbreaks and Shields:

- Install windbreaks or barriers, such as temporary fencing or hoarding, around waste storage areas to reduce wind velocity and prevent the scattering of lightweight materials.
- Use portable screens or mesh panels as shields to protect smaller waste storage units from direct wind exposure.

#### Covered Storage Solutions:

- Utilize covered skips and containers for storing lightweight or fine particulate waste, such as dust or small debris, to prevent dispersion by wind.
- Store volatile or odorous materials in covered or enclosed containers to minimize air contamination and potential odour issues.

#### Strategic Placement of Waste Bins:

- Place waste bins and skips in locations sheltered from prevailing winds, such as behind buildings or natural windbreaks, to reduce wind exposure.
- Avoid placing waste storage units in open, elevated, or unprotected areas where wind impact is greater.

#### Weighting Down Lightweight Waste:

- Use weights or sandbags to secure tarpaulins and other covers, especially in windy conditions, to prevent them from being lifted.

- Place heavier materials on top of lighter waste materials to hold them down and prevent scattering.

#### Waste Compaction:

- Compact waste materials in bins and skips to reduce their volume and make them less likely to be affected by wind.

In addition to that Non-hazardous waste may be discarded in a sink drain. However, this practice should only be done with **small quantities of materials**. Final determination of whether a material is hazardous waste will be made by the hazardous waste coordinator. Waste that can be poured down sink drain include:

- **Non-hazardous:** No radioactive waste, No hazardous chemical waste, No untreated biohazardous waste
- **Liquid:** No solids, sludges, or viscous substances
- **Will not interfere with sewage treatment operations:** No corrosive pH levels, No grease or oil, No hot (150°F or higher) temperatures in volumes of more than 10 gallons

#### 5.1.2.4 Unknown Material

This covers all materials that are not properly identified, such as an unlabelled barrel or other unlabelled container. Containers should be examined, opened and sampled by a qualified individual.

- Open metal containers only with non-sparking tools, in case contents are flammable. Don't smell unknown chemicals, and take precautions to prevent contact with eyes and skin by using safety goggles, face shield, gloves, protective disposable clothes and chemical resistant boots.
- If the container is in poor condition, or damaged in any way, seek expert advice before moving or touching the container. A barrel with the top or sides pushed out, for example, may indicate the contents are under pressure, and opening it could be dangerous.
- If the container is not in good condition, make plans to put the original container into another larger (over-pack) container with the approval of the Environmental Officer.

Record information about the material, such as:

- A description of the container and where it is located;
- Former use of the material and who may have further information.

Assign a number and place a label on the container. The word "caution" must be included on the label, and the name of the person that will be contacted if the container needs to be moved. For example, the first unknown material would be labelled: *"Unknown Material #1, CAUTION: Contact Eng Claude Prior to Moving"*. Information learned about the unknown material can be added to the label.

#### 5.1.2.5 Collection of Waste

A basic collection schedule should be agreed between the contractor and the waste subcontractor on a monthly basis, and any changes should be informed prior by **at least 3 days** unless in the case of emergency or unpredictable events. A basic register can be adopted as follows:

| Waste Collection Register | Month  |        |        |        |
|---------------------------|--------|--------|--------|--------|
|                           | Week 1 | Week 2 | Week 3 | Week 4 |



|                                             | Date | Time | Date | Time | Date | Time | Date | Time |
|---------------------------------------------|------|------|------|------|------|------|------|------|
| <b>Non-Hazardous Waste</b>                  |      |      |      |      |      |      |      |      |
| Bulky Waste                                 |      |      |      |      |      |      |      |      |
| Concrete, Cement Waste                      |      |      |      |      |      |      |      |      |
| Scrap Metal                                 |      |      |      |      |      |      |      |      |
| Plastics and Polythene                      |      |      |      |      |      |      |      |      |
| Non-Hazardous chemicals, liquids and solids |      |      |      |      |      |      |      |      |
| <b>Hazardous Waste</b>                      |      |      |      |      |      |      |      |      |
| Medical Waste                               |      |      |      |      |      |      |      |      |
| Oils and Liquids                            |      |      |      |      |      |      |      |      |
| Tar                                         |      |      |      |      |      |      |      |      |

Furthermore, please note the following instructions for the collection of waste from the construction yard:

- During waste collection, there must be a worker from the E&S team inspecting and supporting.
- Collection of waste must be done in the designated area only.
- Waste tires will be collected separately from other waste streams and will be turned into value via licensed transporters in accordance with relevant regulations. They are to be stored separately on-site where applicable.

#### 5.1.2.6 Truck Drivers

- All truck drivers that will be engaged in waste disposal activities for the project must be adequately **trained, pre-assessed and have passed the appropriate checks.**
- Every truck driver will always maintain an accessible mobile phone with them.
- Truck drivers will ensure that they are:
  - Following the route plan as per the transport management plan for the disposal of solid waste.
  - They must provide the disposal receipt during their next collection trip.
- No truck drivers responsible for the transport of hazardous waste would be engaged with non-hazardous waste and vice versa.
- During collection, truck drivers cannot be alone and must be working with a pair.
- There must be a clear focal point for the coordination for the transport of waste material, that can connect the E&S team with the contractor's drivers.
- At the same time, there must be available channels of communication directly with the drivers if needed.

#### 5.1.2.7 Trucks

- All trucks that will be involved in the collection and disposal of waste must ensure:
  - They have appropriate means for covering their load
  - They have been screened and vehicle number provided to the E&S team





- They are approved in terms of load and capable of navigating the proposed route
- No hazardous and non-hazardous waste can be mixed during disposal by any truck
- Vehicles are not to be overloaded with waste during collection or disposal
- **No waste collection and disposal maintenance for the vehicles should be conducted on-site or on the road** - it must be in designated offsite areas
- The trucks must be appropriately maintained and inspected, even if they are a third-party subcontractor
- Trucks must be licensed and have the appropriate measures to minimize impact from windblown debris

#### 5.1.2.8 Disposal of waste

Waste shall be disposed in accordance with local legislation and IFC EHS General Guidelines. All waste should be appropriately handled, stored and disposed to avoid pollution. The subcontractor will be responsible for this procedure under the supervision of the Environmental Manager by filling the monitoring records everything waste is disposed. Furthermore, they will be providing the disposal receipts as appropriate. The following measures will apply to the disposal of waste:

- Wastes will be transported and disposed to licensed waste management sites or facilities. A due diligence will be undertaken to understand if the sites are materially compliant with project requirements (either by the contractor or a third-party subcontractor)
- All waste transfer notes and hazardous waste consignment notes where the waste is being taken will be recorded.
- Properly and improperly disposed material during earthwork should be placed in the appropriate waste disposal areas straight away – stored separately and in an appropriate manner away from solid waste
- Materials destined for disposal in the landfill include refuse, materials contaminated with waste, or which have been ruined, and materials for which markets do not exist.
- Any waste that can be recycled or reused must be collected and sent for recycling. To identify recycling opportunities, investigation of external markets for recycling by industrial processing operations located in the region. This would allow to reduce the waste to be disposed.
- Treatment and disposal facilities across the road will be designated to receive the waste (composting operators, landfill or incinerators depending on the waste type). The Environmental Manager and the waste disposal company will discuss the options and agree.
- **No waste should be disposed in any location that is not a licensed waste disposal facility and a location that was agreed with the local authority + E&S team.**

## 6. Emergency Responsiveness<sup>3</sup>

This section outlines the procedures to address various emergency scenarios that may occur during the construction phase. The goal is to safeguard workers, local communities, and the environment while ensuring quick and effective responses to emergencies.

### 6.1 Emergency Scenario Identification and Procedures

#### 6.1.1 Falls from Heights:

- **Prevention Measures:** Mandatory use of PPE, such as safety harnesses, lifelines, and guardrails. Regular safety inspections of scaffolding and platforms.
- **Response:** If a fall occurs:
  1. The nearest worker must alert the ERT and apply first aid if trained.
  2. The injured worker must be stabilized using first aid, avoiding unnecessary movement if there are suspected fractures.
  3. Immediate evacuation to the on-site clinic will be carried out, and for serious injuries, transport to the nearest hospital will be initiated.
  4. An investigation into the cause of the fall will follow, with corrective measures implemented.

#### 6.1.2 Fire Outbreaks:

- **Prevention Measures:** Fire extinguishers will be placed throughout the construction site and regularly inspected. Workers will be trained in fire prevention and emergency evacuation.
- **Response:**
  1. In the event of a fire, the ERT will be immediately notified, and nearby workers will use fire extinguishers to control the fire if it is safe to do so.
  2. All personnel will evacuate according to the designated routes.
  3. The ERT, trained in firefighting, will handle the situation until local fire services arrive, if needed.
  4. The incident will be recorded, and a root cause investigation will be conducted.

#### 6.1.3 Hazardous Material Spill:

- **Prevention Measures:** All hazardous materials will be properly stored and handled according to IFC EHS guidelines.
- **Response:**
  1. In the event of a spill, the ERT will cordon off the affected area.
  2. Workers trained in hazardous material handling will wear appropriate PPE and use spill containment kits to control the spill.

---

<sup>3</sup> Refer to the **Emergency Management Plan** standalone document for more details.



3. An assessment will be conducted to determine if further environmental or health risks exist, and external waste management services will be called if necessary.
4. The site will be cleaned, and waste will be disposed of as per the waste management plan.

#### 6.1.4 Marine Incidents

Marine incidents occurring during barge operations between Savaric Yard and the berths will trigger immediate notification of NPA Marine Operations and activation of the PSSCC emergency coordination protocol.

### 6.2 Emergency Response Team (ERT)

The Emergency Response Team (ERT) is a critical component of the project's emergency management system. The team will consist of workers trained in firefighting, first aid, and hazardous material handling. Roles and responsibilities include:

- **Health and Safety Officer:** Overall coordination of the ERT and ensuring the implementation of the emergency response protocols.
- **First Aid Team:** Administer first aid, manage the transfer of injured persons to the on-site clinic, and coordinate with external medical services for transport to the nearest hospital.
- **Fire Marshals:** Trained in firefighting and responsible for controlling small fires before emergency services arrive. They follow instructions of the H&S Officer.
- **Environmental Officer:** Responsible for containment and control of any chemical spills or accidents involving hazardous substances.

The ERT will activate joint response protocols with the NPA Fire Service and external responders for fire, spill, medical, or marine emergencies. Any incident exceeding site-level response capacity will be escalated to the PSSCC for port-wide coordination.

### 6.3 On-Site Medical Clinic

An on-site medical clinic will serve as the first point of care for all injuries and emergencies. The clinic will be staffed with a nurse trained in basic first aid and equipped with necessary supplies in coordination of the Occupational Doctor. In the event of serious injuries or emergencies that cannot be handled on-site, the patient will be evacuated to the nearest hospital.

### 6.4 Emergency Equipment

- **Fire Extinguishers:** Strategically placed throughout the site and inspected weekly for functionality and expiration dates.
- **First Aid Kits:** Located at all project sites and regularly inspected to ensure proper stocking and conditions.
- **Spill Kits:** Positioned near areas where hazardous materials are stored or used.

### 6.5 Incident Reporting and Investigation Procedure

After any emergency, the Incident Report Form (check annexes) will be completed by the Health and Safety Manager. The steps are as follows:

1. **Immediate Reporting:** All incidents, including near misses, must be reported to the Health and Safety Officer immediately.



2. **Investigation:** A thorough investigation will be conducted to determine the cause of the incident, including witness interviews and site inspections.
3. **Corrective Actions:** Based on the findings, corrective actions will be implemented to prevent recurrence.
4. **Reporting:** A detailed incident report, along with recommendations, will be submitted to the project management team and filed as part of the project's safety records.

Table 6-1: Description of Incident Categories

| Category             | Description                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Insignificant</b> | Minor incidents with no injury, damage, or environmental impact. Typically, does not require corrective action.                                                       |
| <b>Minor</b>         | Small-scale incidents causing minor injuries, minimal damage, or slight environmental impact. Requires first aid and basic corrective actions.                        |
| <b>Moderate</b>      | Incidents that result in moderate injuries, damage to property, or environmental impacts. Requires medical treatment and corrective actions to prevent recurrence.    |
| <b>Major</b>         | Incidents causing significant injuries, major damage, or severe environmental impacts. Requires detailed investigation and immediate corrective actions.              |
| <b>Critical</b>      | High-impact incidents with severe injuries, substantial property damage, or widespread environmental harm. Requires urgent intervention, investigation, and response. |
| <b>Fatality</b>      | Incident resulting in the death of a worker or community member. Requires full investigation and compliance with regulatory requirements.                             |

Table 6-2: Description of Incident Types

| Incident Type          | Description                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Injury</b> | An incident involving injury to a worker or third party. It can range from minor injuries to severe injuries requiring medical attention or fatality. |
| <b>Hazard</b>          | A condition or situation that has the potential to cause harm, injury, or environmental damage but has not yet led to an incident.                    |
| <b>Near Miss</b>       | An unplanned event that did not result in injury, damage, or harm but had the potential to do so.                                                     |
| <b>Active/Security</b> | Incidents related to security breaches, theft, or active threats on the construction site, including criminal activity.                               |
| <b>Quality</b>         | Incidents that affect the quality of the construction or materials used, potentially compromising safety, durability, or project standards.           |
| <b>Environmental</b>   | Incidents that result in environmental harm, such as spills, pollution, habitat destruction, or non-compliance with environmental standards.          |



## 6.6 Evacuation and Coordination with External Services

- **Evacuation Plan:** In the event of severe injuries or life-threatening situations, the ERT will initiate evacuation protocols to transfer the injured to the nearest hospital.
- **Coordination with Local Authorities:** The project will establish strong communication channels with local fire services, medical facilities, and other emergency services to ensure swift external support when necessary.

ITB-Hitech will coordinate all emergency response planning and drills with the NPA Fire Service and external emergency responders. Joint simulations will be conducted periodically, especially when construction fronts shift along the quay wall. Emergency access routes will be reviewed jointly prior to the start of works in each new berth zone to guarantee unobstructed access for fire trucks, paramedics, and spill response vehicles

## 6.7 Inspections and Drills

**Weekly** inspections of emergency equipment (e.g., fire extinguishers, first aid kits) will be conducted, and emergency drills will be carried out quarterly to ensure all workers are familiar with response protocols. All emergency drills will be recorded in a registry by the E&S Manager.

## 6.8 Emergency Contact Information

- Fire service: 767 or 112
- Police: 0803 440 1649 (General Lagos State Police Command)
- Ambulance: 0802 288 7777 (Lagos State Ambulance Service - LASAMBUS)

## 7. Security management

All security operations will follow the Voluntary Principles on Security and Human Rights (VPSHR)<sup>4</sup>, ensuring that the security personnel conduct themselves in a manner that respects human dignity and avoids conflicts with workers and nearby local communities. The security will not be subcontracted. Security staff employed by the contractor will oversee the restricted access of project sites and active construction sites within the ports including night shifts for project facilities security. This will be in coordination with the ports security staff to ensure adequate communication and coordination. A standalone **Security Management Plan (SMP)** has been prepared for the project.

## 8. Training

The E&S Team is committed to equipping the project staff with the knowledge and skills required to uphold the CESMP. The training approach is characterized by a systematic and comprehensive strategy that addresses the various dimensions of CESMP implementation, as well as compliance with IFC and World Bank standards, all while adhering to best practices.

### 8.1 Training Strategy

#### 8.1.1 Needs Assessment:

- **Identification of Training Needs:** The process begins with a rigorous assessment conducted by the E&S Team to pinpoint specific training requirements aligned with the CESMP.
- **Stakeholder Engagement:** Active engagement with stakeholders, including project managers, department heads, and the dedicated environmental and social team, takes place during staff meetings and is further enriched through question-and-answer sessions during toolbox talks.

#### 8.1.2 Objectives and Learning Outcomes:

- **Clear Training Objectives:** The E&S Team defines unambiguous and measurable training objectives for each program to ensure participants acquire the essential knowledge and skills.
- **Alignment with CESMP:** All training objectives are in direct alignment with the CESMP

#### 8.1.3 Training Programs:

- **Diverse Training Programs:** The contractor offers a spectrum of training programs, encompassing induction training, toolbox talks, online courses, workshops, and specialized skill-based training.
- **Topic Relevance:** Each training program is meticulously tailored by the E&S Team to address specific facets of CESMP compliance and implementation. These include environmental protection, biodiversity management, health and safety, cultural heritage preservation, and more.

<sup>4</sup> [https://www.miga.org/sites/default/files/archive/Documents/VPSHR\\_Toolkit\\_v3.pdf](https://www.miga.org/sites/default/files/archive/Documents/VPSHR_Toolkit_v3.pdf)





- **Incorporation of IFC and World Bank Standards:** All training content is meticulously designed by the E&S Team to encompass the principles and standards set forth by the IFC and World Bank, ensuring unwavering alignment with international best practices.
- **Responsibility for Training Materials:** For each training or toolbox talk, the E&S Manager prepares the training material and script, which is subsequently used by the Trainer during the sessions. An exception arises when external trainers, such as the Occupational Doctor, Local Health Authority, Waste Contractor, or International Experts, are engaged to train workers in the usage of heavy machinery.

#### 8.1.4 Training Delivery:

- **Qualified Trainers:** The trainers, carefully selected by the E&S team, have extensive experience in their respective domains, including environmental management and health and safety.
- **Interactive Learning:** Various training methods are employed to facilitate interactive learning, including presentations, discussions, hands-on exercises, and online courses. These approaches foster active engagement and enhance learning effectiveness.
- **Location Flexibility:** To cater to varying needs and content types, training sessions are conducted both online and on-site (mainly on-site), with the location chosen for participant convenience and content relevance.

#### 8.1.5 Training Records and Documentation:

- **Training Log:** The E&S team maintains a comprehensive Training Log that meticulously records details of each training session. This includes dates, training names, durations, participant numbers, department information, training types, training providers, locations and reference to the Environmental and Social Management and Monitoring Plan (ESMMP).
- **Workers Log:** A database containing comprehensive worker data, including records of completed training, is maintained. This invaluable resource enables the identification of any missing training during audits.
- **Documentation:** Training records are diligently preserved for auditing and reporting purposes, guaranteeing transparency and compliance. This includes photographic documentation of each session and participant lists with signatures.

#### 8.1.6 Continuous Improvement:

- **Feedback Mechanism:** Active encouragement of feedback from participants is a key component of the approach, ensuring continuous improvement in training programs.
- **Monitoring and Evaluation:** Regular reviews are conducted by the E&S team to assess the effectiveness of training programs. Necessary adjustments are made to ensure alignment with evolving compliance requirements.

The approach to training is engineered to empower all project personnel with the knowledge and skills essential for effective CESMP implementation. By aligning training programs with the CESMP, the contractor and the E&S Team remain steadfast in upholding the highest standards of environmental and social responsibility throughout every phase of the project. The Training Log serves as an indispensable tool for monitoring training activities and their alignment with CESMP objectives, bolstering overall project success and compliance.

Table 8-1: Workers Training Log Template

| Responsible: E&S Manager |      |               |       |                  |            |      | Supervisor: Project Manager |          |        |         |
|--------------------------|------|---------------|-------|------------------|------------|------|-----------------------------|----------|--------|---------|
| No.                      | Date | Training Name | Hours | No. Participants | Department | Type | Training Provider           | Location | MP Ref | Content |
|                          |      |               |       |                  |            |      |                             |          |        |         |



Figure 8-1: E&S Training Process

#### 8.1.7 Types of Training Programs

The Training Management Plan includes three types of training programs, each tailored to different aspects of capacity building and compliance:

1. **Workshops:** Structured, interactive sessions focusing on specific skills or topics, such as hazardous material handling or community engagement. Workshops often involve hands-on exercises, role-playing, and scenario-based learning to ensure practical understanding.
2. **Awareness Campaigns:** Short, focused sessions designed to raise awareness on specific issues such as Gender-Based Violence (GBV), environmental protection, or Personal Protective Equipment (PPE) use. These campaigns may use posters, videos, and group discussions to reach a broad audience.
3. **Training Sessions:** Formal, in-depth training courses delivered by certified trainers. These sessions cover essential topics such as safety at work, emergency preparedness, and occupational health and safety. Training sessions include theoretical learning, practical demonstrations, assessments, and certification.

## 8.2 Training Programme

The Training Programme outlines the different types of training, frequency, target participants, content, and delivery methods for the project's duration. The program is divided into three phases:

1. **Induction Training**
2. **Regular Training**
3. **Specialized Training.**

### 8.2.1 Induction Training

The objective of the EHS induction training is to provide all new workers with an overview of the project, the CESMP, including key policies, procedures, and ESHS standards.

- **Content:**
  - Overview of the Tin Can Port Modernization project
  - Introduction to the CESMP
  - Key ESHS Policies and Standards (IFC Performance Standards)
  - Roles and Responsibilities of Workers
  - Emergency Response Plan
  - Site Rules and Code of Conduct
  - Introduction to Health, Safety, and Environmental (HSE) Risks
- **Frequency:** Mandatory for all new workers upon joining.
- **Delivery Method:** Classroom-style presentations, followed by a site tour.
- **Duration:** 1.5 hours



## 8.2.2 Regular Training

| Training Topic                             | Content Overview                                                                                                      | Target Audience              | Frequency            | Delivery Method                            | Trainer                         |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------|--------------------------------------------|---------------------------------|
| <b>General ESHS Awareness</b>              | ESHS policies, worker rights, environmental protection, social and cultural awareness, waste management.              | All workers                  | <b>Quarterly</b>     | Awareness Campaigns, Workshops             | E&S Manager                     |
| <b>Personal Protective Equipment (PPE)</b> | Types of PPE, correct usage, maintenance, and storage.                                                                | All workers                  | <b>Bi-monthly</b>    | Hands-on demonstrations, Training Sessions | H&S Officer                     |
| <b>Waste Management</b>                    | Waste segregation, disposal procedures, recycling initiatives, and spill response.                                    | All workers                  | <b>Bi-monthly</b>    | Workshops                                  | Environmental Officer           |
| <b>Emergency Response and Evacuation</b>   | Emergency procedures, fire drills, evacuation routes, and roles during emergencies.                                   | All workers                  | <b>Quarterly</b>     | Simulation Drills, Training Sessions       | H&S Officer                     |
| <b>Environmental Stewardship</b>           | Water conservation, biodiversity conservation, dust suppression techniques.                                           | Field and site workers       | <b>Semi-annually</b> | Workshops                                  | Environmental Officer           |
| <b>Community Engagement</b>                | Respectful interaction with local communities, grievance mechanisms, community health and safety.                     | All workers, especially CLOs | <b>Semi-annually</b> | Training Sessions                          | Community Liaison Officer (CLO) |
| <b>Biodiversity Management</b>             | Wildlife conservation, handling wildlife encounters, no hunting, and respecting wildlife habitats near the work site. | All workers                  | <b>Semi-annually</b> | Workshops, Awareness Campaigns             | Environmental Officer           |



| Training Topic                                              | Content Overview                                                                                             | Target Audience | Frequency            | Delivery Method                        | Trainer                         |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------|----------------------|----------------------------------------|---------------------------------|
| <b>Respect for Communities and Socioeconomic Activities</b> | Understanding and respecting local communities' lifestyles, economic activities, and preventing disturbance. | All workers     | <b>Semi-annually</b> | Training Sessions, Awareness Campaigns | Community Liaison Officer (CLO) |

### 8.2.3 Specialized Training

| Specialized Training Topic          | Content Overview                                                                  | Target Audience                        | Frequency       | Delivery Method                             | Trainer                          |
|-------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------|-----------------|---------------------------------------------|----------------------------------|
| <b>First Aid and CPR</b>            | Basic first aid, CPR techniques, handling injuries, maintaining a safe site.      | Designated First Aiders, Supervisors   | <b>Annually</b> | Practical Demonstrations, Training Sessions | Certified First Aid Trainer      |
| <b>Firefighting and Fire Safety</b> | Fire prevention, use of extinguishers, fire control, evacuation procedures.       | All workers, especially fire wardens   | <b>Annually</b> | Practical Drills, Workshops                 | Certified Fire Safety Instructor |
| <b>Working at Heights</b>           | Safety precautions, use of harnesses, fall arrest systems, scaffolding standards. | Construction workers, Maintenance crew | <b>Annually</b> | Practical Demonstrations, Training Sessions | Certified Safety Instructor      |

### 8.2.4 Awareness Campaigns

Awareness campaigns are a critical component of the CESMP, designed to address specific ESHS issues and engage both workers and communities:

- **Gender-Based Violence (GBV) Awareness:** Focused on reducing GBV risks through education on respectful behaviour, reporting mechanisms, and the project's zero-tolerance policies. Campaigns will highlight the importance of gender equality, respect in the workplace, and the consequences of harassment or violence.
- **Water Conservation and Management:** Campaigns to promote responsible water usage, reduce wastage, and protect local water sources. Workers and community members will be educated on the impact of water scarcity, sustainable water practices, and the importance of maintaining clean and accessible water for all.
- **Disease Prevention (e.g., COVID-19, Malaria):** Awareness programs on health and hygiene practices, use of mosquito nets, sanitation practices, vaccination awareness, and general wellness. Campaigns will focus



on preventing the spread of infectious diseases through personal hygiene, social distancing, and preventive measures.

- **Sexually Transmitted Diseases (STDs), Pregnancy, and Contraception Awareness:** In partnership with community medical services, campaigns will focus on educating workers on safe sexual practices, the importance of using contraceptives, and preventing unwanted pregnancies and the spread of STDs. Free contraceptives (such as condoms) will be distributed, and regular free health checks, including tests for STDs, will be provided for all workers. These campaigns will emphasize the importance of reproductive health, consent, and respectful relationships

## 9. Supplier and Subcontractor Management

The contractor must ensure that all third-party entities involved in the project meet the project's regulatory framework's standards in terms of quality, safety, environmental performance, and social responsibility. The key aspects of this plan include:

1. **Contractual Obligations:** All suppliers and subcontractors will enter into formal agreements outlining specific environmental, health, safety, and social obligations. These obligations include adherence to the project's CESMP and all its mitigation measures, including waste management, pollution prevention, biodiversity, safety protocols, and labour rights. The contracts with these subcontractors will include clauses that the subcontractor is to align with the project's requirements.
2. **Performance Monitoring:** Suppliers and subcontractors' performance will be regularly monitored against agreed Key Performance Indicators (KPIs). This includes adherence to environmental and social standards, timely delivery of goods and services, and safety compliance.
3. **Waste Management:** Subcontractors responsible for waste collection and disposal must hold relevant permits, particularly for hazardous and medical waste. Compliance with national regulations and IFC EHS Guidelines will be mandatory, with traceability from collection to disposal being required.
4. **Training and Awareness:** All subcontractors will undergo mandatory environmental and social training before project commencement. This ensures that all parties are aware of the project's E&S obligations and implement them on-site effectively.
5. **Local Content:** The project prioritizes hiring local workers and suppliers wherever possible, contributing to local economic development. A minimum quota for hiring workers from surrounding communities will be established to maximize socioeconomic benefits. This minimum quota is set to be **40%** of the total contractor's project workforce.
6. **Supplier and Subcontractor Evaluation:** A stringent vetting process will be implemented, including evaluating suppliers based on their environmental, social, and technical capabilities. This ensures that only compliant suppliers are engaged.
7. **GRM:** Subcontractors must integrate the project's GRM into their processes to handle complaints and issues raised by workers, communities, or other stakeholders.
8. **Non-Conformance Management:** A structured non-conformance system will be established, where any deviation from project standards results in corrective action, and repeated failures may lead to contract termination.
9. **Coordination and Communication:** The project will establish a communication protocol to ensure ongoing and clear communication between contractors, subcontractors, and the project management team. This includes regular performance review meetings and reports.
10. **Sustainability and Continuous Improvement:** The project promotes sustainability in all subcontracting processes, encouraging suppliers to adopt energy-efficient, low-impact technologies, and continuous improvement in their performance.





## 9.1 Suppliers delivery on site

Before delivery, the Project Manager will inform the supplier of the HSE requirements to be followed while entering site. Deliveries will be made to a specific location according to a defined protocol. Instructions for storage and assembly will be provided in the packages or sent separately to the site by the project team.

When entering the site, suppliers will follow the same instructions as those imposed on visitors or deliverers, particularly with regard to the required PPE.

If any of the equipment ordered may present a hazard, the supplier must provide full information beforehand on the precautions to be taken during use, storage and transport.

## 9.2 Non-Conformity System

To ensure that subcontractors adhere to the contractor requirements, a Non-Conformity System has been established. This system is designed to identify, categorize, resolve, and manage non-conformities effectively to maintain the integrity and success of the project.

### 9.2.1 Identification:

- Non-conformities can be identified through various sources such as internal audits, external audits, site inspections, incident reports, or feedback from stakeholders.
- Regular monitoring activities and reports will help in the early detection of any deviations from the agreed standards and procedures.
- Non-conformities are documented in a standardized format, detailing the nature, location, and potential impact of the issue.

### 9.2.2 Categorization:

Once identified, non-conformities are categorized based on their severity to prioritize resolution efforts:

- **Low Severity:** Minor issues that do not significantly impact project timelines, safety, or quality. These are typically isolated incidents that can be corrected quickly.
- **Medium Severity:** Issues that have a moderate impact on project activities, requiring corrective actions to prevent escalation. These may affect project schedules or pose a moderate risk to safety and quality.
- **High Severity:** Critical issues that pose a significant risk to project success, safety, or compliance. These require immediate attention and resolution to avoid severe consequences.

### 9.2.3 Resolution:

Each category of non-conformity has specific response times for addressing the issues:

- **Low Severity:** Must be addressed **within 30 days**. Corrective actions are documented and reviewed to ensure completion.
- **Medium Severity:** Must be addressed **within 15 days**. A detailed action plan is developed, and progress is monitored closely.
- **High Severity:** Must be addressed **within 7 days**. Immediate corrective actions are taken, and continuous monitoring is implemented until the issue is resolved.

All corrective actions are documented, including the steps taken, responsible parties, and completion dates.

#### 9.2.4 Consequences:

Persistent or unresolved high-severity non-conformities can result in contract termination. The steps include:

- **Initial Notification:** The subcontractor is formally notified of the high-severity issue and given a deadline for resolution.
- **Follow-Up Audit:** A follow-up audit is conducted to verify if the issue has been resolved within the given timeframe.
- **Final Decision:** If the issue remains unresolved or recurs, the contractor may proceed with contract termination. The subcontractor will be informed of the decision and provided with a final notice outlining the reasons for termination.

In addition to contract termination, other potential consequences for non-conformities include:

- Withholding of payments until issues are resolved.
- Imposing penalties as per the contract agreement.
- Requirement for additional training or re-certification of subcontractor staff.
- Increased frequency of audits and inspections until compliance is achieved.

#### 9.2.5 Continuous Improvement:

- Lessons learned from non-conformities are documented and analysed to prevent recurrence.
- The Non-Conformity System is regularly reviewed and updated to incorporate best practices and improve its effectiveness.
- Subcontractors are encouraged to participate in continuous improvement initiatives, providing feedback and suggestions for enhancing compliance and performance.

## 10. Grievance Redress Mechanism<sup>5</sup>

### 10.1 Purpose and Scope

The Grievance Redress Mechanism (GRM) is a structured, transparent, and confidential process designed to receive, assess, and resolve grievances raised by project workers and stakeholders, including communities, civil society, and affected persons, in connection with the project. It applies to all grievances related to project activities, impacts, or relationships with ITB-Hitech, its subcontractors and third parties.

The GRM is a critical component of the project's SEP and ESMS, aligned with IFC Performance Standard 1 and the expectations of Equator Principles lenders. It promotes accountability, protects complainant rights, and builds trust with communities. The mechanism is accessible, with no cost or discrimination against those who submit grievances. It also handles sensitive matters such as human rights violations, child or forced labour, and Gender-Based Violence and Sexual Harassment (GBVSH). The mechanism covers grievances related to:

| Environmental                                                                                                                                                                                           | Social                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Governance                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• <i>Pollution</i></li> <li>• <i>Dust and noise</i></li> <li>• <i>Waste</i></li> <li>• <i>Vibrations</i></li> <li>• <i>Biodiversity and ecosystem</i></li> </ul> | <ul style="list-style-type: none"> <li>• <i>Working conditions</i></li> <li>• <i>Occupational Health and Safety</i></li> <li>• <i>Harassment or discrimination</i></li> <li>• <i>Gender-based violence</i></li> <li>• <i>Child or forced labour</i></li> <li>• <i>Unpaid wages or benefits</i></li> <li>• <i>Welfare facilities and sanitation</i></li> <li>• <i>Loss of job or income</i></li> <li>• <i>Property damage</i></li> <li>• <i>Land use concerns</i></li> <li>• <i>Community safety concerns</i></li> <li>• <i>Compensation concerns</i></li> <li>• <i>Damage to cultural heritage</i></li> <li>• <i>Security issues</i></li> <li>• <i>Human rights concerns</i></li> </ul> | <ul style="list-style-type: none"> <li>• <i>Compliance, Ethics and Transparency</i></li> <li>• <i>Internal or External Communication</i></li> <li>• <i>Governance structure / Management</i></li> <li>• <i>Retaliation after complaint</i></li> </ul> |

#### Key Principles of the GRM

- **Accessibility:** Free and available to all stakeholders, including vulnerable groups, in local languages and via multiple channels.
- **Confidentiality:** All complaints, including anonymous ones, are handled discreetly.
- **Non-Retaliation:** No stakeholder will face reprisal for submitting a grievance.
- **Cultural Sensitivity:** Engagement and processing will respect local norms and customs.
- **Transparency:** Complainants will receive timely updates and explanations.
- **Responsiveness:** All grievances will be acknowledged, tracked, and resolved through a fair process.
- **Appeal Rights:** Complainants dissatisfied with resolutions may escalate the issue further.

<sup>5</sup> Refer to the **Workers Grievance Mechanism** standalone document for more details.

## 10.2 GRM Process

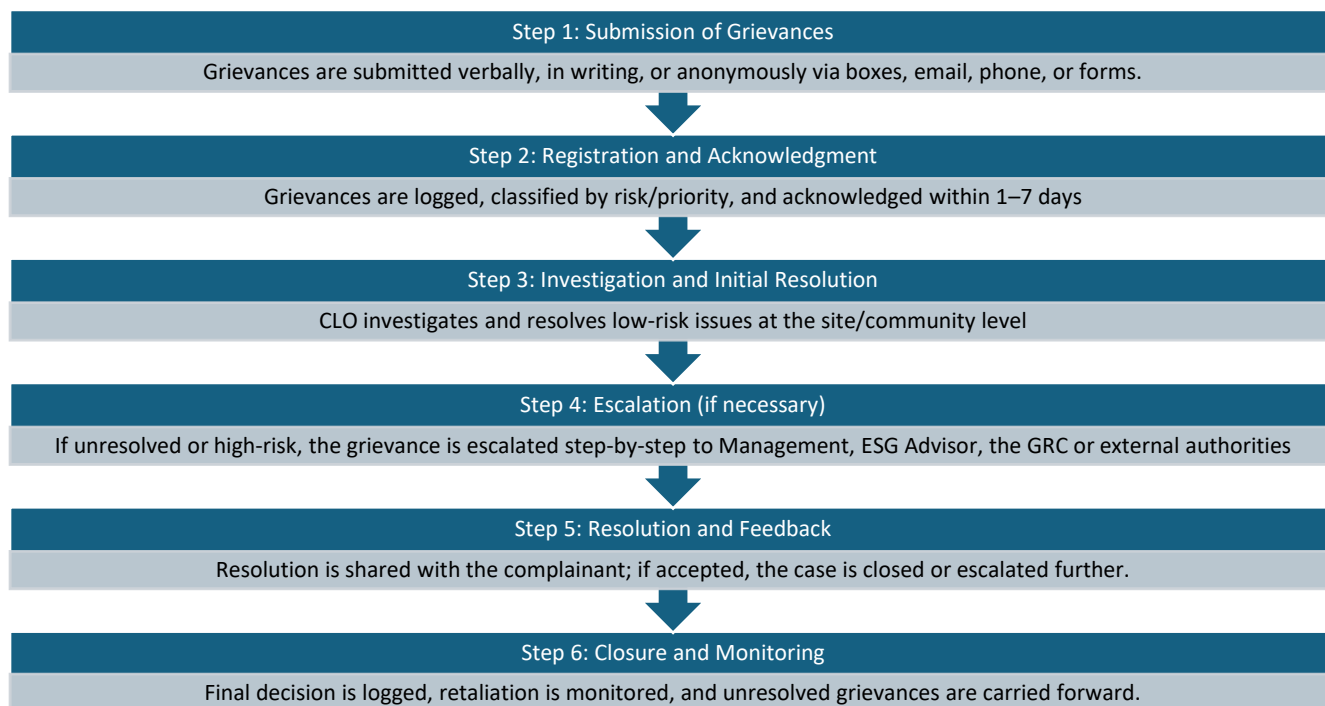


Figure 10-1: GRM Process Overview

**Step 1: Submission of Grievances.** Grievances can be submitted by individuals, groups, or representatives via the following channels:

- Verbally or in writing to the Community Liaison Officer (CLO), or site staff
- Email address to [grievance@itbng.com](mailto:grievance@itbng.com) and hotline 234 812 9999 988
- Grievance Boxes installed at project sites and community halls
- By completing a Grievance Form

Grievances can be submitted in any local language (CLO and HR team are fluent in Pidgin English), and multilingual support will be made available. Also, they can be submitted anonymously with no requirement to include a reply method. However, if the complainant wishes to receive a response, they may optionally include contact details.

**Step 2: Registration and Acknowledgment.** All grievances will be recorded in the Grievance Log with a unique reference number by the CLO. The grievance is classified using:

- **Risk Level Matrix** (High / Medium / Low)
- **Priority Level Matrix** (Urgent / Moderate / Low)

The grievance is acknowledged within:

- **1–2 working days** if high priority
- **7 working days** for all others

### Risk Level Definitions

| Risk Level | Definition |
|------------|------------|
|------------|------------|



|                    |                                                                                                                                                                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>High Risk</b>   | The grievance involves a serious legal violation, threat to life or safety, breach of human rights, reputational damage, or major financial implications. Immediate attention and escalation are required. May require external reporting or independent handling. |
| <b>Medium Risk</b> | The grievance poses a moderate concern that could escalate if not resolved promptly. It may impact worker morale, safety practices, or stakeholder confidence. Resolution should be prioritized within a set timeframe.                                            |
| <b>Low Risk</b>    | The grievance is minor in nature, with limited impact, no legal or safety consequence, and no escalation potential. It can be resolved at site level using routine procedures.                                                                                     |

#### Priority Level Definitions

| Priority Level       | Definition                                                                                                                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>High (Urgent)</b> | The grievance requires immediate action to prevent harm, legal exposure, reputational risk, or stakeholder backlash. Must be addressed within <b>3–5 working days</b> and escalated where needed. |
| <b>Medium</b>        | The grievance requires timely resolution to avoid operational or social disruptions. Should be addressed within <b>7–14 working days</b> .                                                        |
| <b>Low</b>           | The grievance does not require urgent intervention and may be addressed within normal operational cycles. Should be resolved within <b>14–21 days</b> .                                           |

The grievance is assigned to dedicated personnel for its resolution-seeking process.

**Step 3: Investigation and Initial Resolution.** The CLO begins investigation and may consult relevant departments depending on the type of complaint. Investigations will consider the nature, risk, and priority level of the grievance.

- Low-risk, low-priority grievances are resolved at the site/community interface level.
- If unresolved or higher in risk/complexity, the case is escalated as per the table below.

**Step 4: Escalation (if necessary).** Escalation and timelines will be managed as per the table below:

| Step | Step Responsible      | Applies When                                                                                                        | Timeline                   | Timeline (High Priority Grievance)                                            |
|------|-----------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------|
| 1    | CLO                   | Grievance is submitted (verbal, written, box, email); logged, acknowledged. Risk and Priority level are categorized | Within 7 working days      | Acknowledge within 1–2 working days; resolve or escalate within 3–5 days      |
| 2    | HSE Unit              | Grievance is Medium–High Risk, unresolved, or requires policy review                                                | Within 14 working days     | Resolve within 5–7 working days if urgent or safety-related                   |
| 3    | Project Manager       | For complex, cross-departmental cases or Medium–High Risk grievances with potential disruption                      | Within 7 working days      | Respond within 3–5 working days if risk to safety, legal compliance, or trust |
| 4A   | Project Director (PD) | Informed of serious issues; intervenes only in cases of systemic failure, retaliation, or escalation blockage       | As required                | Immediate intake and interim protection within 24–48 hours                    |
| 4B   | External ESG Advisor  | For sensitive grievances (e.g., GBV/SH, human rights, retaliation, child/forced labour)                             | Immediately / Confidential | Actively informed within 48 hours for any reputational, legal, or public risk |



| Step | Step Responsible                                           | Applies When                                                                                                                                                                                                                                                                                                                               | Timeline                     | Timeline (High Priority Grievance)                                                       |
|------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------|
| 4C   | Grievance Resolution Committee (GRC)                       | Grievance involves multiple stakeholders or sensitive community concerns requiring transparency, community trust, or multi-stakeholder oversight. Applicable in cases of unresolved external grievances with potential for public unrest, reputational risk, media attention, or where internal resolution mechanisms have been exhausted. | Within 10–15 working days    | Convene within 5–7 working days if community unrest, media risk, or protests expected    |
| 5    | External Authorities (e.g., Ministry of Labour, IAP, NICN) | If grievance remains unresolved, involves legal/human rights concerns, or is escalated by complainant                                                                                                                                                                                                                                      | <i>Per statutory process</i> | May be referred immediately if risk to life, child labour, or forced labour is confirmed |

**Step 5: Resolution and Feedback.** A proposed resolution is shared with the complainant (where identifiable).

- If accepted, the grievance is closed.
- If not accepted, the grievance moves to the next escalation tier.
- All resolutions are documented, including actions taken and dates.

**Step 6: Closure.** Final status is logged in the Grievance Register. The complainant (if known) is informed of the outcome.

- CLO monitor for possible retaliation or unresolved tension.
- Grievances that are not resolved are carried forward with updated tracking notes.

Any indication of retaliation will trigger an immediate investigation. CLO is required to monitor for post-grievance retaliation and report findings quarterly. Procedures such as follow-up surveys, confidential feedback channels, and whistleblower protections will be implemented to ensure thorough monitoring and prevention of retaliation.



## 11. Monitoring and Reporting

Below is the overview of the reporting of the project CESMP.

Table 11-1: Reporting Overview of the Project CESMP

| Activity                               | Description and Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Responsible         | Reviewing Entity                      | Frequency      |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------|----------------|
| <b>Monthly HSE Performance Report</b>  | <p>A detailed report is prepared to highlight all the progress in E&amp;S monitoring implementation as well as the E&amp;S performance of the project. This includes the following:</p> <ul style="list-style-type: none"> <li>• <i>Status of HSE Unit and responsibilities</i></li> <li>• <i>KPIs of Human Resources Department and HSE Unit</i></li> <li>• <i>E&amp;S Inspections done.</i></li> <li>• <i>Nonconformities received by NESREA</i></li> <li>• <i>Corrective actions in response to the non-Conformities</i></li> <li>• <i>E&amp;S Mitigation Measures in place and its status, E&amp;S Achievement of the month</i></li> <li>• <i>Opportunity Areas and Focus for the next month.</i></li> <li>• <i>Stakeholder Engagement Registry and MOM of each meeting</i></li> <li>• <i>Incidents registry</i></li> <li>• <i>Grievances Registry and HSE Trainings Log</i></li> </ul> | HSE Unit            | Top Management                        | Monthly        |
| <b>Grievances Reporting</b>            | Summarize grievances received, actions taken to address them, and outcomes. Ensure timely and effective resolution of stakeholder concerns.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | CLO and LCO (or HR) | E&S Manager                           | Monthly        |
| <b>Grievances Reporting</b>            | Summarize grievances received, actions taken to address them, and outcomes. Ensure timely and effective resolution of stakeholder concerns.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | CLO and LCO (or HR) | E&S Manager                           | Monthly        |
| <b>E&amp;S Internal Audits</b>         | This is an auditing <b>procedure</b> conducted by the HSE Manager with the support of the HSE Unit, the Project Manager, HR Manager and all managers of the project to verify all procedures of the management plans are in place. This includes the request of any required documentation, site visits in all project sites, 1 to 1 interview with managers, Focus Groups discussions and the request of any needed information or data. The objective is to also ensure all previous IESC recommendations from the previous monitoring visit have been implemented or on track as per the project planning.                                                                                                                                                                                                                                                                               | HSE Manager         | Project Manager and Managing Director | Quarterly      |
| <b>E&amp;S Self-Monitoring Reports</b> | As part of the Lenders Agreement, an E&S Self-Monitoring Report to have a full and clear understanding of all Project E&S activities, monitoring and implementation of the environmental and social                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | E&S Manager         | ESG Advisor and Lenders               | Every 3 months |





| Activity                                                          | Description and Objectives                                                                                                                                                                                                                                                                                                                        | Responsible                                           | Reviewing Entity | Frequency         |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|------------------|-------------------|
|                                                                   | management plans, and compliance with <b>the IFC and Equator Principles IV</b> .                                                                                                                                                                                                                                                                  |                                                       |                  |                   |
| <b>Quarterly Environmental Compliance Reports to FMEnv/NESREA</b> | In line with FMEnv EIA Act and NESREA regulations, these reports document air, noise, water, soil, and waste monitoring results. They demonstrate implementation of corrective actions, national standards compliance, and respond to any FMEnv or NESREA observations. Structured per national guidelines and includes laboratory-verified data. | National Environmental Consultant certified by NESREA | FMEnv and NESREA | Quarterly         |
| <b>Independent Review and External Audit Reports</b>              | External review and audit of environmental and social performance, ensuring compliance with international standards and lender requirements.                                                                                                                                                                                                      | ESG Advisor                                           | Lenders          | <b>Biannually</b> |



## 12. Auditing

As part of the CESMP implementation, the Contractor will carry out ad hoc audits throughout the construction phase relevant to the project activities being carried out. The purpose of auditing and reporting within the CESMP is to systematically assess and verify the implementation of environmental and social measures during the construction phase. Audits provide real-time insights, identify areas for improvement, and ensure compliance with project requirements.

### 12.1 Audit Procedure

**Quarterly** internal audits will be undertaken by the Contractor. The project will carry out periodic internal HSE audits to determine:

- Whether or not the CESMP is being properly implemented and maintained;
- Whether or not there are areas or actions which must be implemented to achieve continual improvement.

**Ad hoc** internal audits may be conducted in response to new processes and / changes to existing processes and based on previous audit results.

#### 12.1.1 Audit Planning

The Project Director, E&S Manager and Project Manager are responsible for defining the scope of each audit. The audit scope is based on lenders and national requirements or other requirements considering the items below:

- Requirements from relevant management standards;
- Legal and contractual requirements;
- Previous audits results;
- Characteristics of the processes and functions to be audited;
- Important changes in workplaces, areas or departments, or analysis of performance data on health and safety at work, environmental, quality and productivity of the areas to audit;
- Occurrence of quality, health and safety and environmental events.

The Audit Program Manager is/ are responsible for assigning the Lead Auditor for each audit in their respective audit program considering the following three main parameters:

- Auditors do not have any conflict of interest in relation to the audited activities to ensure objectivity;
- Auditors are independent from the audited activities to ensure impartiality;
- Knowledge on the scope under audit and knowledge of local language.

The Lead Auditor selects other team members as appropriate, considering the above parameters. Audit planning will be done **at least four weeks** before the audit. During the audit, information relevant to the audit objectives, scope and criteria, including information relating to interfaces between functions, activities and processes shall be collected using appropriate sampling.

#### 12.1.2 Conducting the Audit

Audits shall be conducted such that all relevant aspects of management for achieving the defined objectives (e.g., reducing non-conformance costs) can be reviewed to determine whether the CESMP complies with applicable standards and is being properly implemented. Auditors must also include the verification of effectiveness of previously implemented corrective actions via random reviews.

### 12.1.3 Review of Audit Results

Once the audit has been closed, the Lead Auditor will prepare and distribute an Audit Report and Findings **within two business days**. Audit findings will be classified according to the table below:

Table 12-1: Classification of Audit Findings

| Types of Findings          | Description                                                                                                                                                                                  | Action Required                                                                                           |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Non-Conformity (NC)</b> | Non-compliance of the CESMP or other documented requirements, (system failure: more than one example).                                                                                       | Corrective and preventive action is required.<br>Containment may be required.                             |
| <b>Improvements (I)</b>    | Action to optimize processes or CESMP requirements that have been deemed free of non-conformities.                                                                                           | Corrective action is not mandatory as this is a suggestion, but a response is required in the audit tool. |
| <b>Best Practice (BP)</b>  | Practices that are considered best in class or an excellent implementation of a regulation. Also, to be used when significant improvement in certain areas from prior audits has been shown. | Do not require any formal action but can be shared with other organizations                               |

The audit schedule, audit report, list of findings, attendance record, auditor's note during audit, and any additional evidence used for audit will be retained and a PDF version uploaded in designated digital workspaces.

The Lead Auditor/ Auditor shall assess the corrective action plan during the evaluation including the specified implementation date for completion of the corrective action. Improvement measures do not require an evaluation.

The Lead Auditor/ Auditor shall verify the completion of the corrective action based on the objective evidence. If the meaningfulness of the evidence submitted is insufficient, the lead auditor rejects the action. The effectiveness check can be performed during subsequent audits.

### 13. Other Management Plans

The following management plans are to be prepared as standalone documents. It is crucial to note that the CESMP and these management plans should be read and implemented together, as they are interdependent and complement each other.

- **Dust Management Plan:** This plan identifies sources of dust, outlines measures to control emissions, and includes monitoring strategies to protect air quality and public health.
- **Environmental Construction Method Statement:** This statement describes environmentally friendly construction methods and practices to minimize environmental impacts during project implementation.
- **Emergency Management Plan:** This plan outlines procedures for responding to emergencies, including natural disasters and accidents, to minimize harm to people, property, and the environment.
- **Human Resources Management Plan:** This plan outlines strategies for managing human resources, including recruitment, training, and welfare, to ensure a safe and healthy work environment.
- **Waste Management Plan:** This procedure outlines the proper management of construction and hazardous waste, including storage, handling, transportation, and disposal, to minimize environmental impacts.
- **Wastewater Management Plan:** Establishes controls for the collection, treatment, and disposal of wastewater generated during construction activities to prevent pollution of surface water, drainage systems, and the marine environment.
- **Construction Contaminated Land Management Plan:** This plan outlines procedures for managing contaminated land during construction activities, including remediation measures to protect human health and the environment.
- **Soil and Water Management Plan:** Details sediment control, runoff management, drainage protection, and pollution prevention measures to safeguard soil and surface water resources during construction. Measures relating to sediment control and protection of the marine environment referenced in the CESMP are implemented through the SWMP.
- **Occupational Health and Safety Management Plan:** This plan outlines measures to ensure the health and safety of workers, including training, protective equipment, and monitoring, in accordance with international standards.
- **COVID-19 Management Procedure:** This procedure outlines measures to prevent and control the spread of COVID-19 among workers and the community, including hygiene practices, tests and awareness campaigns.
- **Community Health and Safety Management Plan:** This plan addresses health and safety concerns of local communities near the project site, focusing on mitigating risks from construction activities and promoting community well-being.
- **Contractor and Supplier Management Plan:** This plan outlines procedures for selecting, contracting, and managing contractors and suppliers to ensure compliance with environmental, social, and health and safety requirements.

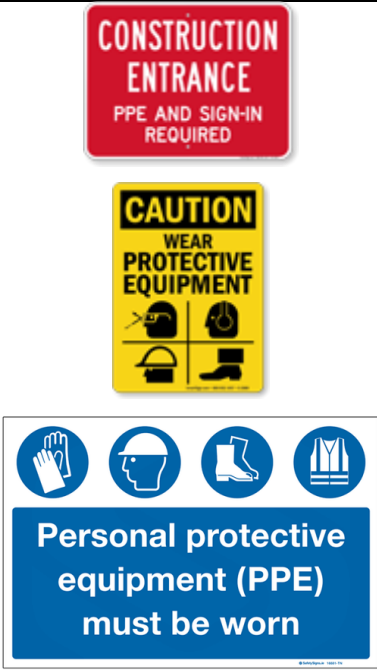


- **Stakeholder Engagement Plan:** This plan outlines strategies for engaging with stakeholders, including local communities, to address their concerns and ensure their participation in decision-making processes. It should include a Community Grievance Mechanism.
- **Community and Workers Grievance Mechanism:** Provides accessible and transparent channels for workers and external stakeholders to raise concerns or complaints related to construction activities, working conditions, or environmental and social impacts, and ensures timely resolution.
- **Traffic Management Plan:** Establishes measures to manage construction traffic, including routing, signage, speed control, and coordination with port authorities, to reduce safety risks and disruption to other port users.
- **Security Management Plan:** Defines security arrangements for the Project, including access control, coordination with public security forces where applicable, and conduct requirements for security personnel, consistent with human rights principles.
- **Interface and Coordination Management Plan:** Describes coordination mechanisms between the Project, port operators, terminal users, and other third parties operating within the port environment to ensure safe and efficient interface management during construction.

## Appendixes

### Appendix A: Safety Signs

| Description                                                                                                                                                                                                                                                                                                                                                                                                                      | Sample                                                                                              | Size<br>W x H [cm] |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------|
| <p>Hazardous Waste Danger Warning</p> <ul style="list-style-type: none"> <li>To be placed near defined hazardous waste storage location</li> <li>The positioning should be based on the flow of workers and be in a visible and accessible area for the storage area in the construction yard project.</li> </ul>                                                                                                                |                   | 30x45 cm           |
| <p>Signs to segregate hazardous waste:</p> <ul style="list-style-type: none"> <li>Please note the provided samples are examples and do not need to be strictly adhered to.</li> <li>The Environmental Officer is to determine the most suitable and necessary signs</li> <li>The frequency of the signs is dependent on the number of storage locations and flow of waste determined and managed by the E&amp;S team.</li> </ul> |                  | 15x30 cm           |
| <p>Signs to guide towards waste storage area:</p> <ul style="list-style-type: none"> <li>This serves as general guidance to all workers for where waste is stored (where the waste drums and skips would be located)</li> </ul>                                                                                                                                                                                                  |                 | 30x45 cm           |
| <p>Basic guideline sign for storage of containers:</p> <ul style="list-style-type: none"> <li>In the storage area</li> </ul>                                                                                                                                                                                                                                                                                                     |  | 40x45cm            |
| <p>Safety reminders around all areas associated with waste storage, handling and disposal</p> <ul style="list-style-type: none"> <li>Provided are examples that serve as refresher for the workers</li> </ul>                                                                                                                                                                                                                    |                 | 30x45 cm           |

| Description                                                                                                                                                                                                                                                                                                | Sample                                                                               | Size<br>W x H [cm] |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------|
| <ul style="list-style-type: none"> <li>They should be strategically placed based on the floor plan for the construction site</li> <li>They should be distributed and not all concentrated in one area</li> <li>There should be an induction to all the signs by the E&amp;S team.</li> </ul>               |    |                    |
| <p>The following signs are based on the ISO 7010 standardized health and safety signs that would be adopted by the project. Not all the ISO signs are provided, rather below are some examples and the project manager is recommended to utilize ISO 7010 as reference for additional signs as needed.</p> |                                                                                      |                    |
| Fire extinguisher                                                                                                                                                                                                                                                                                          |  | 151x151 cm         |
| Fire hose reel                                                                                                                                                                                                                                                                                             |  | 151x151 cm         |
| No spray with water (at each large storage point, a reminder can be placed).                                                                                                                                                                                                                               |  | 50 x 50 cm         |



| Description                                                                                                                              | Sample                                                                               | Size<br>W x H [cm] |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------|
| Refresher sign for no smoking should be placed near waste storage points                                                                 |    | 50 x 50 cm         |
| General warning sign (to be used in combination with other warning signs)                                                                |    | 50 x 50 cm         |
| Example of a biological warning sign in accordance with ISO that would be coupled with the warning sign above                            |    | 50 x 50 cm         |
| Health and Safety reminders should be near waste zones.<br>This is for high visibility clothing especially in areas of waste collection. |   | 50 x 50 cm         |
| No worker is allowed near waste without appropriate PPE, including safety boots.                                                         |  | 50 x 50 cm         |

## Appendix B: Incident Report Form

|                     |  |              |  |
|---------------------|--|--------------|--|
| <b>Report No.:</b>  |  | <b>Date:</b> |  |
| <b>Prepared by:</b> |  | <b>Time:</b> |  |

## Incident / Accident Category

|                                          |                                 |                                    |                                          |                                   |                                        |
|------------------------------------------|---------------------------------|------------------------------------|------------------------------------------|-----------------------------------|----------------------------------------|
| <input type="checkbox"/> Insignificant   | <input type="checkbox"/> Minor  | <input type="checkbox"/> Moderate  | <input type="checkbox"/> Major           | <input type="checkbox"/> Critical | <input type="checkbox"/> Fatality      |
| <input type="checkbox"/> Personal Injury | <input type="checkbox"/> Hazard | <input type="checkbox"/> Near Miss | <input type="checkbox"/> Active/Security | <input type="checkbox"/> Quality  | <input type="checkbox"/> Environmental |

|                                           |                                    |                                    |                                    |                                     |                              |                                |
|-------------------------------------------|------------------------------------|------------------------------------|------------------------------------|-------------------------------------|------------------------------|--------------------------------|
| <b>Patient's Name:</b>                    | <b>Job Position:</b>               |                                    |                                    |                                     |                              |                                |
| <b>Event Location:</b>                    |                                    |                                    |                                    |                                     |                              |                                |
| <b>Event Description:</b>                 |                                    |                                    |                                    |                                     |                              |                                |
| <b>Root Cause:</b>                        | <input type="checkbox"/> Behaviour | <input type="checkbox"/> Procedure | <input type="checkbox"/> Equipment | <input type="checkbox"/> Conditions | <input type="checkbox"/> PPE | <input type="checkbox"/> Other |
| <b>Expected Duration of Work Absence:</b> | <input type="checkbox"/> 0 Days    | <input type="checkbox"/> 1-14 Days | <input type="checkbox"/> 14 Days   | <input type="checkbox"/> Permanent  |                              |                                |

| INJURY TYPE AND COST EVALUATION | Injury |          | Disease |               | Consequence |                 | Damage |           | Costs |        |
|---------------------------------|--------|----------|---------|---------------|-------------|-----------------|--------|-----------|-------|--------|
|                                 |        | Head     |         | Skin          |             | Sprain/Strain   |        | Buildings |       |        |
|                                 |        | Neck     |         | Respiratory   |             | Contusion       |        | Machines  |       |        |
|                                 |        | Eye      |         | Muscular      |             | Maceration      |        | Equipment |       |        |
|                                 |        | Back     |         | Skeletal      |             | Fracture        |        | Tools     |       |        |
|                                 |        | Trunk    |         | Hearing       |             | Burn            |        | Vehicles  |       |        |
|                                 |        | Arm      |         | Neurological  |             | Amputation      |        | Product   |       |        |
|                                 |        | Hand     |         | Psychological |             | Electric Shock  |        | Security  |       |        |
|                                 |        | Finger   |         | Endemic       |             | Asphyxiation    |        | Equipment |       |        |
|                                 |        | Leg      |         | Other         |             | Unconsciousness |        |           |       |        |
|                                 |        | Foot     |         |               |             | Poisoning       |        |           |       |        |
|                                 |        | Toe      |         |               |             | Fatality        |        |           |       |        |
|                                 |        | Internal |         |               |             | Multiple        |        |           |       |        |
|                                 |        | Multiple |         |               |             |                 |        |           |       | Total= |

|                                       |                                      |
|---------------------------------------|--------------------------------------|
| <b>Associated Hazard Description:</b> | <b>Witnesses:</b>                    |
| <b>Resolution of Risks:</b>           | <b>Implemented Control Measures:</b> |
| <b>Lessons Learned:</b>               | <b>Supervisor's Name:</b>            |

| Accident Report Checklist                                           |                                                                 |                                                                                                |
|---------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Testimonies collected and attached?        | <input type="checkbox"/> Medical records received and attached? | <input type="checkbox"/> Photos taken and attached?                                            |
| <input type="checkbox"/> Description includes all relevant details? | <input type="checkbox"/> Hazard resolved?                       | <input type="checkbox"/> If exposure occurred, are concentration and monitoring data attached? |



## Appendix C: COVID-19 Management<sup>6</sup>

Although the region has not experienced severe outbreaks compared to other parts of the world, preventive measures remain essential to ensure the health and safety of all workers and community members associated with the construction project. The project will follow national guidelines and international best practices to minimize the risk of COVID-19 transmission.

### Recognition of COVID-19 Symptoms

- Primary Symptoms: Coughing, fever, shortness of breath, and difficulty breathing.
- Secondary Symptoms: Chills, body aches, sore throat, headache, diarrhea, nausea/vomiting, runny nose.
- Action: Workers will be educated on recognizing COVID-19 symptoms through awareness campaigns led by the Occupational Doctor, the Nurse and the E&S team. If a worker develops any symptoms, they must refrain from coming to work, notify their supervisor, and seek immediate consultation with the on-site nurse.

### Monitoring and Early Detection:

- Temperature Checks: Daily temperature checks will be conducted at all site entry points for workers and visitors.
- Temperature Threshold: Any individual with a temperature exceeding 37.8°C (100°F), as per World Health Organization (WHO) guidelines, will be referred immediately to the on-site nurse for evaluation.
- Further Evaluation: If the nurse identifies COVID-19 symptoms, the individual will be sent to the nearest hospital for a COVID-19 test and additional medical examination.

### Preventive Measures and Worker Awareness:

- Awareness Campaigns: Regular awareness sessions will be conducted by the Occupational Doctor and the E&S team, educating workers on symptoms, preventive actions, and the importance of early detection and reporting.
- Hygiene Practices: Workers and visitors will be encouraged to practice good hygiene, including frequent handwashing and using hand sanitizers provided throughout the site.
- Personal Protective Equipment (PPE): Masks will be mandatory for workers in enclosed spaces or when social distancing cannot be maintained. Sanitizers and handwashing stations will be installed at strategic locations across the project site.

### Procedures for Suspected COVID-19 Cases:

- Immediate Action: If a worker or visitor shows symptoms of COVID-19, the on-site nurse will conduct an evaluation.
- Hospital Referral: If the nurse suspects a potential COVID-19 case based on symptoms and the nurse's training, the individual will be referred to the nearest hospital for testing and further evaluation.
- Isolation Protocol: While awaiting test results, the individual will be advised to self-isolate to prevent possible spread.

---

<sup>6</sup> Refer to the **COVID Management Procedure** standalone document for more details.



**Adjustments to the Work Environment:**

- **Social Distancing:** Work activities will be adjusted to reduce close contact among workers. Social distancing measures will be implemented wherever feasible, particularly in break areas, meetings, and transportation arrangements.
- **Remote Work:** Remote work may be arranged for administrative staff if necessary, depending on the evolving pandemic context.
- **Cleaning and Disinfection:** Regular cleaning and disinfection of high-contact surfaces such as door handles, workstations, and common areas will be enforced.

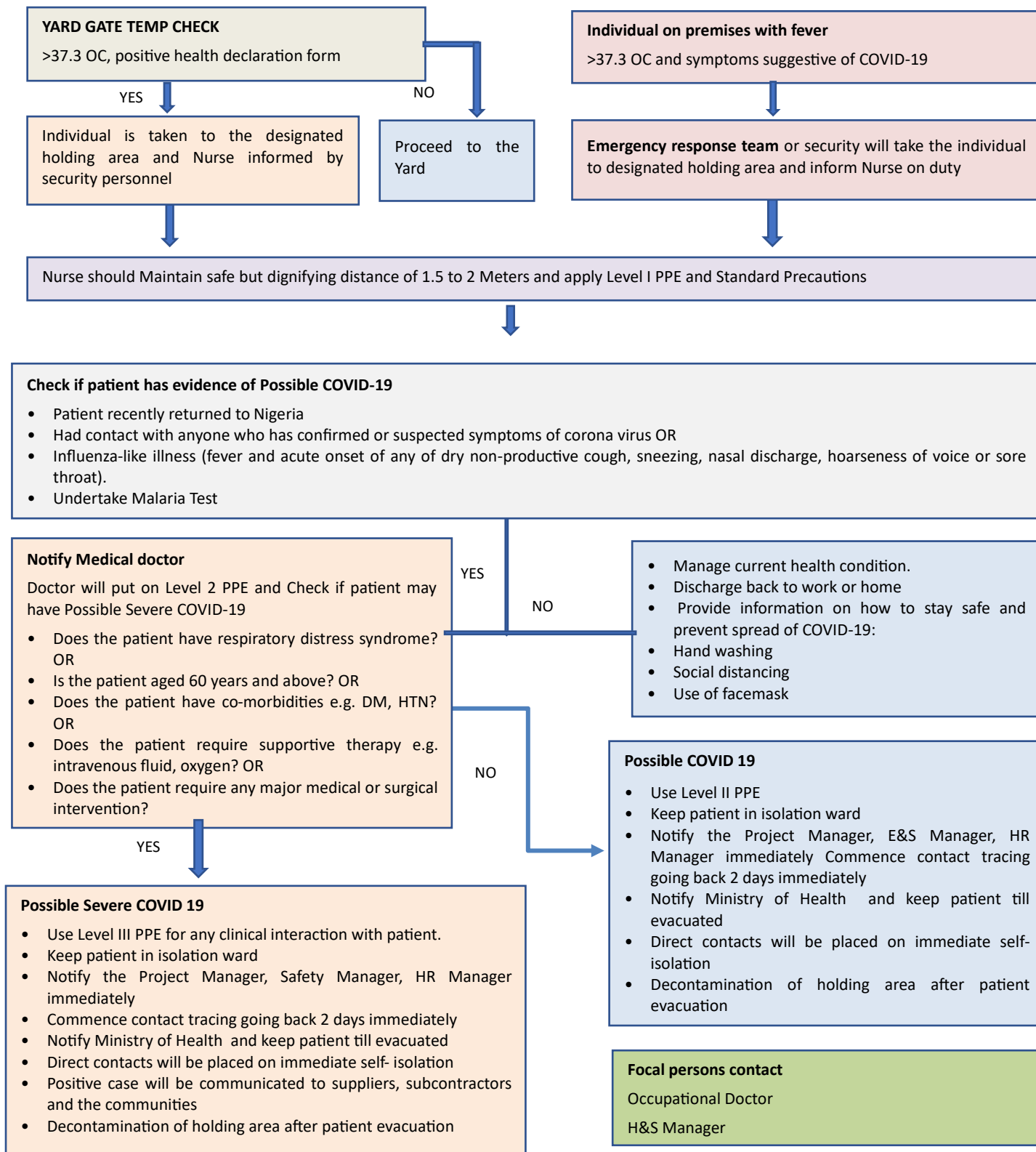
**Additional Protective Measures:**

- **Workforce Segmentation:** Where possible, workers will be grouped into smaller teams to limit widespread exposure. If a COVID-19 case is confirmed, it will allow for quicker isolation of affected teams.
- **Visitor Restrictions:** Access to the construction site will be limited to essential personnel only. Visitors will undergo the same screening and temperature checks as workers.
- **Travel Restrictions:** Workers returning from high-risk areas will be required to follow quarantine procedures in line with national health guidelines.

**Collaboration with Local Health Authorities:**

- The project will maintain close coordination with local health authorities in Lagos to stay informed about any changes in the pandemic situation. All COVID-19 cases or suspected cases will be reported to authorities, and the project will follow any additional guidance provided by health officials.

The Tin Can Port Modernization Project in Lagos is committed to maintaining a safe work environment amid the ongoing global pandemic. By implementing preventive measures, regular monitoring, and awareness programs, the project will ensure the safety of its workers, visitors, and the broader community, allowing operations to continue with minimal disruption.

**COVID - 19 SITE EMERGENCY RESPONSE**

## Appendix D: Waste Inventory Log

The standard waste log is expected to be updated on a **daily/weekly** basis.

| Tin Can Port Modernization Project |            |                      |                     |                                                       |                       |          |                                   |      |            |                                                              |
|------------------------------------|------------|----------------------|---------------------|-------------------------------------------------------|-----------------------|----------|-----------------------------------|------|------------|--------------------------------------------------------------|
| Waste Inventory Log                |            |                      |                     |                                                       |                       |          |                                   |      |            |                                                              |
| Period                             |            |                      |                     |                                                       | Data Logger           |          |                                   |      |            |                                                              |
| Issued on                          |            |                      |                     |                                                       | Location<br>Site Code |          |                                   |      |            |                                                              |
| Record                             |            |                      |                     |                                                       |                       |          |                                   |      |            |                                                              |
| N <sup>7</sup>                     | Date       | Type ID <sup>8</sup> | Waste Type          | Description                                           | Source                | Quantity | Storage                           | MSDS | Logger     | Comments                                                     |
| 1                                  | 01/01/2024 | 3                    | Construction Debris | Cement and concrete debris from concrete mixing works | Project Site A        | 500 kg   | LV - Onsite drum (s) <sup>9</sup> | N/A  | Eng Claude | This is residual cement only and is appropriately segregated |
| 2                                  |            |                      |                     |                                                       |                       |          |                                   |      |            |                                                              |
| 3                                  |            |                      |                     |                                                       |                       |          |                                   |      |            |                                                              |
| 4                                  |            |                      |                     |                                                       |                       |          |                                   |      |            |                                                              |
| 5                                  |            |                      |                     |                                                       |                       |          |                                   |      |            |                                                              |

<sup>7</sup> A separate numbering system could be adopted once the project is in execution phase with the subcontractor

<sup>8</sup> In the developed excel sheet/database for this tool, please developed ID numbers for the items under each type

<sup>9</sup> (s) would be a reference to segregated or mixed as an indicator for the current status of the storage onsite and LV would be the code for a designated location



## Appendix E: Waste Disposal Log

| Tin Can Port Modernization Project |      |            |                      |                                           |                              |                       |                                     |                            |                                    |                                          |
|------------------------------------|------|------------|----------------------|-------------------------------------------|------------------------------|-----------------------|-------------------------------------|----------------------------|------------------------------------|------------------------------------------|
| Waste Disposal Log                 |      |            |                      |                                           |                              |                       |                                     |                            |                                    |                                          |
| Period                             |      |            |                      |                                           | Data<br>Logger               |                       |                                     |                            |                                    |                                          |
| Issued<br>on                       |      |            |                      |                                           | Location<br><i>Site Code</i> |                       |                                     |                            |                                    |                                          |
| Record                             |      |            |                      |                                           |                              |                       |                                     |                            |                                    |                                          |
| N                                  | Date | Waste Type | Quantity<br>of Waste | Confirmation<br>of Disposal<br>Contractor | Time<br>logged               | Time of<br>Collection | Project<br>Coordinator<br>Signature | Subcontractor<br>Signature | Disposal<br>Location and<br>Method | Vehicle Registration<br>Number + Contact |
| 1                                  |      |            |                      |                                           |                              |                       |                                     |                            |                                    |                                          |
| 2                                  |      |            |                      |                                           |                              |                       |                                     |                            |                                    |                                          |
| 3                                  |      |            |                      |                                           |                              |                       |                                     |                            |                                    |                                          |
| 4                                  |      |            |                      |                                           |                              |                       |                                     |                            |                                    |                                          |
| 5                                  |      |            |                      |                                           |                              |                       |                                     |                            |                                    |                                          |
| 6                                  |      |            |                      |                                           |                              |                       |                                     |                            |                                    |                                          |
| 7                                  |      |            |                      |                                           |                              |                       |                                     |                            |                                    |                                          |
| 8                                  |      |            |                      |                                           |                              |                       |                                     |                            |                                    |                                          |





[illegible]

## Appendix G: Hazardous Waste Log

| Tin Can Port Modernization Project<br>Hazardous Waste Monitoring Log |                                          |                                |          |                 |                    |          |                       |                         |
|----------------------------------------------------------------------|------------------------------------------|--------------------------------|----------|-----------------|--------------------|----------|-----------------------|-------------------------|
| Period                                                               |                                          | Inspector                      |          |                 |                    |          |                       |                         |
| Issued on                                                            |                                          | Location<br>Site Code (if any) |          |                 |                    |          |                       |                         |
| Record                                                               |                                          |                                |          |                 |                    |          |                       |                         |
| N                                                                    | Item Name + Identification Number (MSDS) | Physical State                 | Quantity | Date of sorting | Date of collection | Location | Coordinator Signature | Subcontractor Signature |
| 1                                                                    |                                          |                                |          |                 |                    |          |                       |                         |
| 2                                                                    |                                          |                                |          |                 |                    |          |                       |                         |
| 3                                                                    |                                          |                                |          |                 |                    |          |                       |                         |
| 4                                                                    |                                          |                                |          |                 |                    |          |                       |                         |
| 5                                                                    |                                          |                                |          |                 |                    |          |                       |                         |
| 6                                                                    |                                          |                                |          |                 |                    |          |                       |                         |
| 7                                                                    |                                          |                                |          |                 |                    |          |                       |                         |
| 8                                                                    |                                          |                                |          |                 |                    |          |                       |                         |



## Appendix H: Training Log

| Tin Can Port Modernization Project                                 |      |      |                |               |                                |               |          |        |          |                     |                     |
|--------------------------------------------------------------------|------|------|----------------|---------------|--------------------------------|---------------|----------|--------|----------|---------------------|---------------------|
| Construction Environmental and Social Management Plan Training Log |      |      |                |               |                                |               |          |        |          |                     |                     |
| Period                                                             |      |      |                |               | OHS Personnel                  |               |          |        |          |                     |                     |
| Issued on                                                          |      |      |                |               | Location<br>Site Code (if any) |               |          |        |          |                     |                     |
| Record                                                             |      |      |                |               |                                |               |          |        |          |                     |                     |
| #                                                                  | Date | Time | Training Title | Attendee/Name | Role                           | Training Type | Duration | Status | Comments | Signature (trainer) | Signature (trainee) |
| 1                                                                  |      |      |                |               |                                |               |          |        |          |                     |                     |
| 2                                                                  |      |      |                |               |                                |               |          |        |          |                     |                     |
| 3                                                                  |      |      |                |               |                                |               |          |        |          |                     |                     |
| 4                                                                  |      |      |                |               |                                |               |          |        |          |                     |                     |
| 5                                                                  |      |      |                |               |                                |               |          |        |          |                     |                     |
| 6                                                                  |      |      |                |               |                                |               |          |        |          |                     |                     |

A printed format should be provided for workers to sign on their respective status

Annex I - Sample Contractual Clauses



- “The Subcontractor shall comply with all requirements of the Environmental and Social Management Plan (ESMP) and Construction ESMP (CESMP), which form part of this Contract.”
- “The Subcontractor shall perform all works in accordance with the IFC Performance Standards on Environmental and Social Sustainability and the World Bank Group EHS Guidelines.”
- “The Subcontractor shall submit monthly E&S performance reports, including incidents, training conducted, and corrective actions taken.”
- “The Contractor reserves the right to conduct E&S audits of the Subcontractor’s operations at any time. Findings shall be addressed within 14 days.”
- “Failure to comply with E&S obligations may result in suspension of work, financial penalties, or termination of this Contract.”

